X-S32N79-RDB-S
REV A

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Revisions

Rev	Description	Date	Approved
X1	**Reference Schematics - 94936(Rev-B2)** 01.U15 pin 6 is connected to the output net VDD_3.3V_DBG. 02.Added PU R137(4.7 kO) on U57 pin 4 03.J28 default position moved from 2-3 to 1-2 04.R996 DNP, mounted R997(4.7K) 05.Added C1359(1uf) 06.DNP R540 and U93 pin3 connected to VSS via R1358. 07.Added U155 circuit to enable D68 LED 08.I2C_CNTL_CH1_SCL/SDA connected to FS5600(U51). 09.R1025 made DNP. 10.U156 circuit added to either select PRPE from J19(CEM) or J3 (MCIO) 11.U63 APN updated to 315-10810 12.U49 APN updated to 315-10763 13.Mount default: R1294/R1295 14.Added R1355, R1356 on unused VMON pin of U49. 15.U50 Footprint Changed to Soldered 16.R766-769 changed to 0E and connect to GND 17.R801-R804 made PU and DNP 18.L21 Part# changed 19.BH1, BH2, BH3 Part number updated 20.R1350, R1351 added optional as DNP	26 Feb 26	
X2	Updated Review Comments. R898.1 connected to R889.1 R835.1 connected to R830.1 R1363 added as DNP U41, U46, U56 made DNP TP12 Deleted	06 Mar 26	
X3	DNP Page-13 Components A070 Release U15 changed to higher I/P-V limit part TP190 deleted U157, U158 circuit added U159,U162 circuit added for U27 POcycle U160,U161,U163 circuit added for FSSWKUP Q53, U153 Circuit deleted Changed U60 Part number	13 Mar 26 17 Mar 26 24 Mar 26 15 Apr 26 23 Apr 26 29 Apr 26	
A	U149 Part changed to SDINBDA6-64G-ZA Final Release- A085	04 May 26 07 May 26	

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<Variant Name>

Automotive Product Group
6501 William Cannon Drive West
Austin, TX 78735-8598

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Designer: SUMIT	Drawing Title: S32N79-RDB-S		
Drawn by: SUMIT	Page Title: 01-TITLE		
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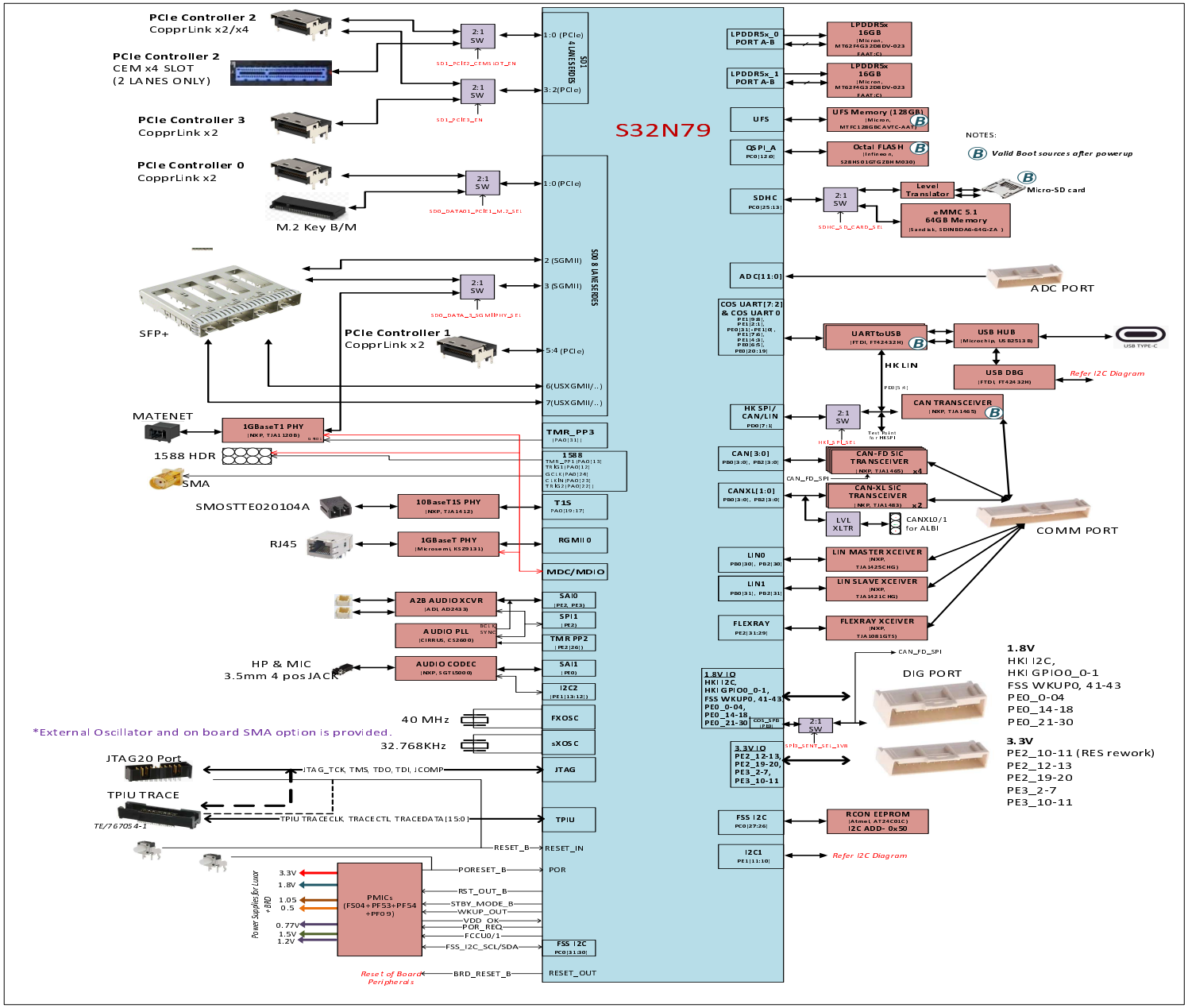
BLOCK DIAGRAM



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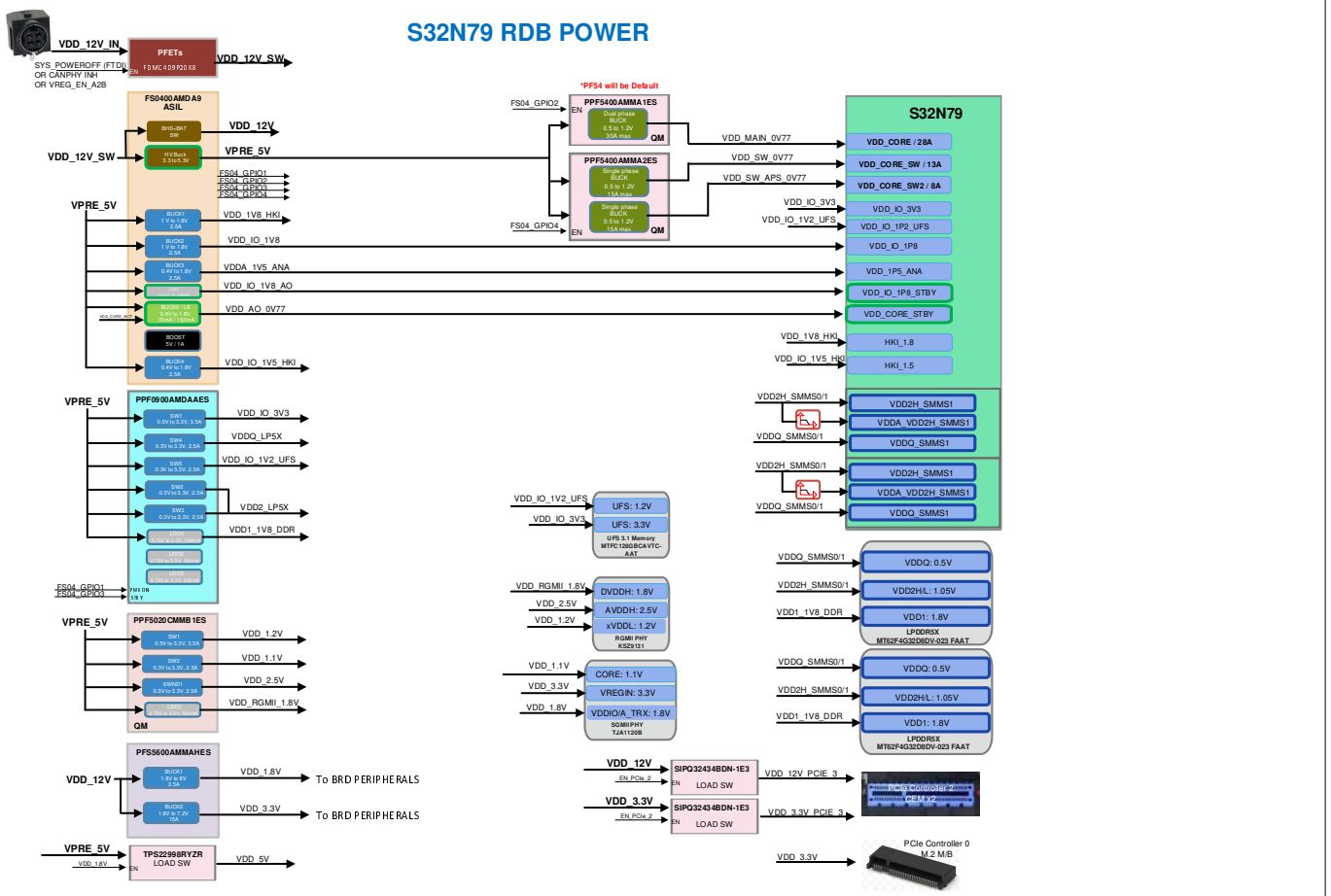
S32N79 RDB POWER



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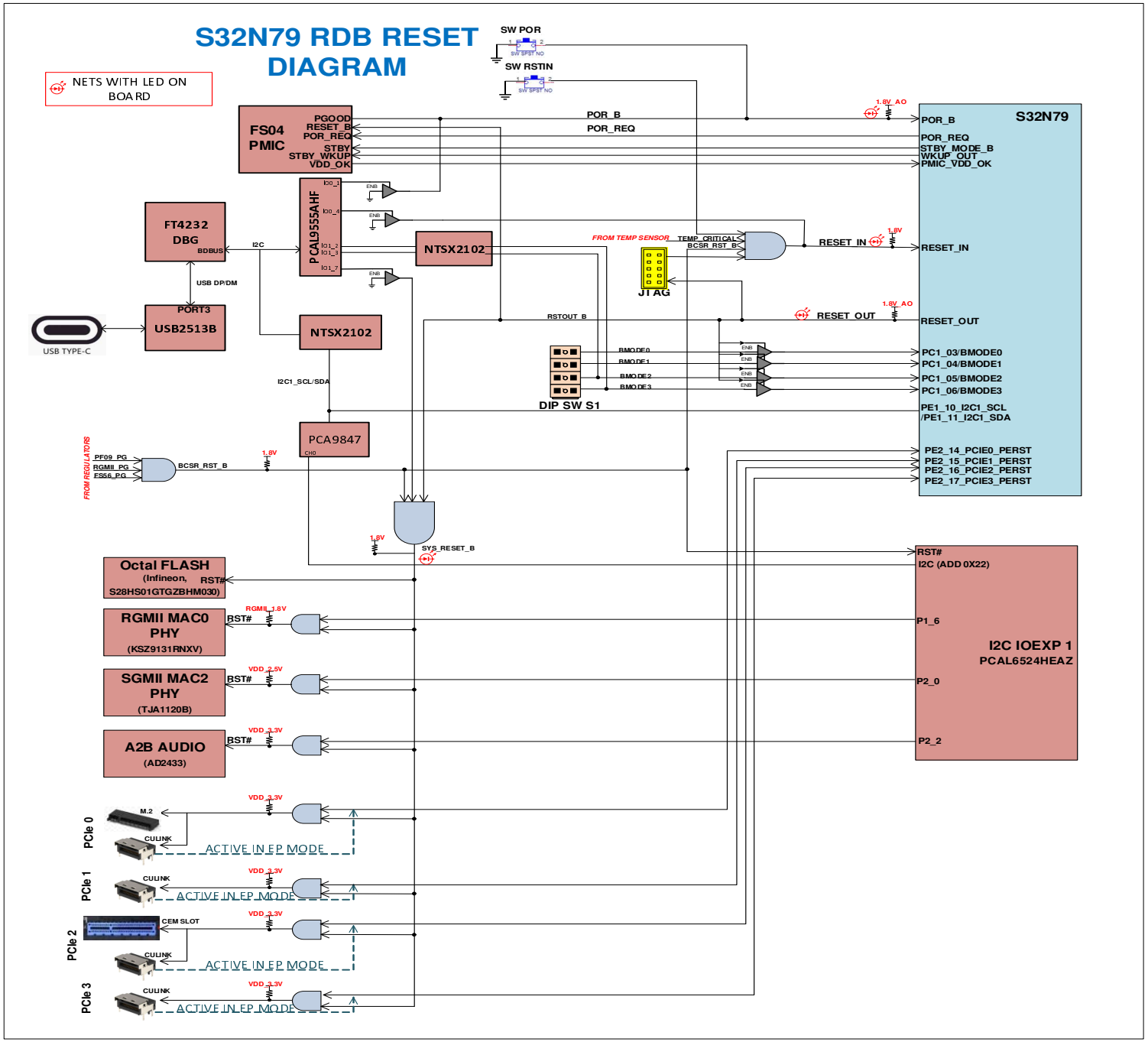
S32N79 RDB RESET DIAGRAM



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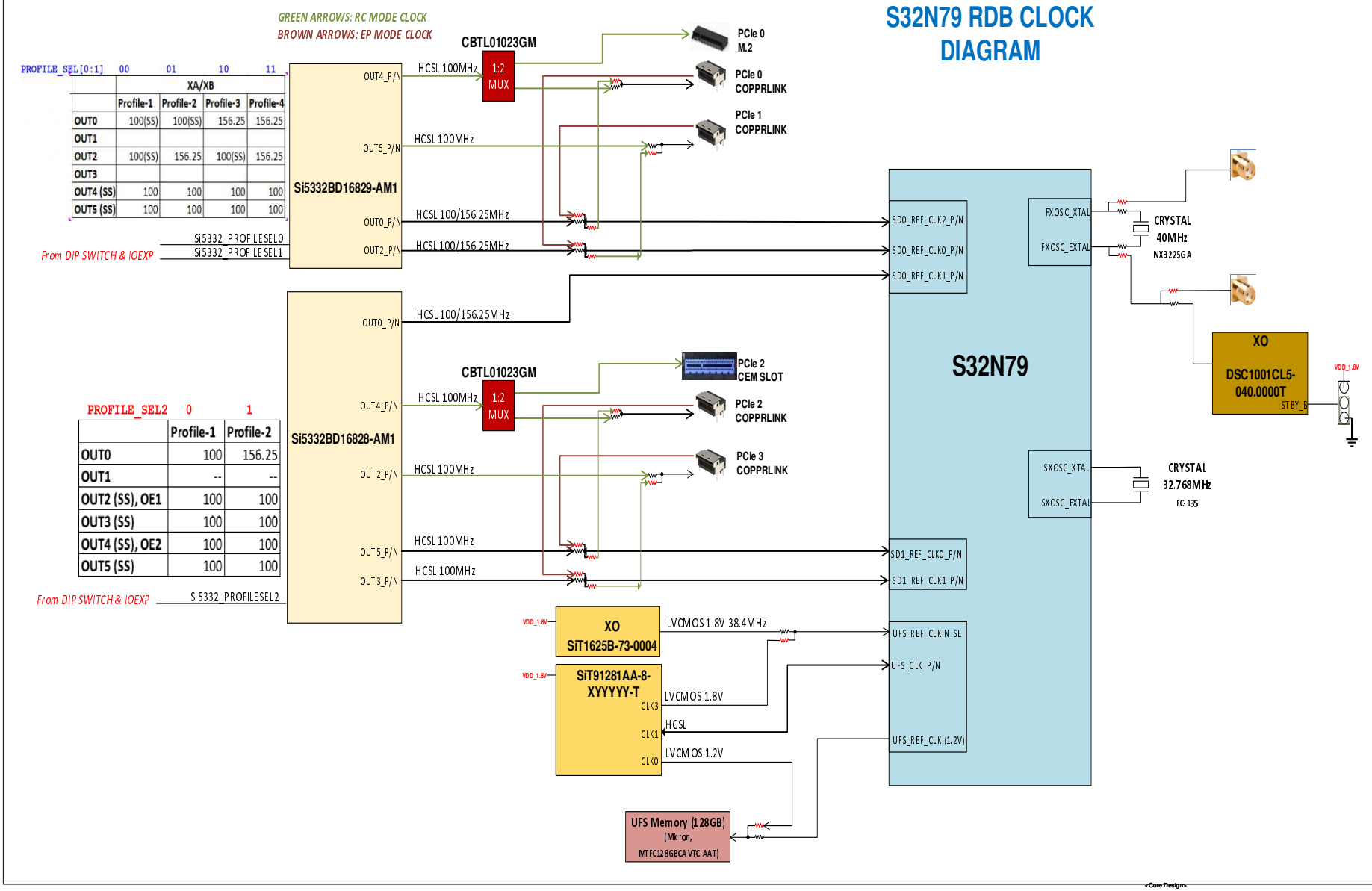

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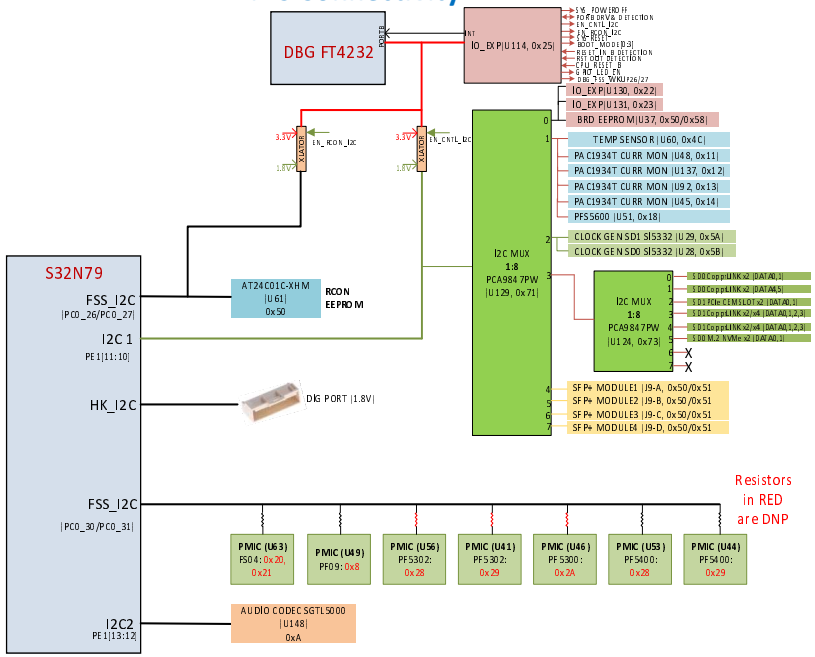
I2C Connectivity



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DIP SWITCH CONTROLS FOR BOARD

S NO	MUX SELECT IOEXPANDER Signal name	DIPSW	DEFAULT	IOEXPANDER ADDR/PORT	When Signal = 0	When Signal = 1
1	BMODE0	S3[1]	0		Mode selection for XOSC BMODE1 BMODE0 MODE **0 0 CRYSTAL MODE** 0 1 DIFFERENTIAL BYPASS MODE 1 0 SINGLE ENDED BYPASS MODE 1 1 RESERVED	
2	BMODE1	S3[2]	0			
3	BMODE2	S3[3]	0	0x25/IO1[2]	Mode selection for BOOT BMODE3 BMODE2 MODE 0 0 SERIAL BOOT 0 1 SERIAL BOOT **1 0 STANDARD BOOT** 1 1 RESERVED/TEST MODE	
4	BMODE3	S3[4]	1	0x25/IO1[3]		
5	SDHC_SD_CARD_SEL	S3[5]	1	0x23/P0[0]	eMMC	SD Card
6	SD0_DATA01_PCIE1_M.2_SEL	S3[6]	0	0x23/P0[1]	SD0 SLOT1 REFCLK -> CopprLINK SLOT1	SD0 SLOT1 REFCLK -> M.2 SLOT
7	SD0_DATA_3_SGMIIPHY_SEL	S3[7]	0	0x23/P0[2]	SD0_LANE3 -> SFP3	SD0_LANE3 -> TJA1120B PHY
8	HKI_SPI_SEL_B	S3[8]	1	0x23/P0[6]	Enable HKI SPI connection to CAN PHYs.	Enable Serial LIN and CAN BOOT
9	JTAG_FTDI_SEL	S3[9]	0	0x23/P0[4]	JTAG -> JTAG ARM Connector	JTAG -> FTDI DBG Port A
10	EN_PORB_BMODE_FRM_FTDI_B	S3[10]	1	0x23/P0[5]	DBG FTDI can not control BMODE2,3 pins	BMODE2,3 can be controlled from DBG FTDI.
11	SD1_PCIE2_CEMSLOT_EN	S2[1]	0	0x23/P0[7]	SD1 LANE0/1 -> CopprLink x2/x4	SD1 LANE0/1 -> PCIe CEM Slot (x4 but only 2 lanes routed)
12	SD1_PCIE3_EN	S2[2]	0	0x23/P1[0]	SD1 LANE2/3 -> CopprLink x4	SD1 LANE2/3 -> CopprLink x2
13	SD0_SLOT1_EP_SEL	S2[3]	0	0x22/P0[4]	PCIE1 CopprLink set as RC	PCIE1 CopprLink set as EP
14	SD0_SLOT2_EP_SEL	S2[4]	0	0x22/P0[5]	PCIE2 CopprLink set as RC	PCIE2 CopprLink set as EP
15	SD0_SLOT3_EP_SEL	S2[5]	0	0x22/P0[6]	PCIE3 CopprLink set as RC	PCIE3 CopprLink set as EP
16	SD0_SLOT4_EP_SEL	S2[6]	0	0x22/P0[7]	PCIE4 CopprLink set as RC	PCIE4 CopprLink set as EP
17	SFP1_TX_DISABLE	S2[7]	1	0x22/P1[0]	Enable TX of SFP1 (Caution: setting TX as ON enables the laser driver on optical modules)	Disable TX of SFP1
18	SFP2_TX_DISABLE	S2[8]	1	0x22/P1[4]	Enable TX of SFP2 (Caution: setting TX as ON enables the laser driver on optical modules)	Disable TX of SFP2
19	SFP3_TX_DISABLE	S2[9]	1	0x22/P2[0]	Enable TX of SFP3 (Caution: setting TX as ON enables the laser driver on optical modules)	Disable TX of SFP3
20	SFP4_TX_DISABLE	S2[10]	1	0x22/P2[4]	Enable TX of SFP4 (Caution: setting TX as ON enables the laser driver on optical modules)	Disable TX of SFP4
21	Si5332_PROFILESEL0	S1[1]	0	0x22/P0[0]	Profile selection for Clock generator 0 [SERDES0 Instance only]. PROFILESEL[0:1] **00** 01 10 11 OUT0 (REFCLK2) **100** 100 156.25 156.25 OUT2 (REFCLK0) **100** 156.25 100 156.25 OUT4 (SLOT1) **100** 100 100 100 OUT5 (SLOT2) **100** 100 100 100 **All values in MHz. 100MHz clock can support SSthrough I2C accesible register.	
22	Si5332_PROFILESEL1	S1[2]	0	0x22/P0[1]		
23	Si5332_PROFILESEL2	S1[3]	0	0x22/P0[2]	Profile selection for clock generator 1 (SERDES0 REFCLK1 & SERDES1) PROFILESEL[2] **0** 1 OUT0 (SD0 REFCLK1) **100** 156.25 * Rest of the clocks on clock gen 2 is set as 100MHz.	
24	SPI3_SENT_SEL	S1[4]	1	0x23/P0[3]	MUX SEL Definition For COS_SPI3 routing to CAN_FD SPI3_SENT_SEL **0** 1 OUT0 (SD0 REFCLK1) 1.8V DIG PORT **SPI3 (DEFAULT)**	
25	OE0	S4[1]	0	NA	UFS CLK GEN U94 CLK0 -> **Disable**	UFS CLK GEN U94 CLK0 -> Enable
26	OE1	S4[2]	0	NA	UFS CLK GEN U94 CLK0 -> **Disable**	UFS CLK GEN U94 CLK0 -> Enable
27	FS0	S4[3]	0	NA	UFS CLOCK FREQUENCY OUTPUT SELECTION FS1 FS0 CLK0, CLK1, CLK3 Frequency [MHz] 0 0 19.2 0 1 26.0 **1 0 38.4(DEFAULT)** 1 1 52.0	
28	FS1	S4[4]	1	NA		

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BOARD CONTROL AND STATUS SIGNALS ASSIGNED TO IOEXPANDER

IOEXPANDER (I2C ADDR)	PORT[BIT]	SIGNAL	DIR	When Signal = 0	When Signal = 1	DEFAULT
U131 (0x23)	P0[0]	SDHC_SD_CARD_SEL	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[1]	SD0_DATA01_PCIE1_M.2_SEL	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[2]	SD0_DATA_3_SGMIIPHY_SEL	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[3]	SPI3_SENT_SEL	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[4]	JTAG_FTDI_SEL	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[5]	EN_PORB_BMODE_FRM_FTDI_	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[6]	HKI_SPI_SEL_B	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P0[7]	SD1_PCIE2_CEMSLOT_EN	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P1[0]	SD1_PCIE3_EN	IN/OUT	REFER DIPSW TABLE		
U131 (0x23)	P1[1]	CANXL0_EN_3.3V	OUT	CANXL PHY0 disabled (in sleep mode)	CANXL PHY0 enabled	1
U131 (0x23)	P1[2]	CANXL1_EN_3.3V	OUT	CANXL PHY1 disabled (in sleep mode)	CANXL PHY1 enabled	1
U131 (0x23)	P1[3]	FR0_A_EN	OUT	FLEXRAY PHY disabled	FLEXRAY PHY enabled	1
U131 (0x23)	P1[4]	LIN0_SLP_B_3.3V	OUT	LIN0 PHY in sleep mode	LIN0 PHY enabled	1
U131 (0x23)	P1[5]	LIN1_SLP_B_3.3V	OUT	LIN1 PHY in sleep mode	LIN1 PHY enabled	1
U131 (0x23)	P1[6]	KSZ_A_RST	OUT	RESET RGMII 1 PHY (KSZ9131)	RGMII 1 PHY (KSZ9131) out of reset	1
U131 (0x23)	P1[7]	KSZ_A_INT_3.3V_B	INPUT	Interrupt from RGMII 1 PHY (KSZ9131)	No interrupt from RGMII 1 PHY (KSZ9131)	1
U131 (0x23)	P2[0]	TJA1120B_PHY_RST_B	OUT	RESET SGMII MAC2 PHY (TJA1120B)	SGMII MAC2 PHY (TJA1120B) out of reset	1
U131 (0x23)	P2[1]	TJA1020B_PHY_INT_3.3V_B	INPUT	Interrupt from SGMII MAC2 PHY (TJA1120B)	No interrupt from SGMII MAC2 PHY (TJA1120B)	1
U131 (0x23)	P2[2]	AD2433_RST_B	OUT	RESET A2B AUDIO PHY (AD2433)	A2B AUDIO PHY (AD2433) out of reset	1
U131 (0x23)	P2[3]	AD2433_INT_B	INPUT	Interrupt from A2B AUDIO PHY (AD2433)	No interrupt from A2B AUDIO PHY (AD2433)	1
U131 (0x23)	P2[5]	BRD_VER0	INPUT			0
U131 (0x23)	P2[6]	BRD_VER1	INPUT	Revision of Board (VER[2:0])		0
U131 (0x23)	P2[7]	BRD_VER2	INPUT	000 - REV A		0
U130 (0x22)	P0[0]	SI5332_PROFILESEL0	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[1]	SI5332_PROFILESEL1	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[2]	SI5332_PROFILESEL2	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[3]	TEST POINT				
U130 (0x22)	P0[4]	SD0_SLOT1_EP_SEL	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[5]	SD0_SLOT2_EP_SEL	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[6]	SD1_SLOT3_EP_SEL	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P0[7]	SD1_SLOT4_EP_SEL	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P1[0]	SFP1_TX_DISABLE	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P1[1]	SFP1_TX_FAULT	IN	SFP1 Transmitter is operational	SFP1 Transmitter is in fault	0*
U130 (0x22)	P1[2]	SFP1_RX_LOS	IN	SFP1 Receiver is operational	Loss Of Signal on SFP1 Receiver	0*
U130 (0x22)	P1[3]	SFP1_MOD_ABS	IN	SFP1 Module is present	SFP1 module is not detected	0*
U130 (0x22)	P1[4]	SFP2_TX_DISABLE	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P1[5]	SFP2_TX_FAULT		SFP2 Transmitter is operational	SFP2 Transmitter is in fault	0*
U130 (0x22)	P1[6]	SFP2_RX_LOS		SFP2 Receiver is operational	Loss Of Signal on SFP2 Receiver	0*
U130 (0x22)	P1[7]	SFP2_MOD_ABS		SFP2 Module is present	SFP2 module is not detected	0*
U130 (0x22)	P2[0]	SFP3_TX_DISABLE	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P2[1]	SFP3_TX_FAULT		SFP3 Transmitter is operational	SFP3 Transmitter is in fault	0*
U130 (0x22)	P2[2]	SFP3_RX_LOS		SFP3 Receiver is operational	Loss Of Signal on SFP3 Receiver	0*
U130 (0x22)	P2[3]	SFP3_MOD_ABS		SFP3 Module is present	SFP3 module is not detected	0*
U130 (0x22)	P2[4]	SFP4_TX_DISABLE	IN/OUT	REFER DIPSW TABLE		
U130 (0x22)	P2[5]	SFP4_TX_FAULT		SFP4 Transmitter is operational	SFP4 Transmitter is in fault	0*
U130 (0x22)	P2[6]	SFP4_RX_LOS		SFP4 Receiver is operational	Loss Of Signal on SFP4 Receiver	0*
U130 (0x22)	P2[7]	SFP4_MOD_ABS		SFP4 Module is present	SFP4 module is not detected	0*
U114 (0x25)	P0[0]	SYS POWEROFF	OUTPUT	SYSTEM POWER IS ENABLED	SYSTEM POWER IS OFF	0
U114 (0x25)	P0[1]	POR_DRV_B	OUTPUT	Assert Power-on-reset (PORB) to SoC	Deassert Power-on-reset (PORB) to SoC	1
U114 (0x25)	P0[2]	EN_CNTL_I2C	OUTPUT	Disable FTDI I2C access to system I2C	Enable FTDI I2C access to system I2C	0
U114 (0x25)	P0[3]	EN_RCON_I2C	OUTPUT	Disable FTDI I2C access to RCON EEPROM	Enable FTDI I2C access to RCON EEPROM	0
U114 (0x25)	P0[5]	PORB_DETECTION	INPUT	Status of PORB signal to Soc is asserted.	Status of PORB signal to Soc is de-asserted.	
U114 (0x25)	P0[6]	RESET_IN_B_DETECTION	INPUT	Status of RESET_IN signal to Soc is asserted.	Status of RESET_IN signal to Soc is de-asserted.	
U114 (0x25)	P0[7]	RESET_OUT_B_DETECTION	INPUT	Status of RESET_OUT signal from Soc is asserted.	Status of RESET_OUT signal from Soc is de-asserted.	
U114 (0x25)	P1[0]					
U114 (0x25)	P1[1]					
U114 (0x25)	P1[2]	DBG_BMODE2	IN/OUT	REFER DIPSW TABLE		
U114 (0x25)	P1[3]	DBG_BMODE3	IN/OUT	REFER DIPSW TABLE		
U114 (0x25)	P1[4]					
U114 (0x25)	P1[5]					
U114 (0x25)	P1[6]	GPIO_LED_EN	OUTPUT	FTDI DEBUG LED Enable. (assert LED For FTDI identification)	FTDI DEBUG LED Disable (Default)	0
U114 (0x25)	P1[7]	SYS_RST_B	OUTPUT	Assert reset to the devices on board except IOEXPANDERS.	De-assert reset to the devices on board except IOEXPANDERS.	1

*SFP module signals status default is with modules plugged in SFP cages and Tx enabled. For No module all signals are HIGH (1) by default

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STACKUP

Part Number: 94936 Rev
Program: N/A
TTM #

Target Thickness: 78.740 (± 10%)
Total Thickness: 77.580
Site: HY

Over mask on plated copper

Lyr	Lyr Type	Image	Foil Wt	Thk (mil)	Cu Thk (mil)	Dk@1 GHz	Generic Name	Construction	Material Family	Df @ 1GHz
tsmp				0.600			PSR4000 G23K			
1	Sig		0.333		1.800		1/3 oz HTE			
				2.680		3.2	1078		TU-883P	0.003
2	Pln		0.333		0.800		1/3 oz HTE			
				2.860		3.09	1078		TU-883P	0.003
3	Sig		0.333		0.800		1/3 oz HTE			
				2.900		2.89	1067		TU-883P	0.0029
4	Pln		0.333		1.000		1/3 oz HTE			
				2.860		2.89	1067		TU-883P	0.0029
5	Sig		0.5	5.000	0.550	3.45	Core 0.127 mm 0.5oz HVLP/2oz RTF	2116	TU-883	0.0031
6	Pln		2		2.400					
				7.040		3.42	1080		TU-883P	0.0029
						3.42	1080			
				15.000		3.45	Core 0.380 mm Filling core	3x2116	TU-883	0.0031
				7.040		3.42	1080		TU-883P	0.0029
						3.42	1080			
7	Pln		2	5.000	2.400	3.45	Core 0.127 mm 0.5oz HVLP/2oz RTF	2116	TU-883	0.0031
8	Sig		0.5		0.550					
				2.860		2.89	1067		TU-883P	0.0029
9	Pln		0.333		1.000		1/3 oz HTE			
				2.900		2.89	1067		TU-883P	0.0029
10	Sig		0.333		0.800		1/3 oz HTE			
				2.860		3.09	1078		TU-883P	0.003
11	Pln		0.333		0.800		1/3 oz HTE			
				2.680		3.2	1078		TU-883P	0.003
12	Sig		0.333		1.800		1/3 oz HTE			
bsmp				0.600						

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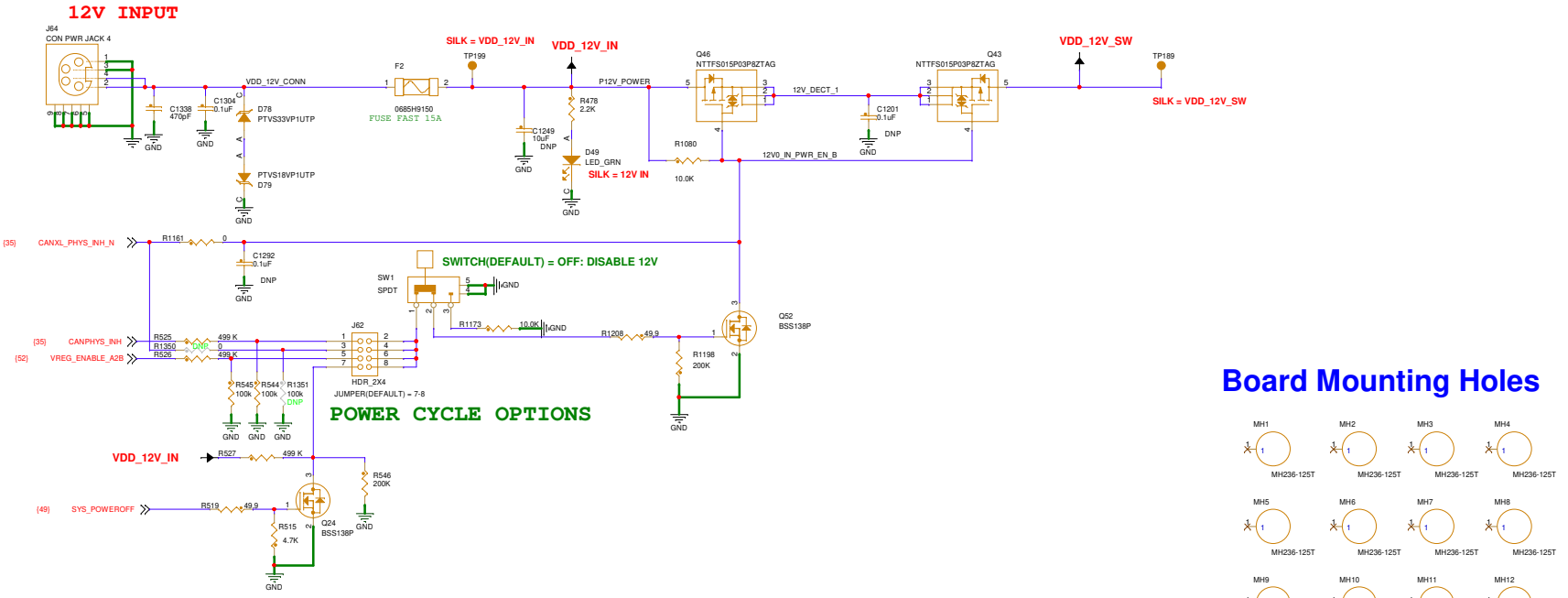
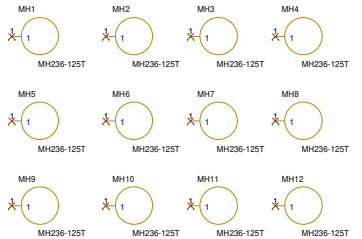
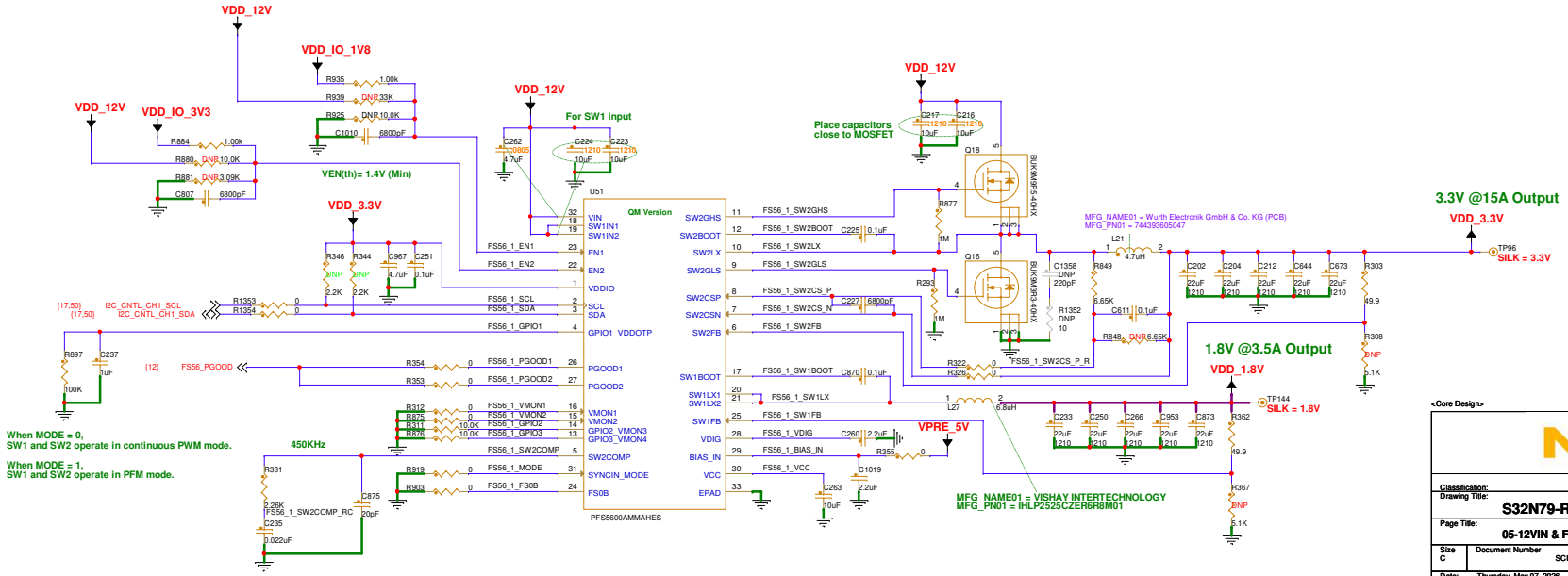
POWER INPUT



Board Mounting Holes



PMIC FS5600



<Core Design>

Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **05-12VIN & FS04**

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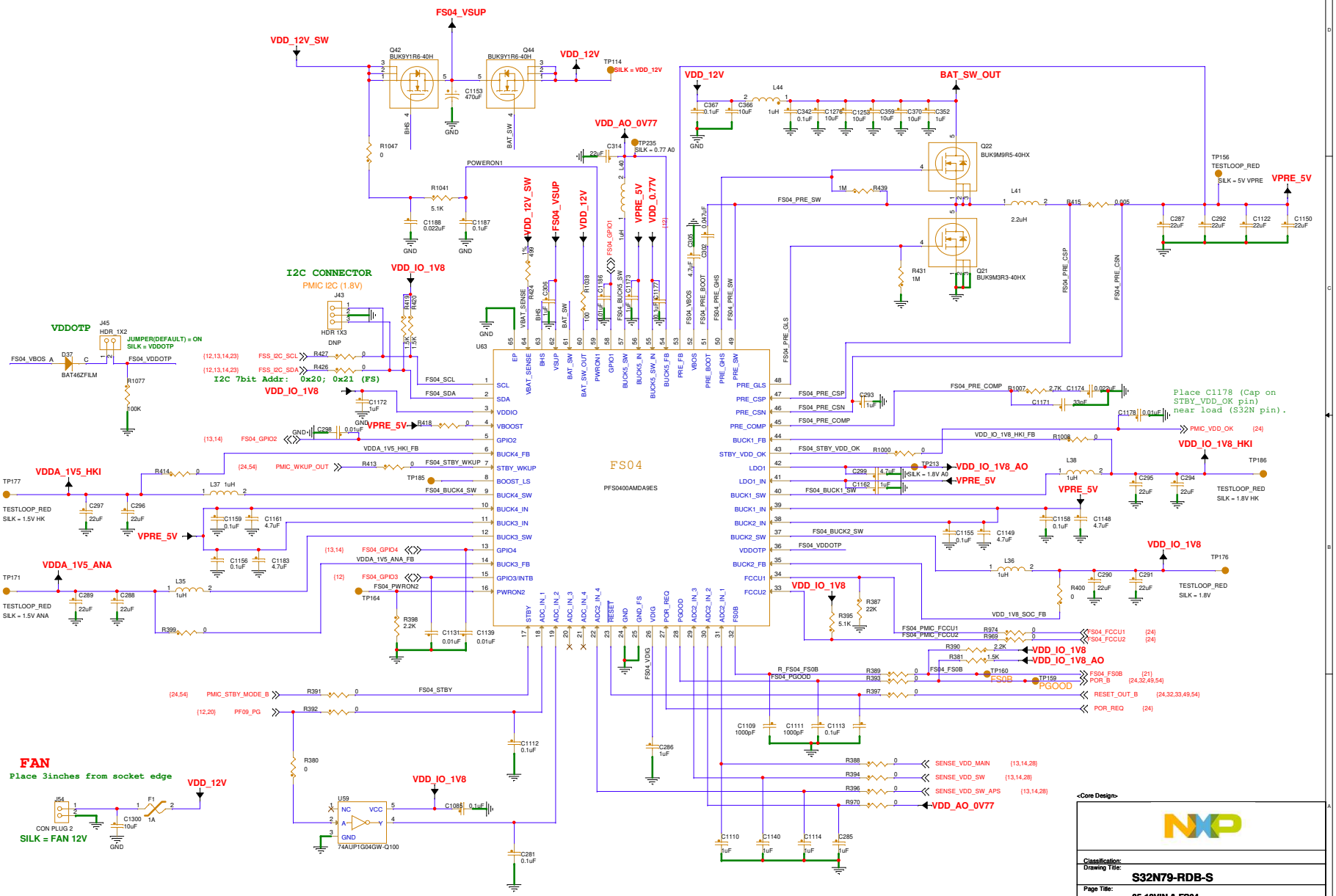
FS04


FS04_VSUP
VDD_12V_SW
VDD_12V
VDD_AO_0V77
VDD_IO_1V8
VPRE_5V
BAT_SW_OUT
I2C CONNECTOR
PMIC I2C (1.8V)
I2C 7bit Addr: 0x20; 0x21 (FS)
VDDOTP
FS04
PFS0400AMDA9ES
FAN
Place 3inches from socket edge
SILK = FAN 12V
VDDA_1V5_HKI
VDDA_1V5_ANA
VDD_IO_1V8_AO
VDD_IO_1V8_HKI
Place C1178 (Cap on STBY_VDD_OK pin) near load (S32N pin).
<Core Design>
Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
05-12VIN & FS04
Size C
Document Number
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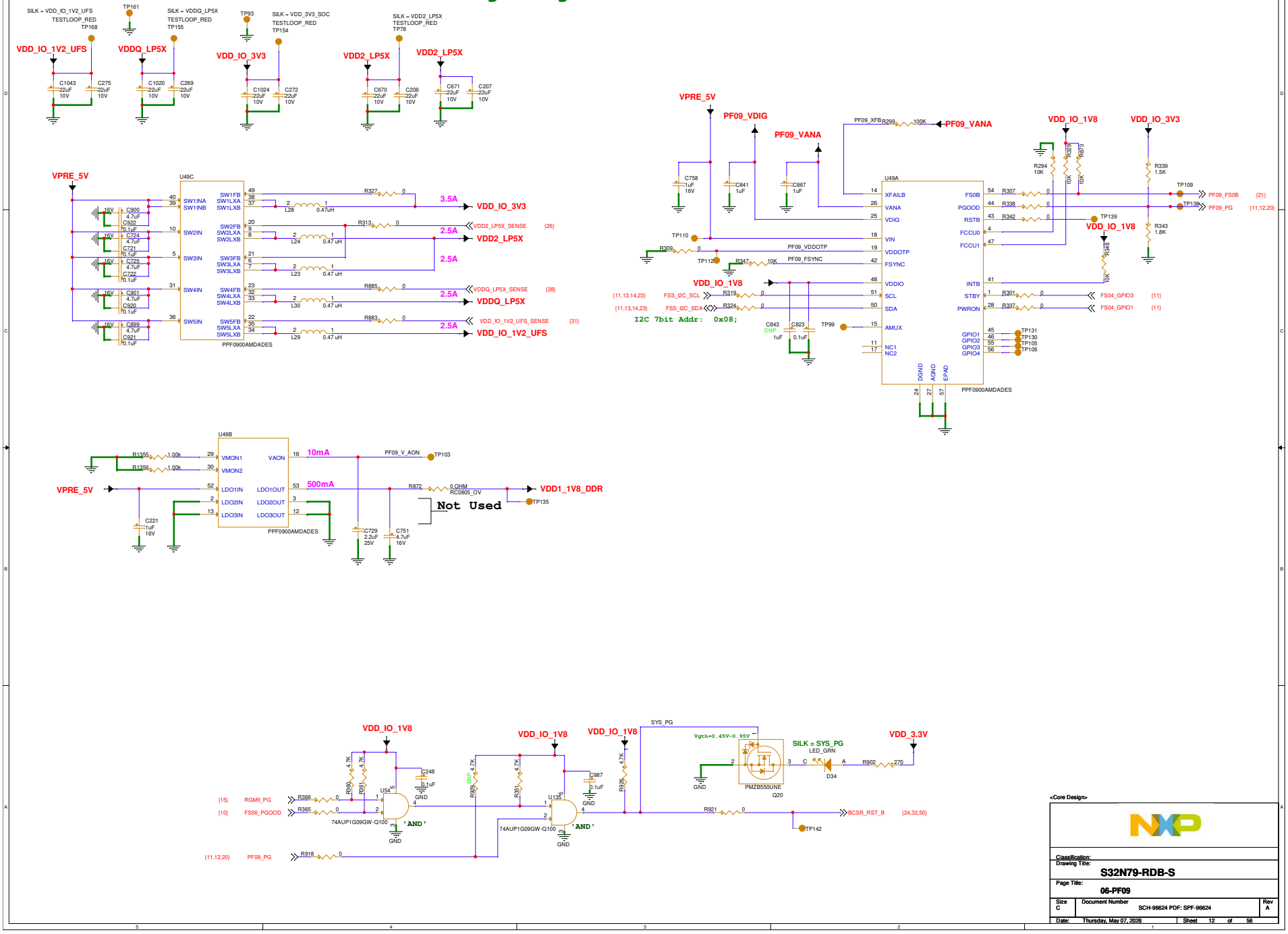
Voltage Regulators – LPDDR5X & Misc.



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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **06-PF09**

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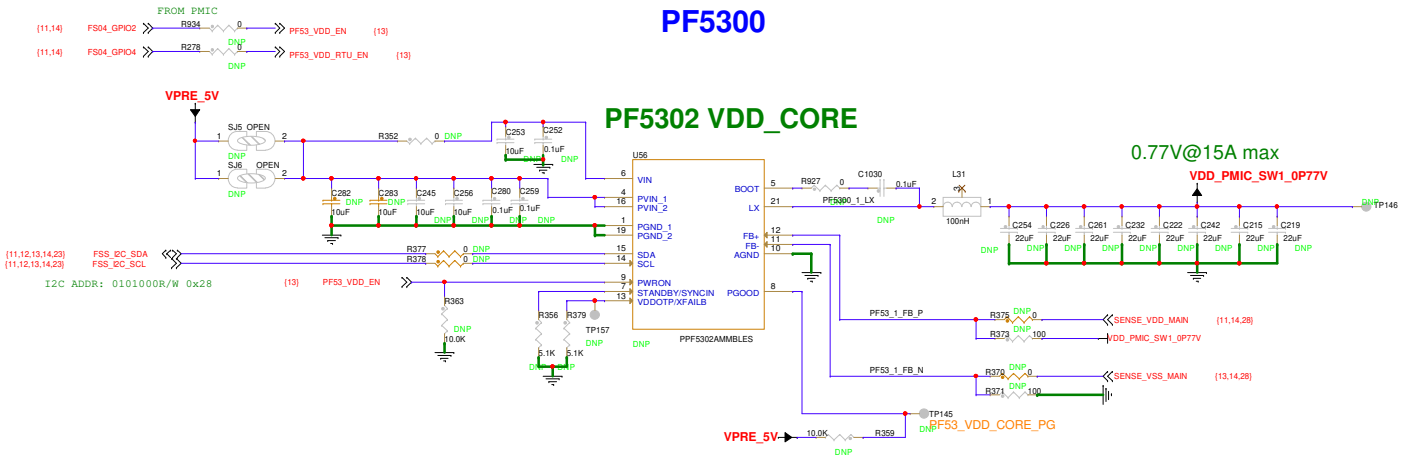
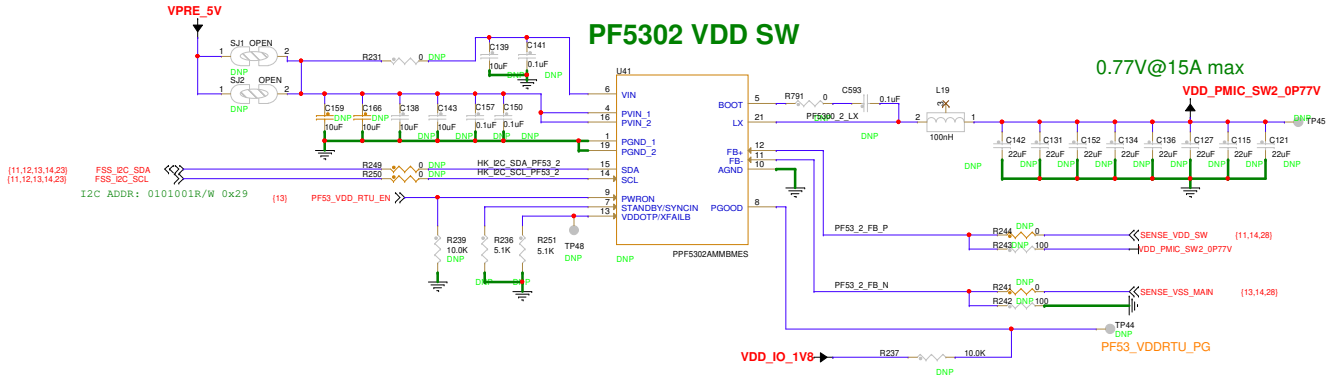
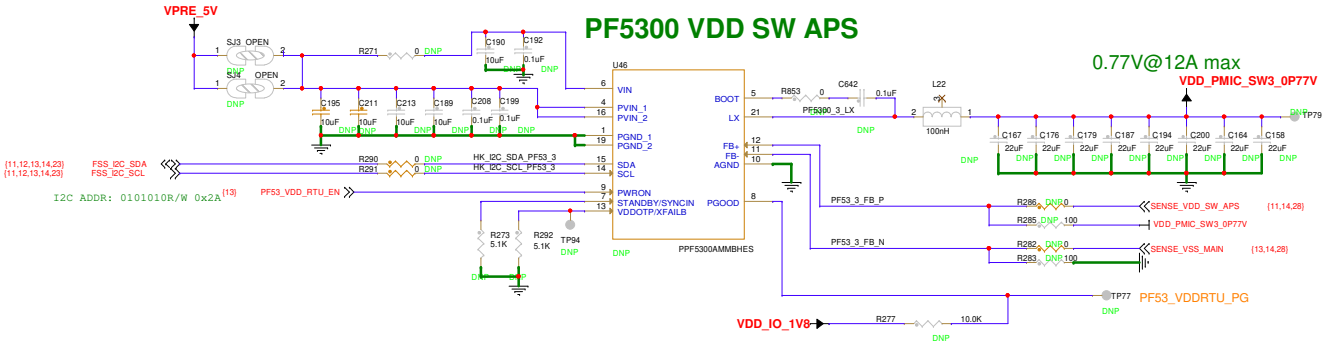
PF5300







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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **07-PF53**

Size	Document Number	Rev
C	SCH-96624 PDF: SPF-96624	A

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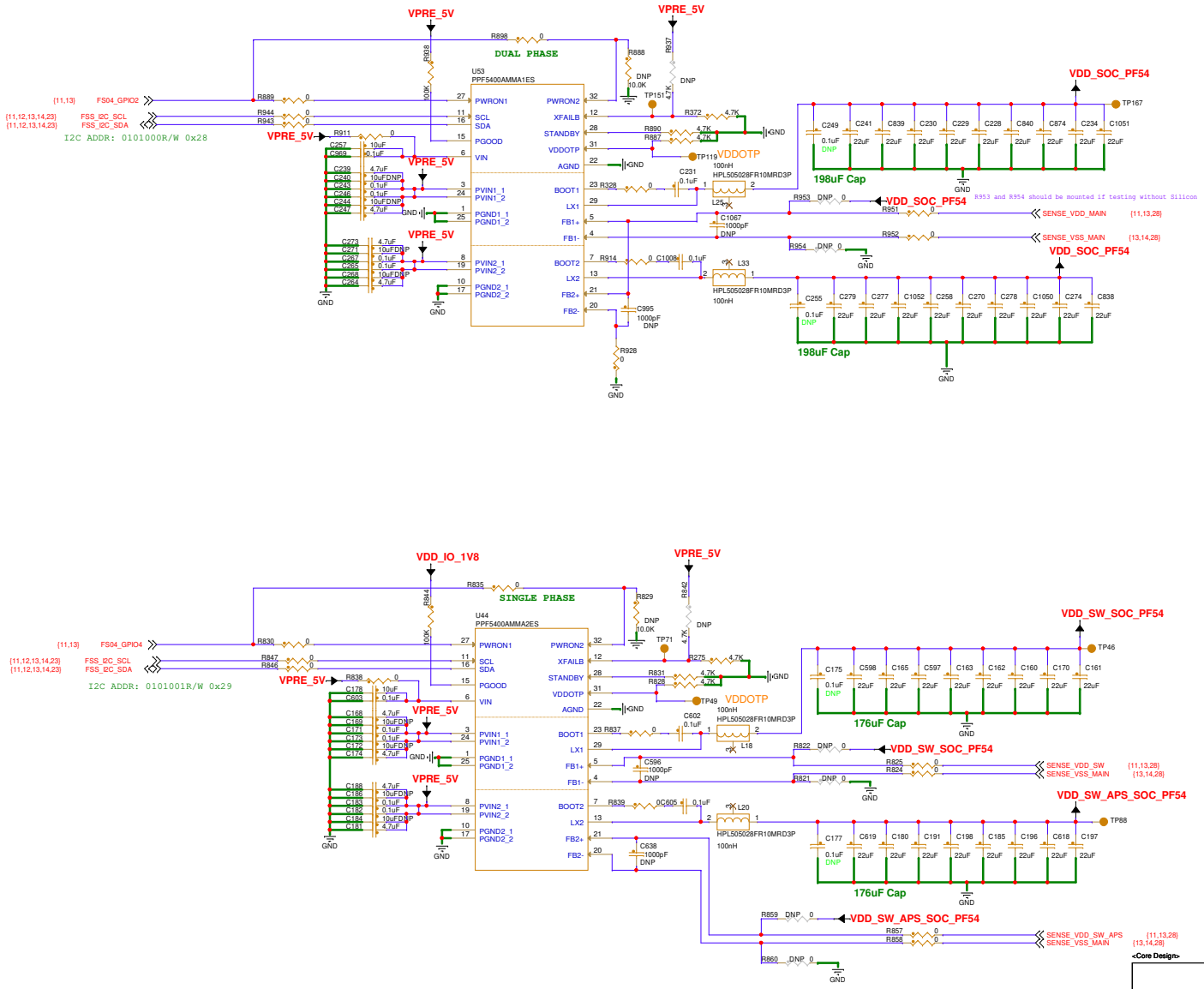
PF5400



Classification:	
Drawing Title:	S32N79-RDB-S
Page Title:	08-PF5400
Size: C	Document Number: SCH-96624 PDF: SPF-96624 Rev: A
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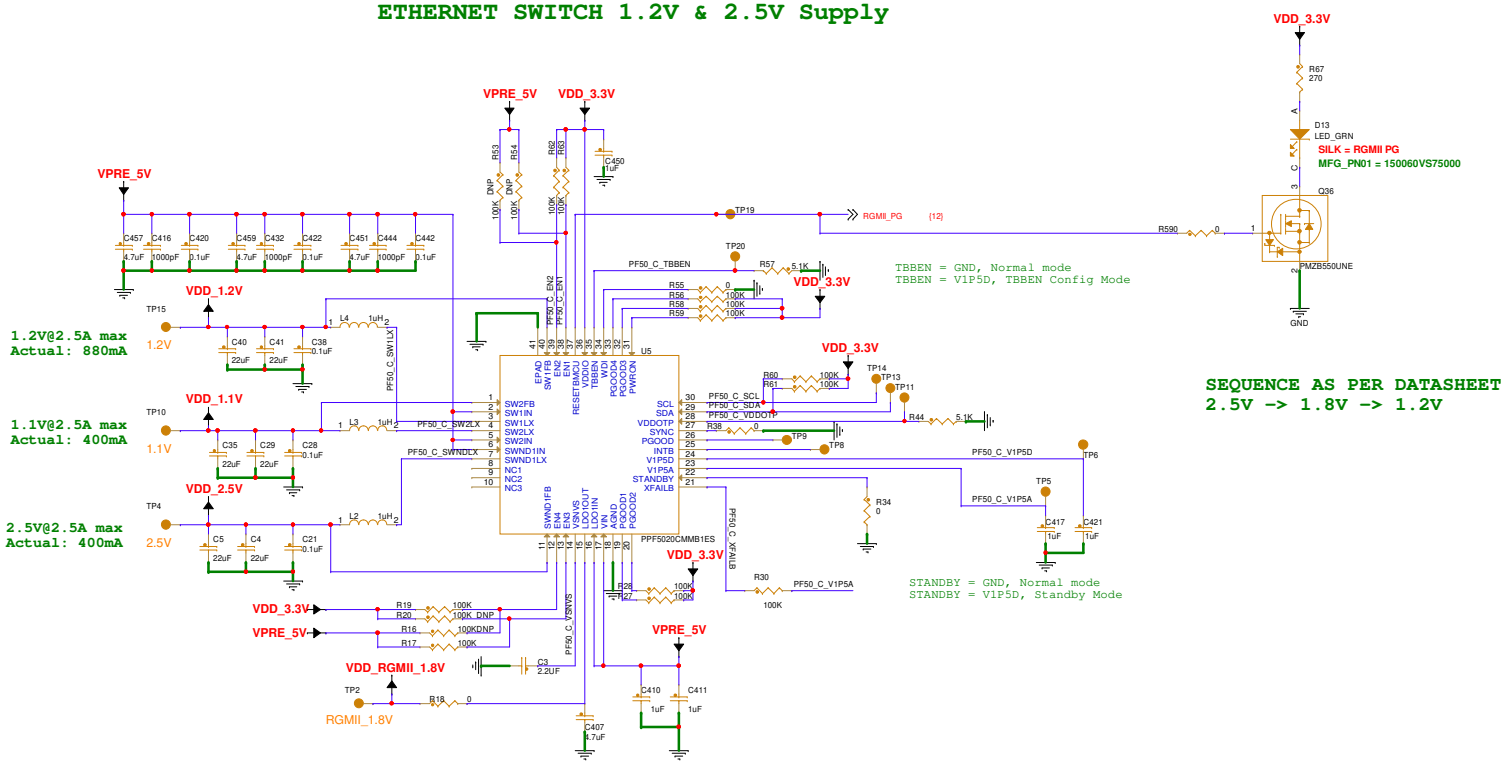
ETHERNET SWITCH 1.2V & 2.5V Supply



<Core Design>

Classification:
Drawing Title: S32N79-RDB-S
Page Title: 10-PF5020 ETH PHY PWR
Size C | Document Number SCH-96824 PDF: SPF-96824 | Rev A
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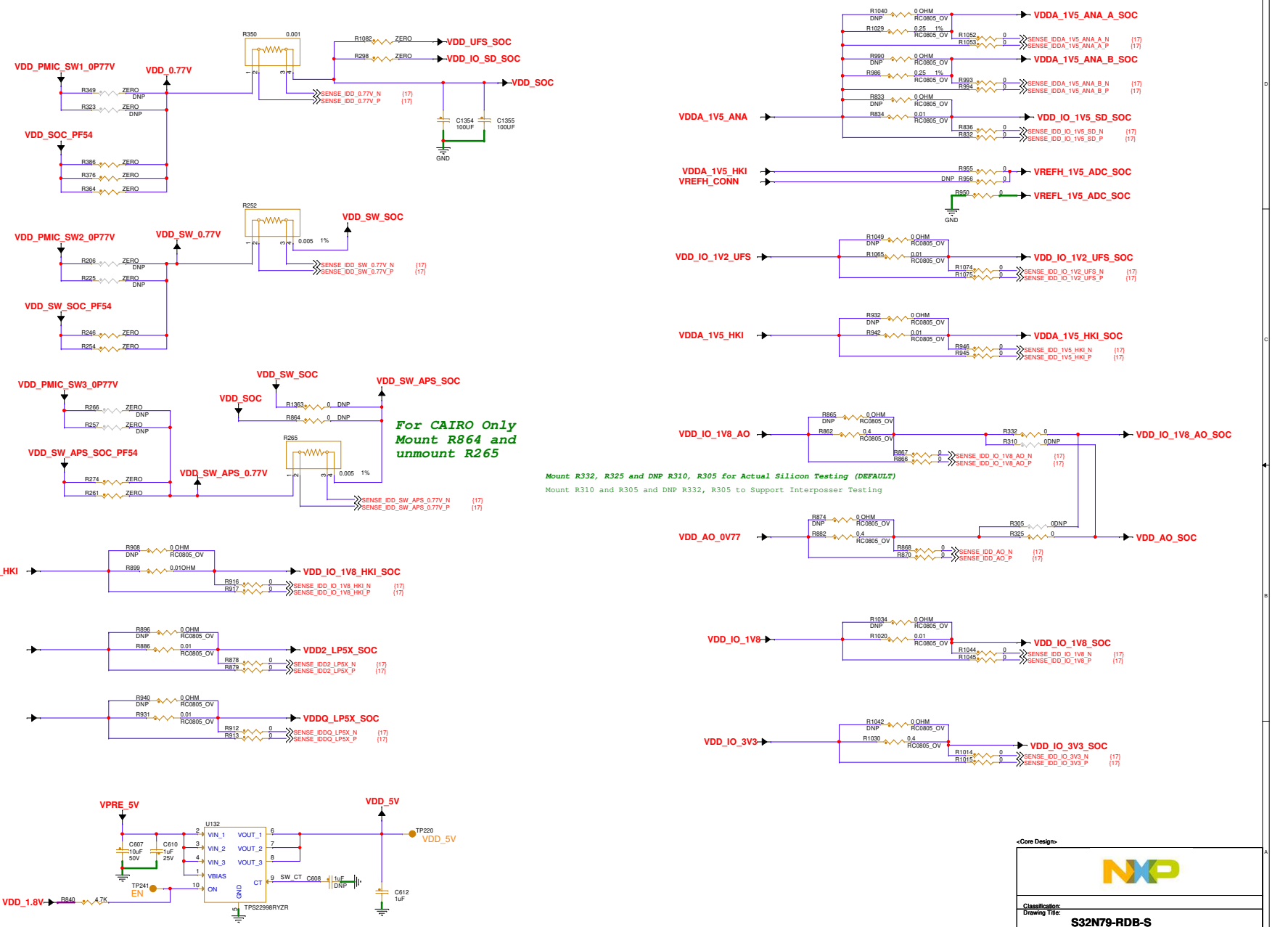

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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **11-POWER RMUX & CUR SENSE R**

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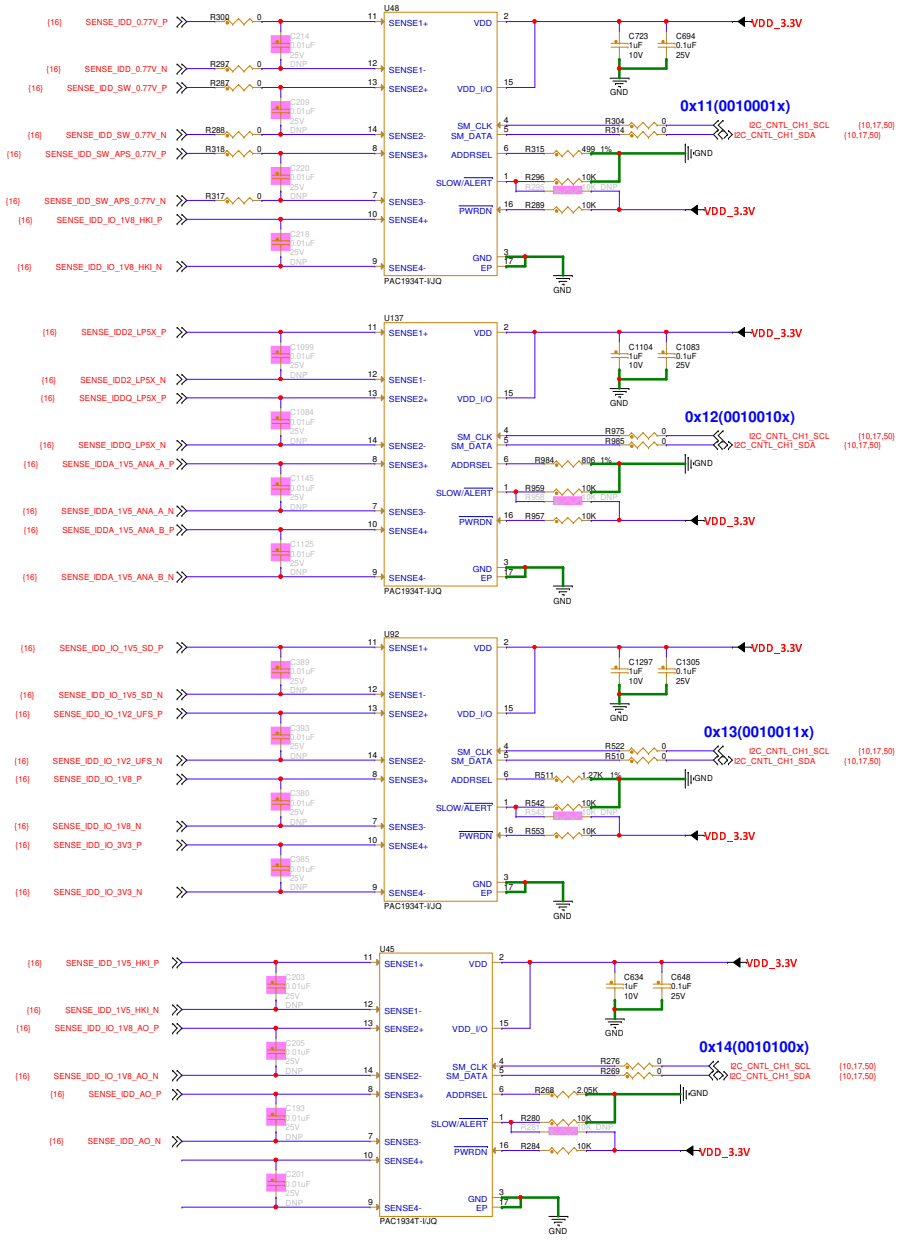
Curr Monitors



<Core Design>

Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **12-CUR MONITORS**

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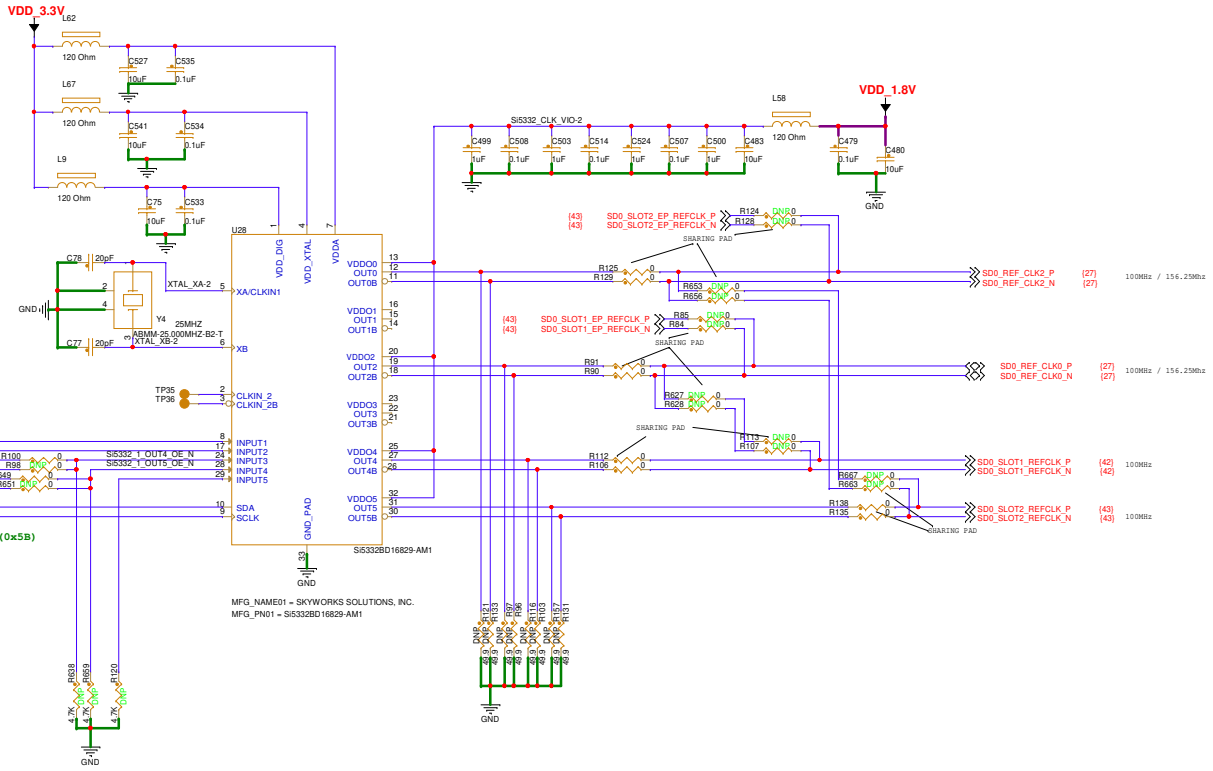
SERDES-0 CLOCKING INTERFACE



PROFILE_SEL[0:1] 00 01 10 11

	XA/XB			
	Profile-1	**Profile-2**	**Profile-3**	**Profile-4**
OUT0	100(SS)	100(SS)	156.25	156.25
OUT1				
OUT2	100(SS)	156.25	100(SS)	156.25
OUT3				
OUT4 (SS)	100	100	100	100
OUT5 (SS)	100	100	100	100

<Core Design>

Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **13-CLOCK GENERATOR I**
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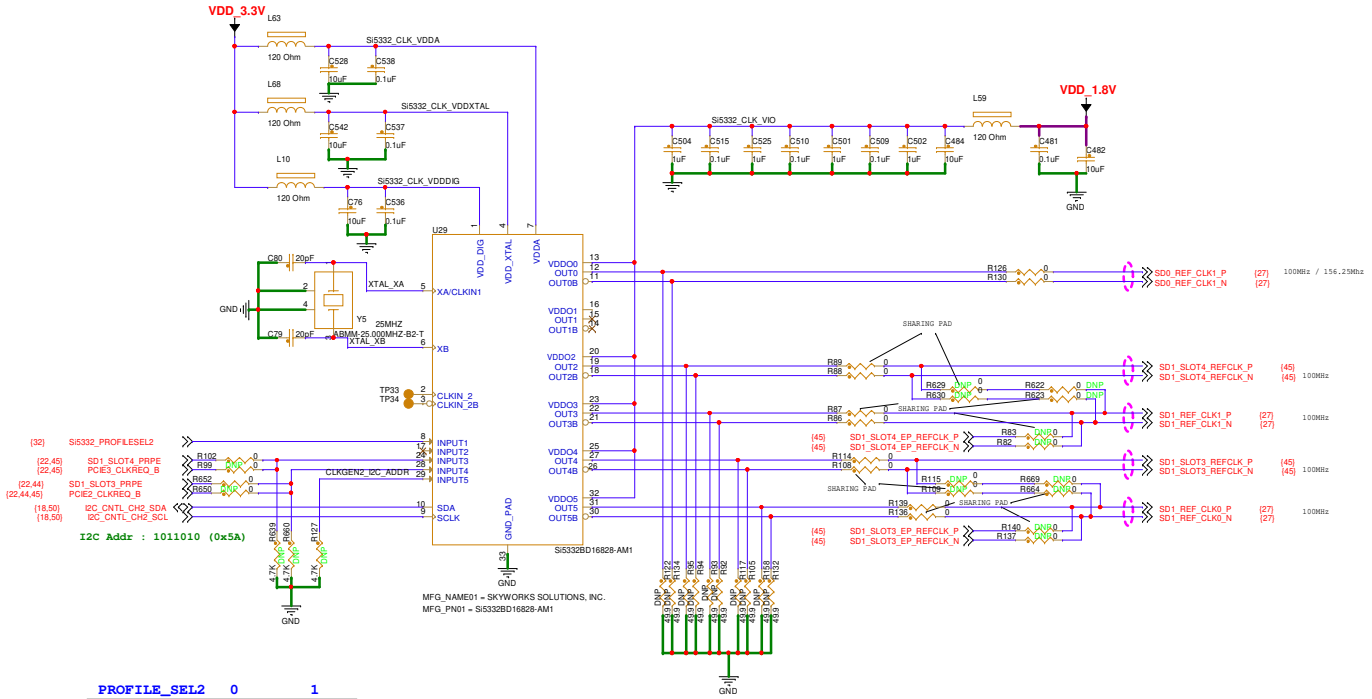
SERDES-1 CLOCKING INTERFACE



PROFILE_SEL2 0 1

	Profile-1	Profile-2
OUT0	100	156.25
OUT1	--	--
OUT2 (SS), OE1	100	100
OUT3 (SS)	100	100
OUT4 (SS), OE2	100	100
OUT5 (SS)	100	100

<Core Design>

Classification: Company Confidential

Drawing Title: S32N79-RDB-S

Page Title: 14-CLOCK GENERATOR II

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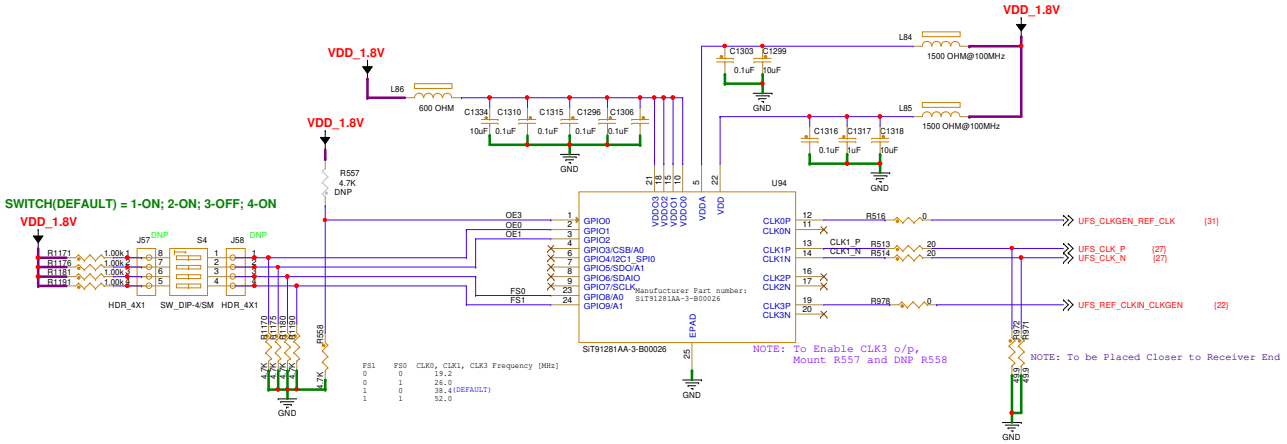
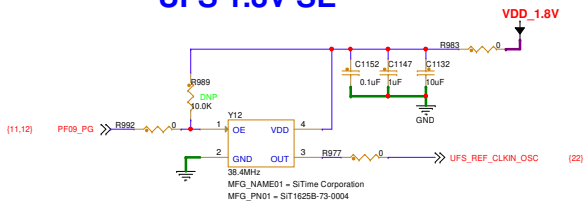
UFS CLK GENERATOR (OPTIONAL)

SiT91281



UFS 1.8V SE



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Classification: <Classification>

Drawing Title: **S32N79-RDB-S**

Page Title: **UFS CLOCK GENERATOR**

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LUXOR PA

U50C

REV 0.8

PA0_00/NWS_ETH_0_RGMII_TXD[0]/NWS_GPIO0[0]/NWS_ETH_0_MII_TXD[0]/NWS_ETH_0_RMII_TXD[0]/NWS_TST_RGMII_RXD0[0]
PA0_01/NWS_ETH_0_RGMII_TXD[1]/NWS_GPIO0[1]/NWS_ETH_0_MII_TXD[1]/NWS_ETH_0_RMII_TXD[1]/NWS_TST_RGMII_RXD0[1]
PA0_02/NWS_ETH_0_RGMII_TXD[2]/NWS_GPIO0[2]/NWS_ETH_0_MII_TXD[2]/NWS_ETH_1_PMD_TX/NWS_TST_RGMII_RXD0[2]
PA0_03/NWS_ETH_0_RGMII_TXD[3]/NWS_GPIO0[3]/NWS_ETH_0_MII_TXD[3]/NWS_ETH_1_PMD_RX/NWS_TST_RGMII_RXD0[3]
PA0_04/NWS_ETH_0_RGMII_TXCTL/NWS_GPIO0[4]/NWS_ETH_0_MII_TXEN/NWS_ETH_0_RMII_TXEN/NWS_ETH_0_PMD_TX/NWS_TST_RGMII_RX_CTL0
PA0_05/NWS_ETH_0_RGMII_TXC/NWS_GPIO0[5]/NWS_ETH_0_MII_TXCLK/NWS_ETH_0_RMII_REF_CLK/NWS_ETH_0_PMD_RX/NWS_TST_RGMII_RXC0
PA0_06/NWS_ETH_0_RGMII_RXD[0]/NWS_GPIO0[6]/NWS_ETH_0_MII_RXD[0]/NWS_ETH_0_RMII_RXD[0]/NWS_ETH_0_PMD_ED/NWS_TST_RGMII_TXD0[0]
PA0_07/NWS_ETH_0_RGMII_RXD[1]/NWS_GPIO0[7]/NWS_ETH_0_MII_RXD[1]/NWS_ETH_0_RMII_RXD[1]/NWS_TST_RGMII_TXD0[1]
PA0_08/NWS_ETH_0_RGMII_RXD[2]/NWS_GPIO0[8]/NWS_ETH_0_MII_RXD[2]/NWS_ETH_0_RMII_RXER/NWS_TST_RGMII_TXD0[2]
PA0_09/NWS_ETH_0_RGMII_RXD[3]/NWS_GPIO0[9]/NWS_ETH_0_MII_RXD[3]/NWS_ETH_1_PMD_ED/NWS_TST_RGMII_TXD0[3]
PA0_10/NWS_ETH_0_RGMII_RXCTL/NWS_GPIO0[10]/NWS_ETH_0_MII_RXDV/NWS_ETH_0_RMII_CRS_DV/NWS_TST_RGMII_TXCTL0
PA0_11/NWS_ETH_0_RGMII_RXC/NWS_GPIO0[11]/NWS_ETH_0_MII_RXCLK/NWS_TST_RGMII_TXC0
PA0_12/NWS_ETH_1_RGMII_TXD[0]/NWS_GPIO0[12]/NWS_ETH_1_MII_TXD[0]/NWS_ETH_3_RMII_TXD[0]/NWS_TST_RGMII_RXD1[0]/NWS_ETH_3_RGMII_TXD[0]/NWS_TMR_TRIG[1]/NWS_TST_RGMII_RXD3[0]
PA0_13/NWS_ETH_1_RGMII_TXD[1]/NWS_GPIO0[13]/NWS_ETH_1_MII_TXD[1]/NWS_ETH_3_RMII_TXD[1]/NWS_TST_RGMII_RXD1[1]/NWS_ETH_3_RGMII_TXD[1]/NWS_TMR_PP[1]/NWS_TST_RGMII_RXD3[1]
PA0_14/NWS_ETH_1_RGMII_TXD[2]/NWS_GPIO0[14]/NWS_ETH_0_MII_TXER/NWS_TST_RGMII_RXD1[2]/NWS_ETH_3_RGMII_TXD[2]/NWS_TMR_PP[2]/NWS_TST_RGMII_RXD3[2]
PA0_15/NWS_ETH_1_RGMII_TXD[3]/NWS_GPIO0[15]/NWS_ETH_0_MII_COL/NWS_TST_RGMII_RXD1[3]/NWS_ETH_3_RGMII_TXD[3]/NWS_TMR_ALARM[1]/NWS_TST_RGMII_RXD3[3]
PA0_16/NWS_ETH_1_RGMII_TXCTL/NWS_GPIO0[16]/NWS_ETH_1_RMII_TXEN/NWS_ETH_3_RMII_TXEN/NWS_TST_RGMII_RX_CTL1/NWS_ETH_3_RGMII_TXCTL/NWS_TMR_ALARM[2]/NWS_TST_RGMII_RX_CTL3
PA0_17/NWS_ETH_1_RGMII_TXC/NWS_GPIO0[17]/NWS_ETH_1_PMD_TX/NWS_ETH_1_RMII_REF_CLK/NWS_ETH_3_RMII_REF_CLK/NWS_TST_RGMII_RXC1/NWS_ETH_3_RGMII_TXC/NWS_TMR_GCLK/NWS_TST_RGMII_RXC3
PA0_18/NWS_ETH_1_RGMII_RXD[0]/NWS_GPIO0[18]/NWS_ETH_1_PMD_RX/NWS_ETH_1_RMII_RXD[0]/NWS_ETH_3_RMII_RXD[0]/NWS_TST_RGMII_TXD1[0]/NWS_ETH_3_RGMII_RXD[0]/NWS_TMR_MEDIA[0]/NWS_TST_RGMII_TXD3[0]
PA0_19/NWS_ETH_1_RGMII_RXD[1]/NWS_GPIO0[19]/NWS_ETH_1_PMD_ED/NWS_ETH_1_RMII_RXD[1]/NWS_ETH_3_RMII_RXD[1]/NWS_TST_RGMII_TXD1[1]/NWS_ETH_3_RGMII_RXD[1]/NWS_TMR_MEDIA[1]/NWS_TST_RGMII_TXD3[1]
PA0_20/NWS_ETH_1_RGMII_RXD[2]/NWS_GPIO0[20]/NWS_ETH_1_RMII_RXER/NWS_ETH_3_RMII_RXER/NWS_TST_RGMII_TXD1[2]/NWS_ETH_3_RGMII_RXD[2]/NWS_TMR_MEDIA[2]/NWS_TST_RGMII_TXD3[2]
PA0_21/NWS_ETH_1_RGMII_RXD[3]/NWS_GPIO0[21]/NWS_ETH_1_MII_RXER/NWS_TST_RGMII_TXD1[3]/NWS_ETH_3_RGMII_RXD[3]/NWS_TMR_MEDIA[3]/NWS_TST_RGMII_TXD3[3]
PA0_22/NWS_ETH_1_RGMII_RXCTL/NWS_GPIO0[22]/NWS_ETH_1_RMII_CRS_DV/NWS_ETH_3_RMII_CRS_DV/NWS_TST_RGMII_TXCTL1/NWS_ETH_3_RGMII_RXCTL/NWS_TST_RGMII_TXCTL3
PA0_23/NWS_ETH_1_RGMII_RXC/NWS_GPIO0[23]/NWS_ETH_0_MII_CRS/NWS_TST_RGMII_TXC1/NWS_ETH_3_RGMII_RXC/NWS_1588_CLK/NWS_TST_RGMII_TXC3
PA0_24/NWS_ETH_0_SLV_MDIO/NWS_GPIO0[24]/NWS_ETH_1_SLV_MDIO/NWS_ETH_3_SLV_MDIO/NWS_TMR_GCLK
PA0_25/NWS_ETH_0_SLV_MDC/NWS_GPIO0[25]/NWS_ETH_1_SLV_MDC/NWS_ETH_3_SLV_MDC/NWS_TMR_TRIG[1]
PA0_26/NWS_AUXCLKIN/NWS_GPIO0[26]/NWS_1588_CLK
PA0_27/NWS_AUXCLKOUT0/NWS_GPIO0[27]/HSC_AUX_CLK0/NWS_ETH_3_SLV_MDIO/NWS_TMR_MEDIA[0]
PA0_28/NWS_AUXCLKOUT1/NWS_GPIO0[28]/HSC_AUX_CLK1/NWS_ETH_3_SLV_MDC/NWS_TMR_MEDIA[1]
PA0_29/NWS_AUXCLKOUT2/NWS_GPIO0[29]/HSC_AUX_CLK2/NWS_ETH_3_SLV_MDIO/NWS_ETH_0_SLV_MDIO/NWS_ETH_1_SLV_MDIO/NWS_TMR_MEDIA[2]
PA0_30/NWS_AUXCLKOUT3/NWS_GPIO0[30]/HSC_AUX_CLK3/NWS_ETH_3_SLV_MDC/NWS_ETH_0_SLV_MDC/NWS_ETH_1_SLV_MDC/NWS_TMR_MEDIA[3]
PA0_31/NWS_TMR_PP[3]/NWS_GPIO0[31]/NWS_PP_SYNC_OUT

MDIO0/NWS_MDIO_DATA0/NWS_GPIO1[0]
MDC0/NWS_MDC0/NWS_GPIO1[1]

pinmux rev1.37

Luxor_23mm Rev 0.8

LUXOR PB

U50D

REV 0.8

PB0_00/RCU_CAN_0_TX/RCU_GPIO0[0]/RCU_LIN24_TX
PB0_01/RCU_CAN_1_TX/RCU_GPIO0[1]/RCU_LIN25_TX
PB0_02/RCU_CAN_2_TX/RCU_GPIO0[2]/RCU_LIN26_TX
PB0_03/RCU_CAN_3_TX/RCU_GPIO0[3]/RCU_LIN27_TX
PB0_04/RCU_CAN_4_TX/RCU_GPIO0[4]/RCU_LIN28_TX
PB0_05/RCU_CAN_5_TX/RCU_GPIO0[5]/RCU_LIN29_TX
PB0_06/RCU_CAN_6_TX/RCU_GPIO0[6]/RCU_LIN30_TX
PB0_07/RCU_CAN_7_TX/RCU_GPIO0[7]/RCU_LIN31_TX
PB0_08/RCU_CAN_8_TX/RCU_GPIO0[8]
PB0_09/RCU_CAN_9_TX/RCU_GPIO0[9]
PB0_10/RCU_CAN_10_TX/RCU_GPIO0[10]
PB0_11/RCU_CAN_11_TX/RCU_GPIO0[11]
PB0_12/RCU_CAN_12_TX/RCU_GPIO0[12]
PB0_13/RCU_CAN_13_TX/RCU_GPIO0[13]
PB0_14/RCU_CAN_14_TX/RCU_GPIO0[14]
PB0_15/RCU_CAN_15_TX/RCU_GPIO0[15]
PB0_16/RCU_CAN_16_TX/RCU_GPIO0[16]/RCU_LIN10_TX
PB0_17/RCU_CAN_17_TX/RCU_GPIO0[17]/RCU_LIN11_TX
PB0_18/RCU_CAN_18_TX/RCU_GPIO0[18]/RCU_LIN12_TX
PB0_19/RCU_CAN_19_TX/RCU_GPIO0[19]/RCU_LIN13_TX
PB0_20/RCU_CAN_20_TX/RCU_GPIO0[20]/RCU_LIN14_TX
PB0_21/RCU_CAN_21_TX/RCU_GPIO0[21]/RCU_LIN15_TX
PB0_22/RCU_CAN_22_TX/RCU_GPIO0[22]/RCU_LIN16_TX
PB0_23/RCU_CAN_23_TX/RCU_GPIO0[23]/RCU_LIN17_TX
PB0_24/RCU_CANXL_0_TX/RCU_GPIO0[24]/RCU_LIN18_TX
PB0_25/RCU_CANXL_1_TX/RCU_GPIO0[25]/RCU_LIN19_TX
PB0_26/RCU_CANXL_2_TX/RCU_GPIO0[26]/RCU_LIN20_TX
PB0_27/RCU_CANXL_3_TX/RCU_GPIO0[27]/RCU_LIN21_TX
PB0_28/RCU_CANXL_4_TX/RCU_GPIO0[28]/RCU_LIN22_TX
PB0_29/RCU_CANXL_5_TX/RCU_GPIO0[29]/RCU_LIN23_TX
PB0_30/RCU_LIN0_TX/RCU_GPIO0[30]
PB0_31/RCU_LIN1_TX/RCU_GPIO0[31]

pinmux rev1.37

Luxor_23mm Rev 0.8

U50E

REV 0.8

PB1_00/RCU_LIN2_TX/RCU_GPIO1[0]
PB1_01/RCU_LIN3_TX/RCU_GPIO1[1]
PB1_02/RCU_LIN4_TX/RCU_GPIO1[2]
PB1_03/RCU_LIN5_TX/RCU_GPIO1[3]
PB1_04/RCU_LIN6_TX/RCU_GPIO1[4]
PB1_05/RCU_LIN7_TX/RCU_GPIO1[5]
PB1_06/RCU_LIN8_TX/RCU_GPIO1[6]
PB1_07/RCU_LIN9_TX/RCU_GPIO1[7]

PB2_00/RCU_CAN_0_RX/RCU_GPI2[0]/RCU_LIN24_RX/COS_SENT_0_RX[0]/FSS_WKUP[4]
PB2_01/RCU_CAN_1_RX/RCU_GPI2[1]/RCU_LIN25_RX/COS_SENT_0_RX[1]/FSS_WKUP[5]
PB2_02/RCU_CAN_2_RX/RCU_GPI2[2]/RCU_LIN26_RX/COS_SENT_0_RX[2]/FSS_WKUP[6]
PB2_03/RCU_CAN_3_RX/RCU_GPI2[3]/RCU_LIN27_RX/COS_SENT_0_RX[3]/FSS_WKUP[7]
PB2_04/RCU_CAN_4_RX/RCU_GPI2[4]/RCU_LIN28_RX/COS_SENT_0_RX[4]/FSS_WKUP[8]
PB2_05/RCU_CAN_5_RX/RCU_GPI2[5]/RCU_LIN29_RX/COS_SENT_0_RX[5]/FSS_WKUP[9]
PB2_06/RCU_CAN_6_RX/RCU_GPI2[6]/RCU_LIN30_RX/COS_SENT_0_RX[6]/FSS_WKUP[10]
PB2_07/RCU_CAN_7_RX/RCU_GPI2[7]/RCU_LIN31_RX/COS_SENT_0_RX[7]/FSS_WKUP[11]
PB2_08/RCU_CAN_8_RX/RCU_GPI2[8]/COS_SENT_1_RX[0]/FSS_WKUP[12]
PB2_09/RCU_CAN_9_RX/RCU_GPI2[9]/COS_SENT_1_RX[1]/FSS_WKUP[13]
PB2_10/RCU_CAN_10_RX/RCU_GPI2[10]/COS_SENT_1_RX[2]/FSS_WKUP[14] AO
PB2_11/RCU_CAN_11_RX/RCU_GPI2[11]/COS_SENT_1_RX[3]/FSS_WKUP[15]
PB2_12/RCU_CAN_12_RX/RCU_GPI2[12]/COS_SENT_1_RX[4]/FSS_WKUP[16]
PB2_13/RCU_CAN_13_RX/RCU_GPI2[13]/COS_SENT_1_RX[5]/FSS_WKUP[17]
PB2_14/RCU_CAN_14_RX/RCU_GPI2[14]/COS_SENT_1_RX[6]/FSS_WKUP[18]
PB2_15/RCU_CAN_15_RX/RCU_GPI2[15]/COS_SENT_1_RX[7]/FSS_WKUP[19]
PB2_16/RCU_CAN_16_RX/RCU_GPI2[16]/RCU_LIN10_RX/FSS_WKUP[20]
PB2_17/RCU_CAN_17_RX/RCU_GPI2[17]/RCU_LIN11_RX/FSS_WKUP[21]
PB2_18/RCU_CAN_18_RX/RCU_GPI2[18]/RCU_LIN12_RX/FSS_WKUP[22]
PB2_19/RCU_CAN_19_RX/RCU_GPI2[19]/RCU_LIN13_RX/FSS_WKUP[23]
PB2_20/RCU_CAN_20_RX/RCU_GPI2[20]/RCU_LIN14_RX/FSS_WKUP[24]
PB2_21/RCU_CAN_21_RX/RCU_GPI2[21]/RCU_LIN15_RX/FSS_WKUP[25]
PB2_22/RCU_CAN_22_RX/RCU_GPI2[22]/RCU_LIN16_RX/FSS_WKUP[26]
PB2_23/RCU_CAN_23_RX/RCU_GPI2[23]/RCU_LIN17_RX/FSS_WKUP[27]
PB2_24/RCU_CANXL_0_RX/RCU_GPI2[24]/RCU_LIN18_RX/FSS_WKUP[28]
PB2_25/RCU_CANXL_1_RX/RCU_GPI2[25]/RCU_LIN19_RX/FSS_WKUP[29]
PB2_26/RCU_CANXL_2_RX/RCU_GPI2[26]/RCU_LIN20_RX/FSS_WKUP[30]
PB2_27/RCU_CANXL_3_RX/RCU_GPI2[27]/RCU_LIN21_RX/FSS_WKUP[31]
PB2_28/RCU_CANXL_4_RX/RCU_GPI2[28]/RCU_LIN22_RX/FSS_WKUP[32]
PB2_29/RCU_CANXL_5_RX/RCU_GPI2[29]/RCU_LIN23_RX/FSS_WKUP[33]
PB2_30/RCU_LIN0_RX/RCU_GPI2[30]/FSS_WKUP[34]
PB2_31/RCU_LIN1_RX/RCU_GPI2[31]/FSS_WKUP[35]

PB3_00/RCU_LIN2_RX/RCU_GPI3[0]/FSS_WKUP[36]
PB3_01/RCU_LIN3_RX/RCU_GPI3[1]/FSS_WKUP[37]
PB3_02/RCU_LIN4_RX/RCU_GPI3[2]/FSS_WKUP[38]
PB3_03/RCU_LIN5_RX/RCU_GPI3[3]/FSS_WKUP[39] AO
PB3_04/RCU_LIN6_RX/RCU_GPI3[4]/FSS_WKUP[40]
PB3_05/RCU_LIN7_RX/RCU_GPI3[5]/FSS_WKUP[41]
PB3_06/RCU_LIN8_RX/RCU_GPI3[6]/FSS_WKUP[42]
PB3_07/RCU_LIN9_RX/RCU_GPI3[7]/FSS_WKUP[43]
PB3_08/FSS_WKUP[0]/RCU_GPI3[8]

pinmux rev1.37

Luxor_23mm Rev 0.8

<Core Design>

Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
15-S32N79 PA-PB

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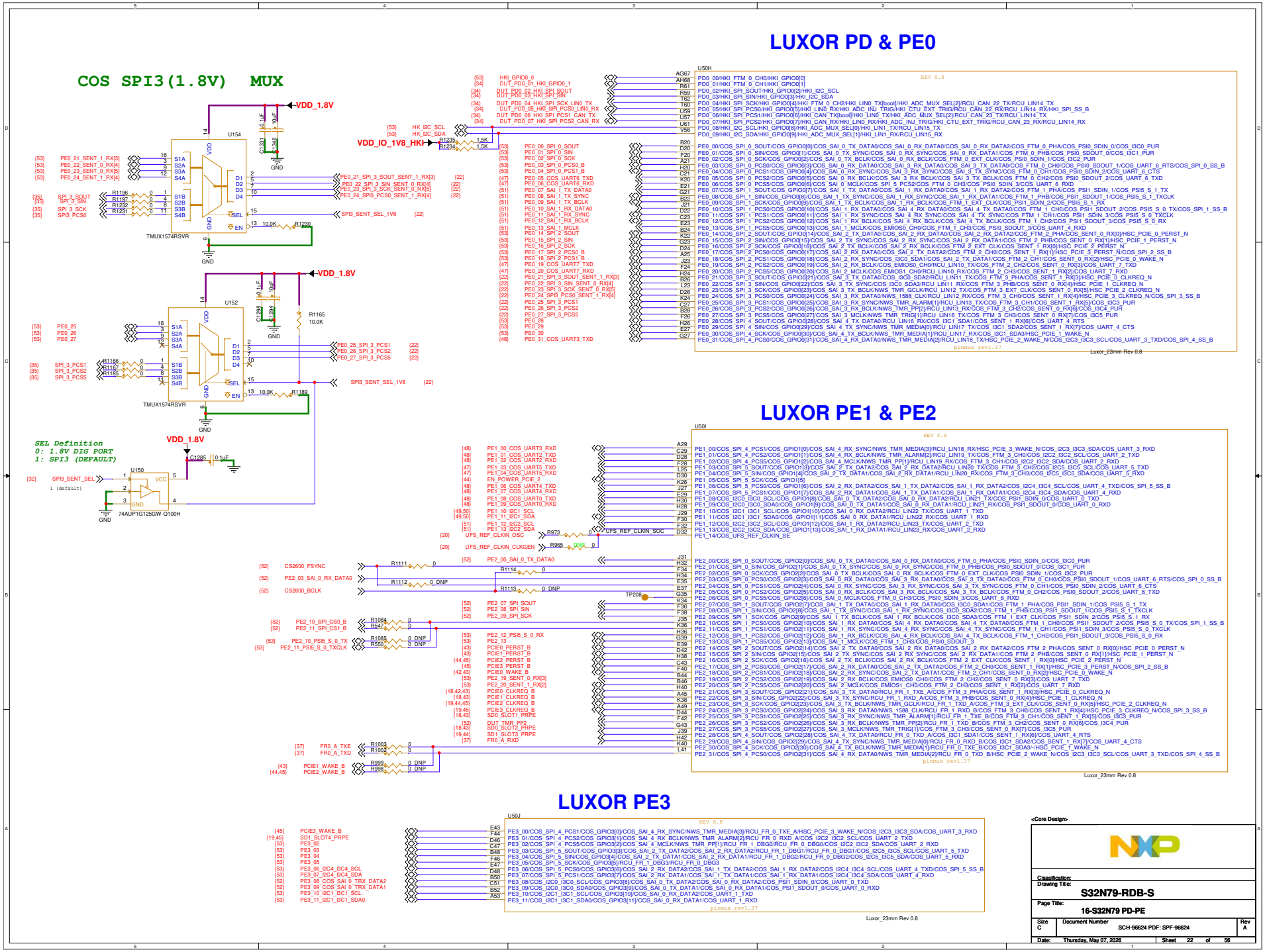

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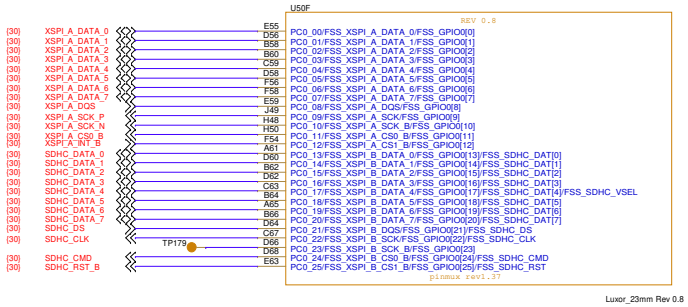
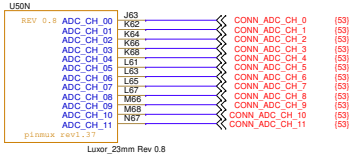
NXP
Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
16-S32N79 PD-PE
Size C
Document Number SCH-96624 PDF: SPF-96624
Rev A
Date: Thursday, May 07, 2026
Sheet 22 of 56

LUXOR PC



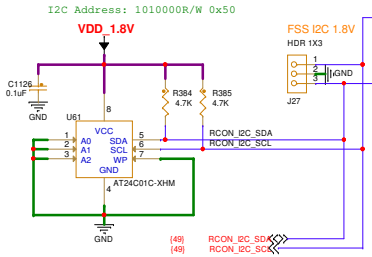
Luxor_23mm Rev 0.8

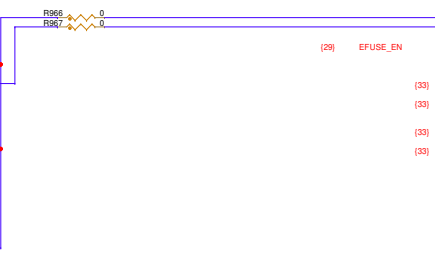
LUXOR ADC

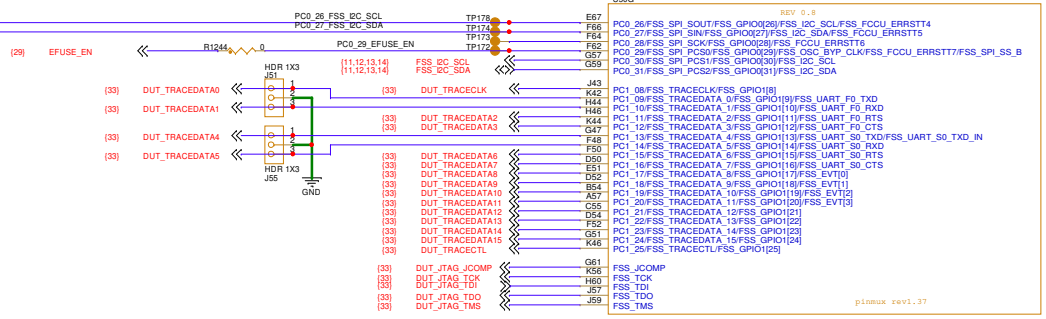


Luxor_23mm Rev 0.8

RCON EEPROM







LUXOR PC

Luxor_23mm Rev 0.8

<Core Design>

Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **17-S32N79 XSPI & ADC**

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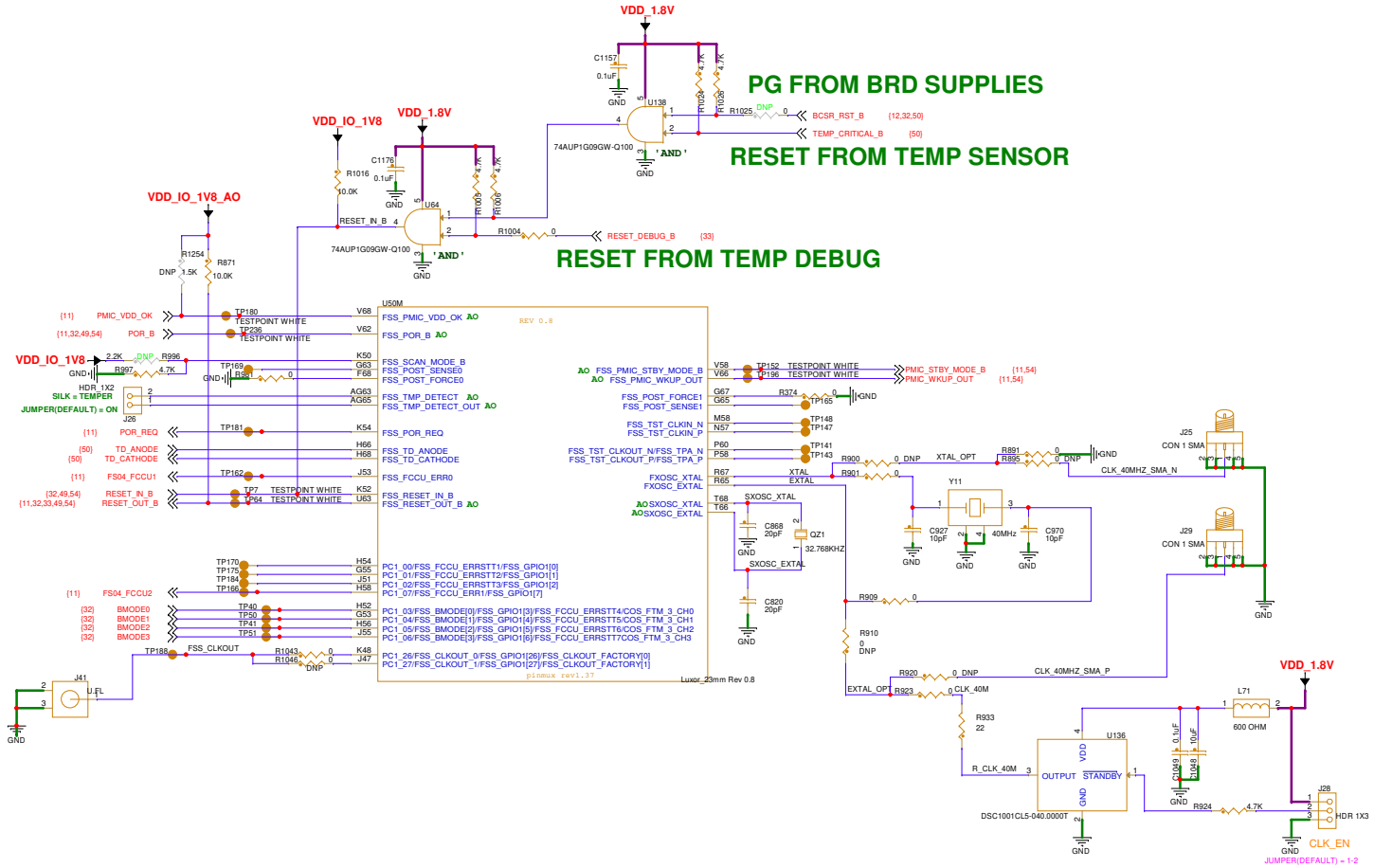
LUXOR MISC (RESET, JTAG, XTAL...)



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Classification:	
Drawing Title:	S32N79-RDB-S
Page Title:	18-S32N79 JTAG & MISC
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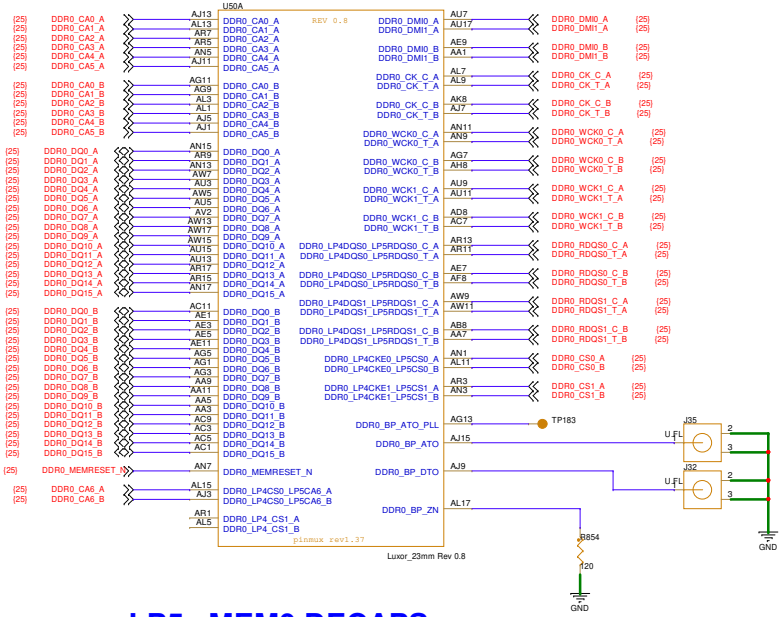
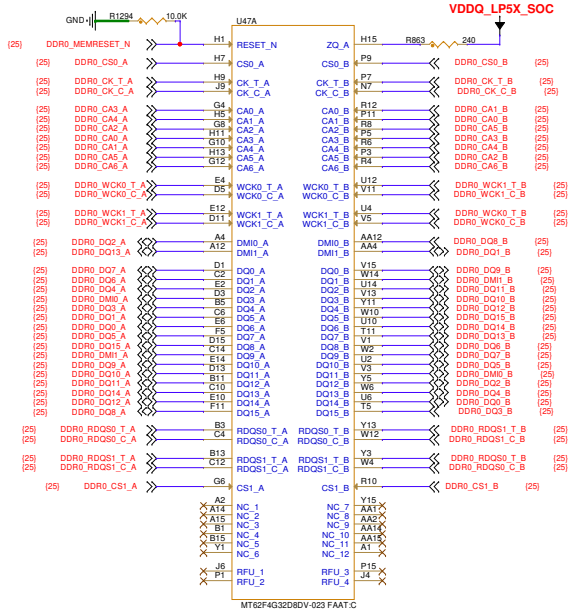
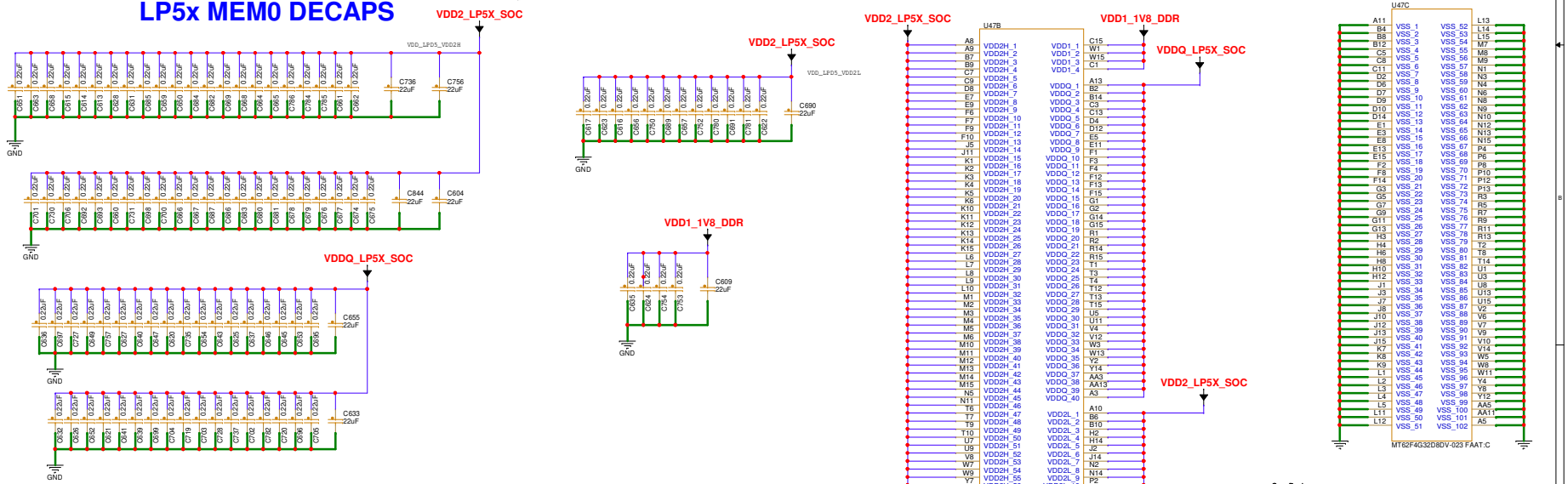
LUXOR DDRC0



LP5x MEM0



LP5x MEM0 DECAPS



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Classification:

Drawing Title:

S32N79-RDB-S

Page Title:

19-S32N79 LPDDR5x[1]

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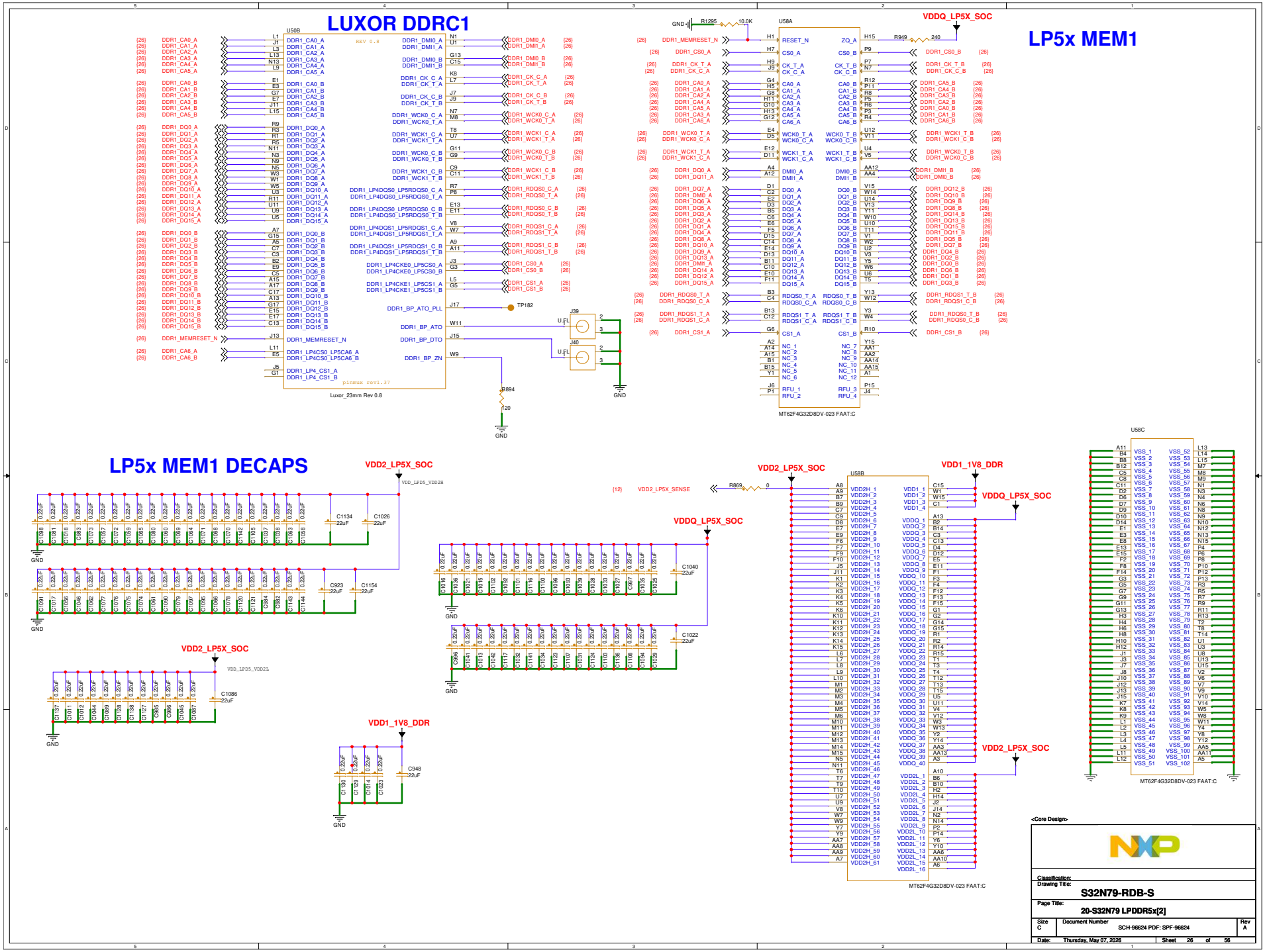
LUXOR DDRC1
LP5x MEM1
LP5x MEM1 DECAPS
U50B
Luxor_23mm Rev 0.8
U58A
MT62F4G32D8DV-023 FAAT:C
U58B
MT62F4G32D8DV-023 FAAT:C
U58C
MT62F4G32D8DV-023 FAAT:C
VDDQ_LP5X_SOC
VDD2_LP5X_SOC
VDD1_1V8_DDR
VDD2_LP5X_SENSE
R1295
10.0K
R949
240
R869
0
R894
120
TP182
J39
J40
U.FL
GND
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Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
20-S32N79 LPDDR5x[2]
Size C
Document Number
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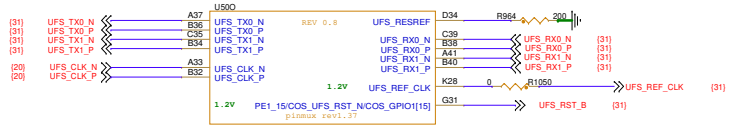
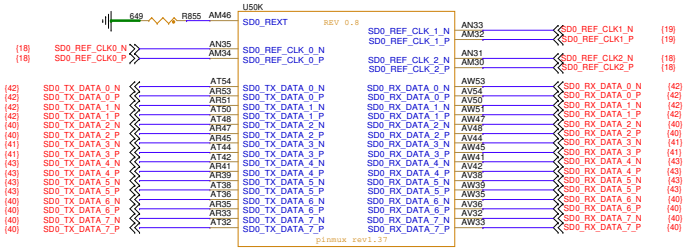
LUXOR UFS



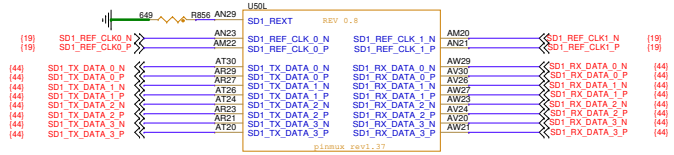
Luxor_23mm Rev 0.8

LUXOR SERDES0



Luxor_23mm Rev 0.8

LUXOR SERDES1



Luxor_23mm Rev 0.8

<Core Design>

Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **21-S32N79 SERDES**

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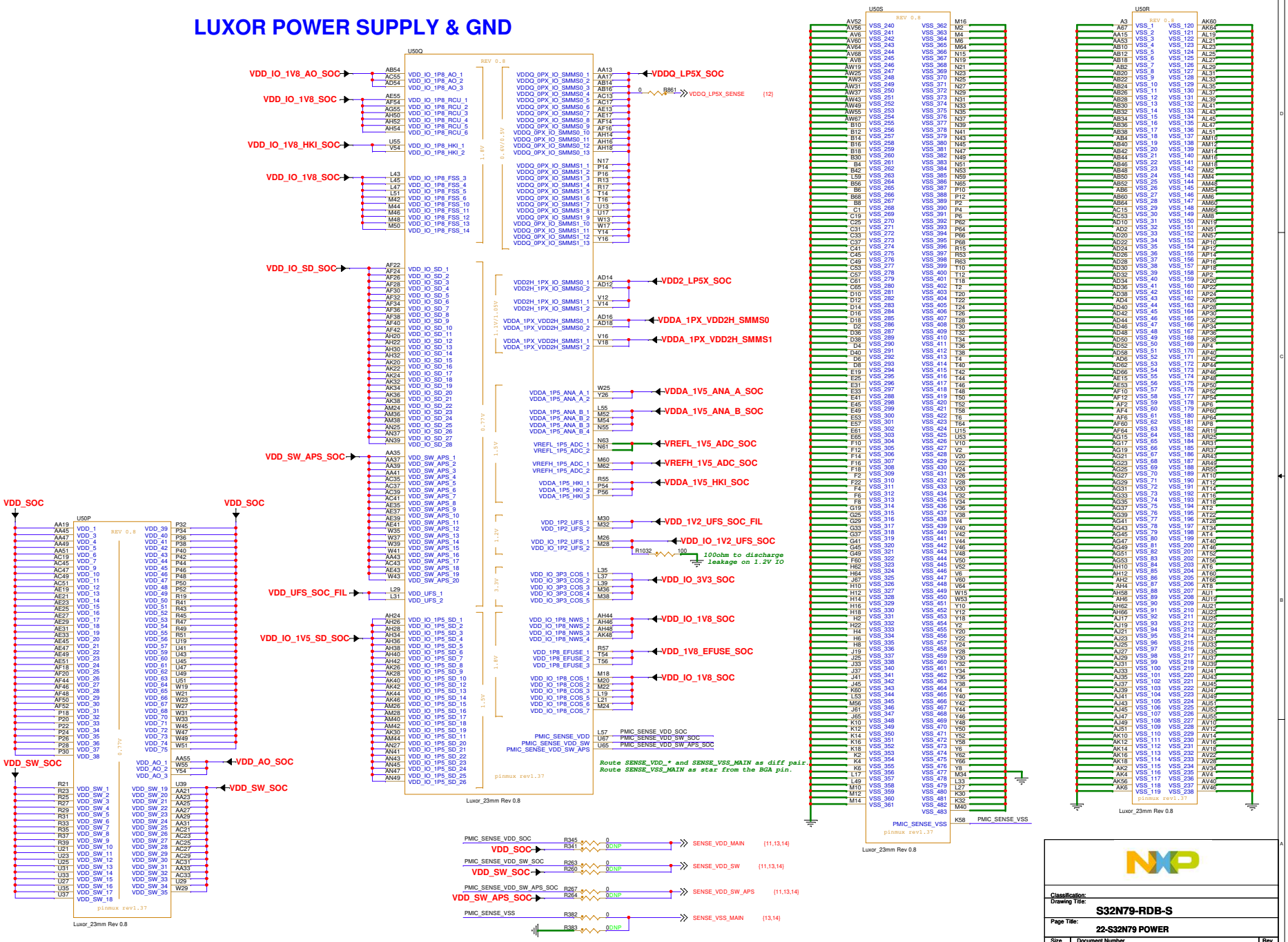
LUXOR POWER SUPPLY & GND



Classification:
Drawing Title:

S32N79-RDB-S

Page Title:

22-S32N79 POWER

Size	Document Number	Rev
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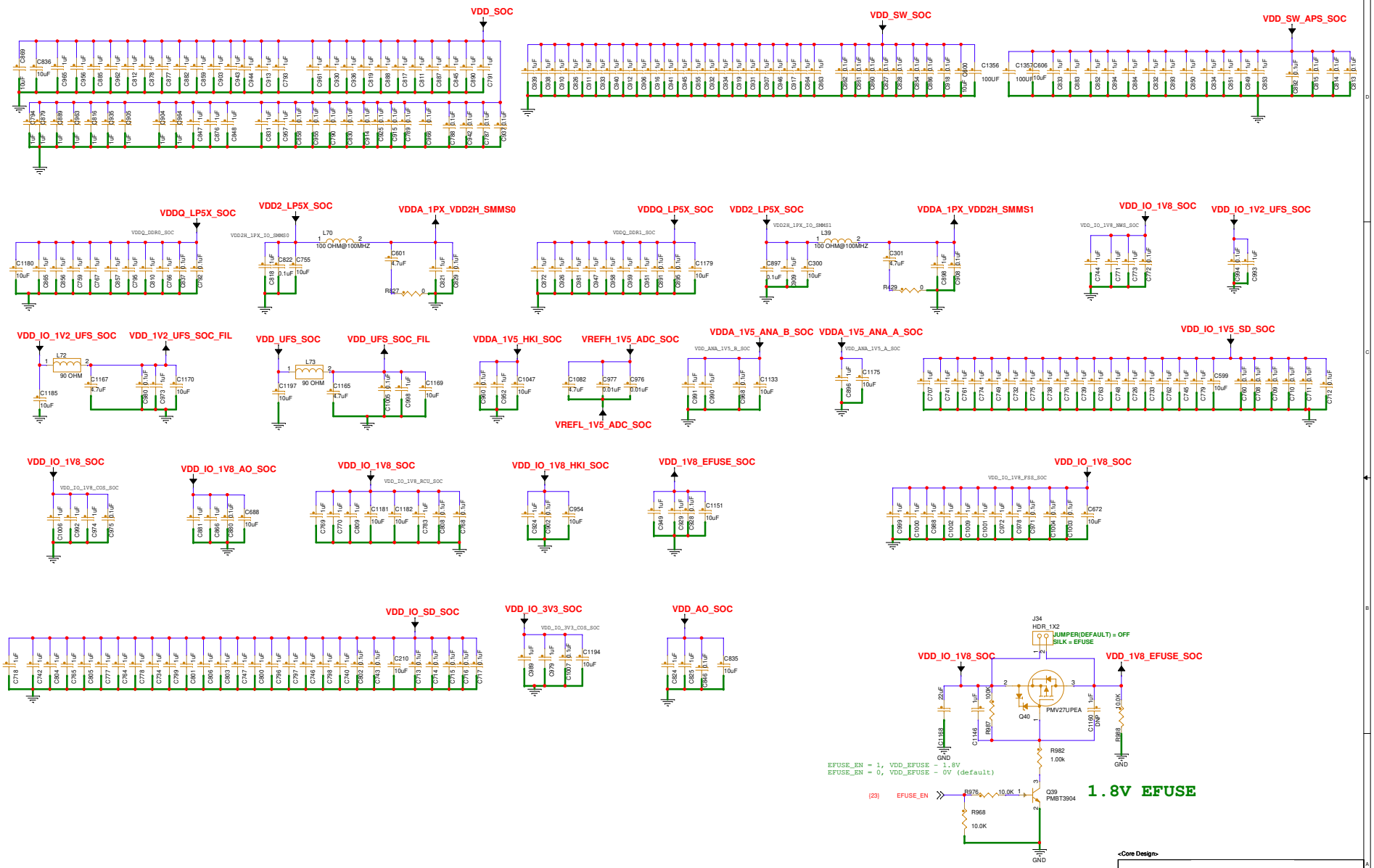

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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **23-S32N79 DECAPS**

Size	Document Number	Rev
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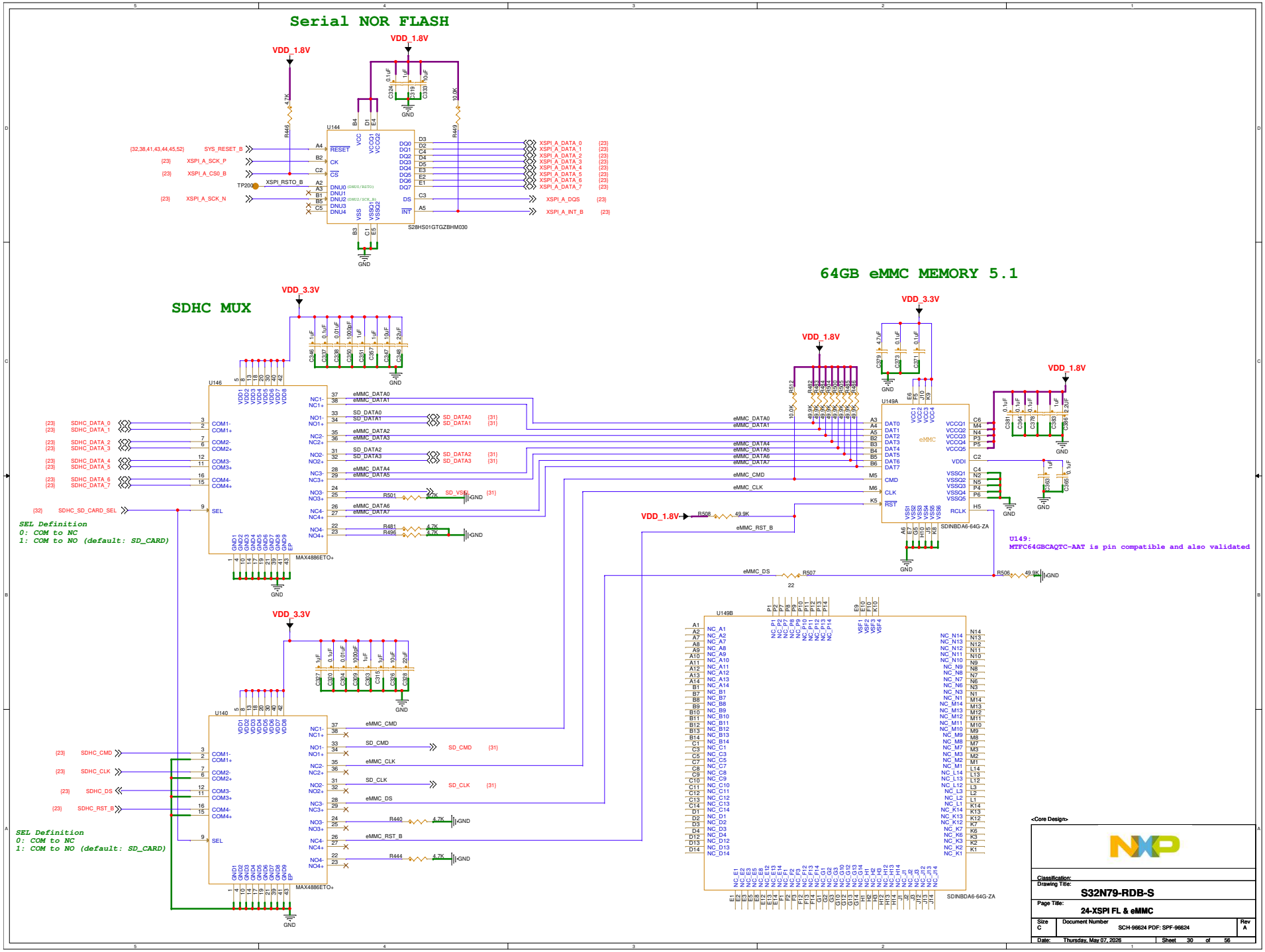
Serial NOR FLASH
VDD_1.8V
U144
S28HS01GTGZBHM030
SYS_RESET_B
XSPI_A_SCK_P
XSPI_A_CS0_B
XSPI_RSTO_B
XSPI_A_SCK_N
XSPI_A_DATA_0
XSPI_A_DATA_1
XSPI_A_DATA_2
XSPI_A_DATA_3
XSPI_A_DATA_4
XSPI_A_DATA_5
XSPI_A_DATA_6
XSPI_A_DATA_7
XSPI_A_DQS
XSPI_A_INT_B
SDHC MUX
VDD_3.3V
U146
U140
MAX4886ETO+
SDHC_DATA_0
SDHC_DATA_1
SDHC_DATA_2
SDHC_DATA_3
SDHC_DATA_4
SDHC_DATA_5
SDHC_DATA_6
SDHC_DATA_7
SDHC_SD_CARD_SEL
SDHC_CMD
SDHC_CLK
SDHC_DS
SDHC_RST_B
SEL Definition
0: COM to NC
1: COM to NO (default: SD_CARD)
64GB eMMC MEMORY 5.1
U149A
U149B
SDINBDA6-64G-ZA
eMMC_DATA0
eMMC_DATA1
eMMC_DATA2
eMMC_DATA3
eMMC_DATA4
eMMC_DATA5
eMMC_DATA6
eMMC_DATA7
eMMC_CMD
eMMC_CLK
eMMC_RST_B
eMMC_DS
U149:
MTFC64GBCAQTC-AAT is pin compatible and also validated
<Core Design>
Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
24-XSPI FL & eMMC
Size C
Document Number
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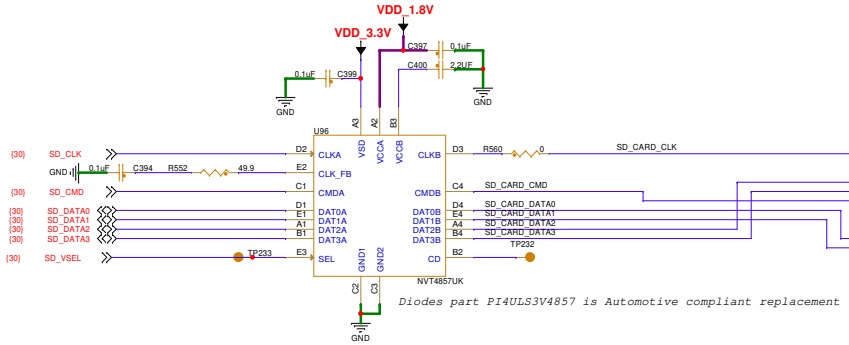
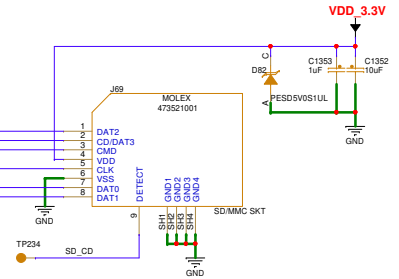
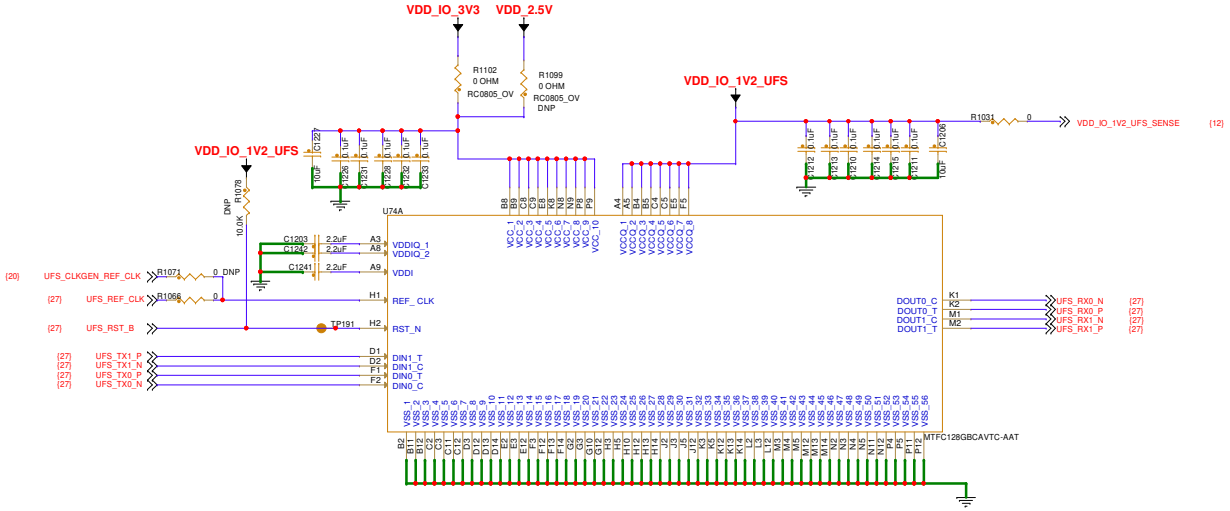
SD Level Translator

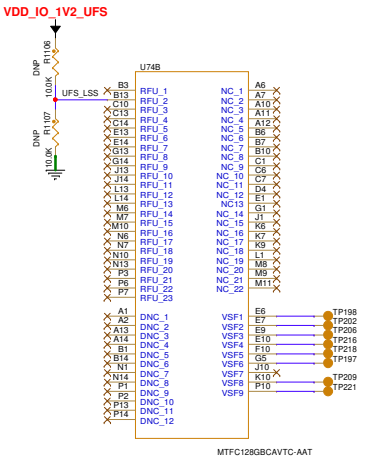


MICRO SD CARD SOCKET



UFS MEMORY





Classification:		
Drawing Title:	S32N79-RDB-S	
Page Title:	25-SD CARD & UFS MEM	
Size: C	Document Number: SCH-96824 PDF: SPF-96824	Rev: A
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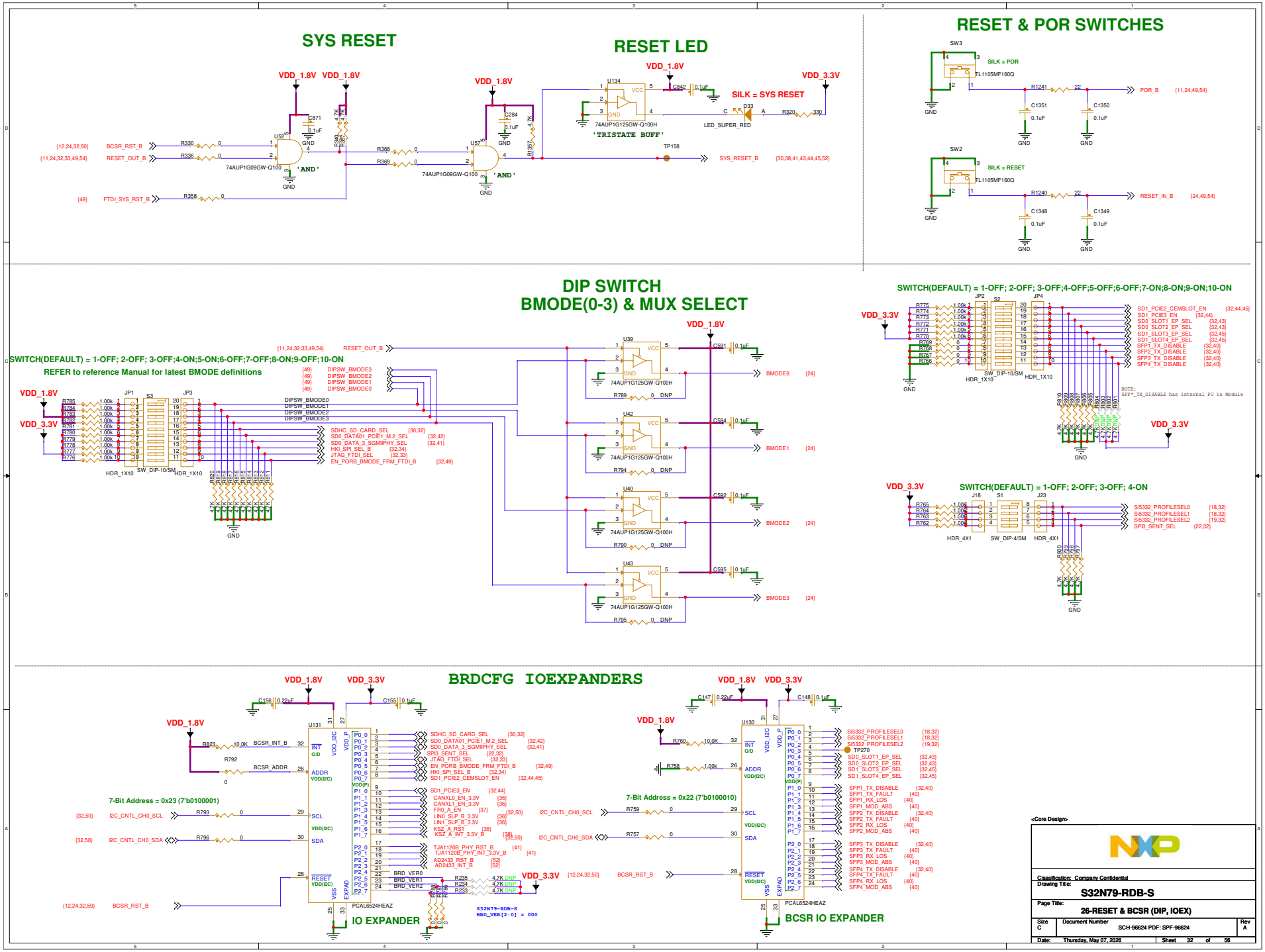
SYS RESET
RESET LED
RESET & POR SWITCHES
DIP SWITCH
BMODE(0-3) & MUX SELECT
BRDCFG IOEXPANDERS
SILK = SYS RESET
SILK = POR
SILK = RESET
'TRISTATE BUFF'
'AND'
SWITCH(DEFAULT) = 1-OFF; 2-OFF; 3-OFF;4-ON;5-ON;6-OFF;7-OFF;8-ON;9-OFF;10-ON
REFER to reference Manual for latest BMODE definitions
SWITCH(DEFAULT) = 1-OFF; 2-OFF; 3-OFF;4-OFF;5-OFF;6-OFF;7-ON;8-ON;9-ON;10-ON
SWITCH(DEFAULT) = 1-OFF; 2-OFF; 3-OFF; 4-ON
7-Bit Address = 0x23 (7'b0100001)
7-Bit Address = 0x22 (7'b0100010)
IO EXPANDER
BCSR IO EXPANDER
S32N79-RDB-S
BRD_VER[2:0] = 000
<Core Design>
Classification: Company Confidential
Drawing Title:
S32N79-RDB-S
Page Title:
26-RESET & BCSR (DIP, IOEX)
Size C
Document Number SCH-96824 PDF: SPF-96824
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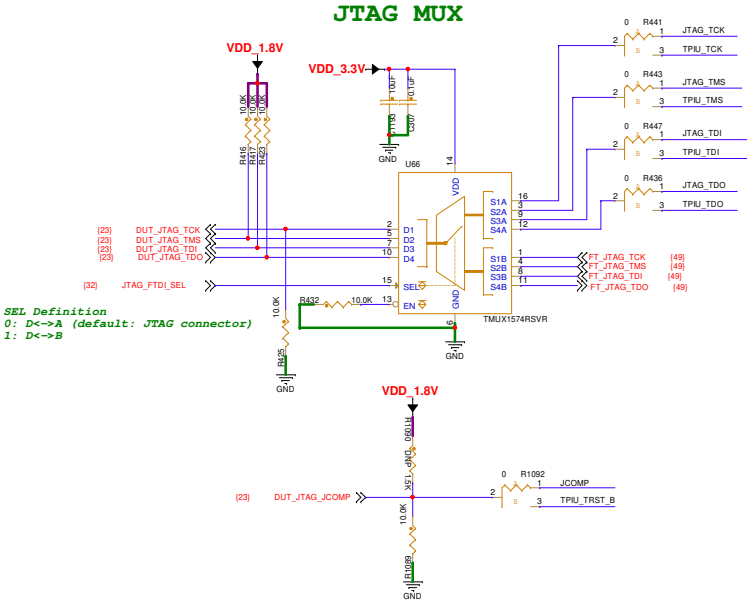
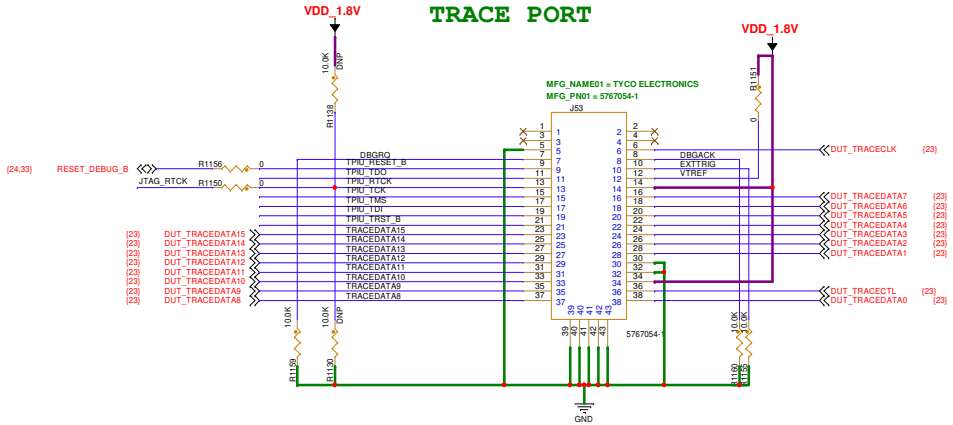
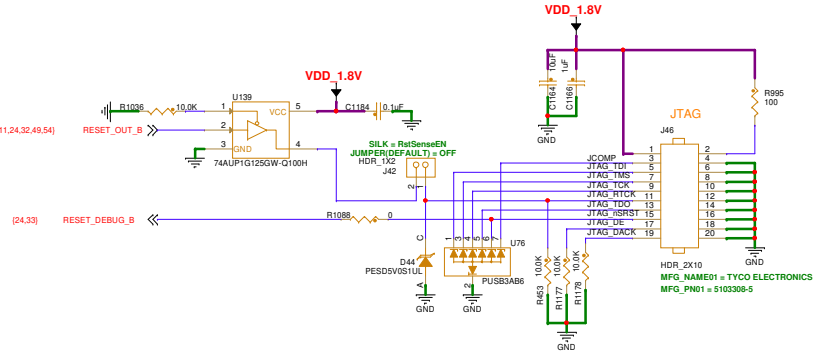
JTAG MUX



TRACE PORT



20-Pin JTAG



<Core Design>

Classification:	
Drawing Title:	S32N79-RDB-S
Page Title:	27 JTAG & TPIU
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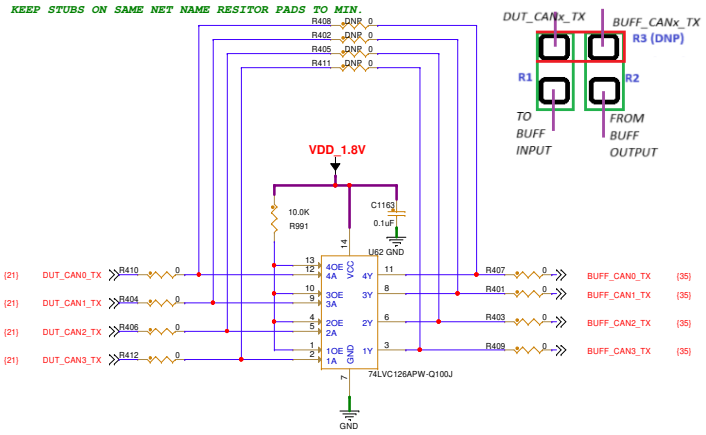
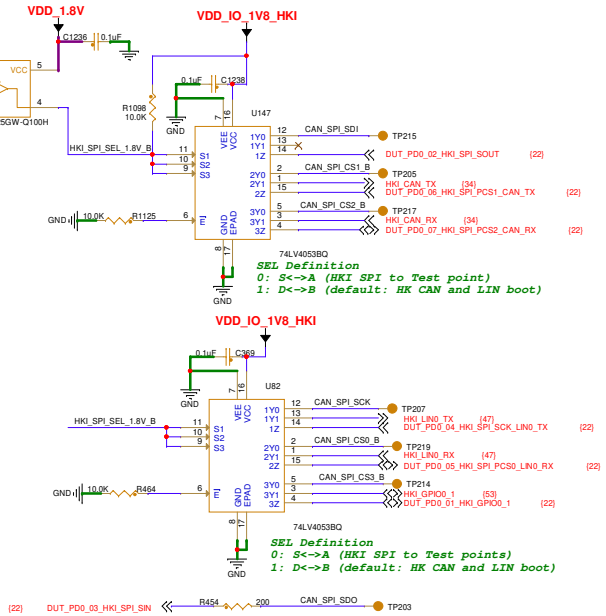
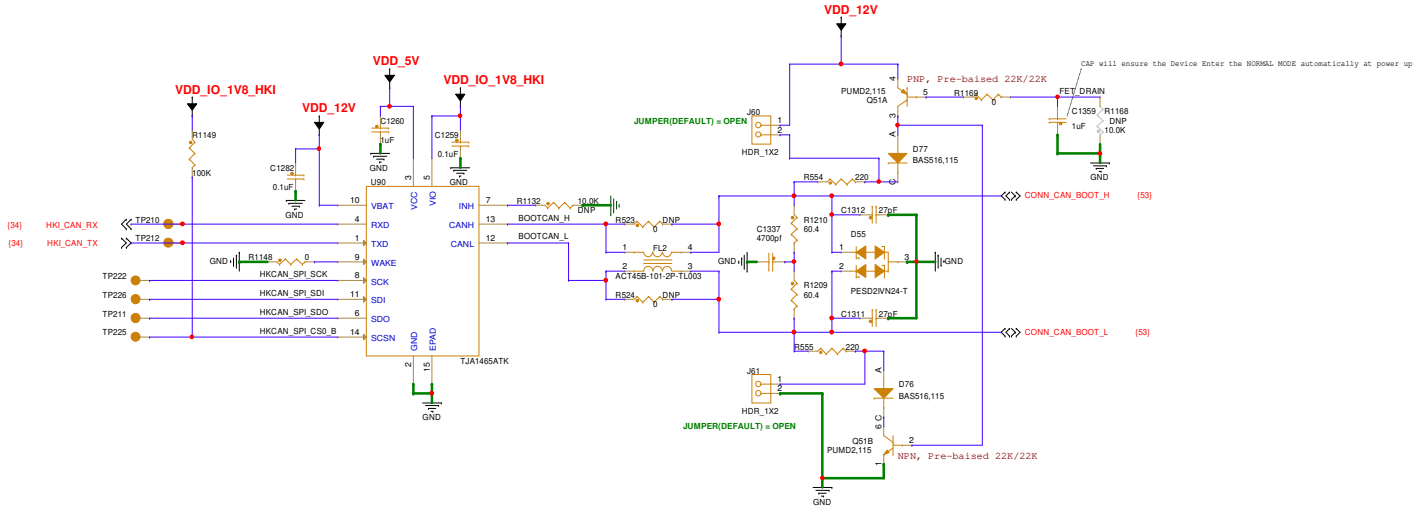
CAN TX BUFFER



HKI SPI MUX



BOOT FROM CAN



<Core Design>

Classification: Company Confidential

Drawing Title: **S32N79-RDB-S**

Page Title: **28-CAN BOOT & HIZ BUFF**

Size	Document Number	Rev
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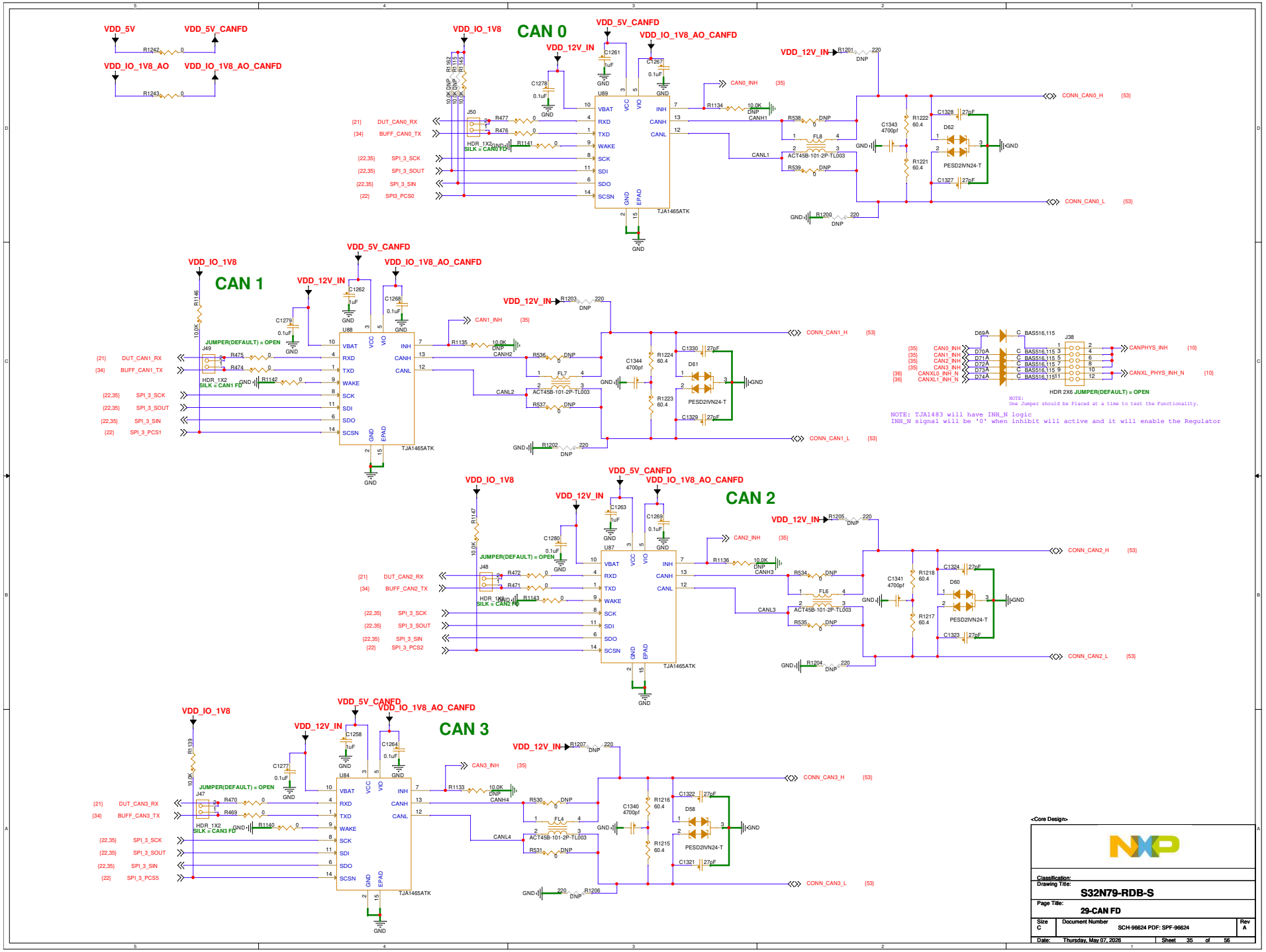
VDD_5V
VDD_5V_CANFD
VDD_IO_1V8_AO
VDD_IO_1V8_AO_CANFD
CAN 0
CAN 1
CAN 2
CAN 3
VDD_IO_1V8
VDD_12V_IN
TJA1465ATK
JUMPER(DEFAULT) = OPEN
HDR 2X6 JUMPER(DEFAULT) = OPEN
NOTE: TJA1483 will have INH_N logic
INH_N signal will be '0' when inhibit will active and it will enable the Regulator
<Core Design>
Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
29-CAN FD
Size C
Document Number SCH-96624 PDF: SPF-96624
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Date: Thursday, May 07, 2026
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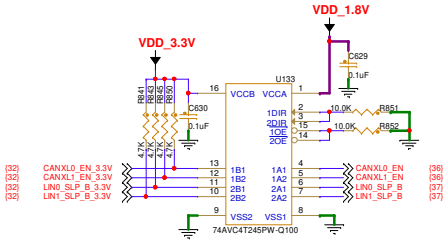
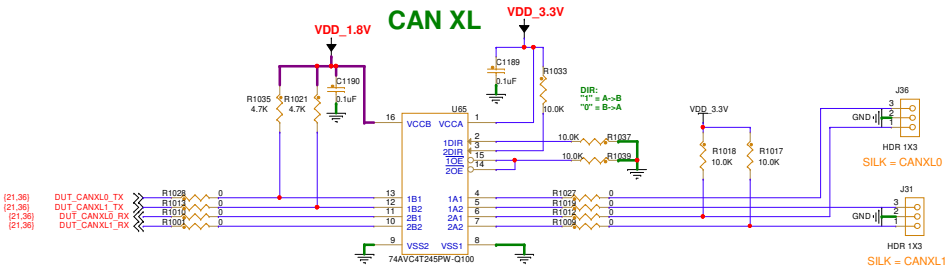
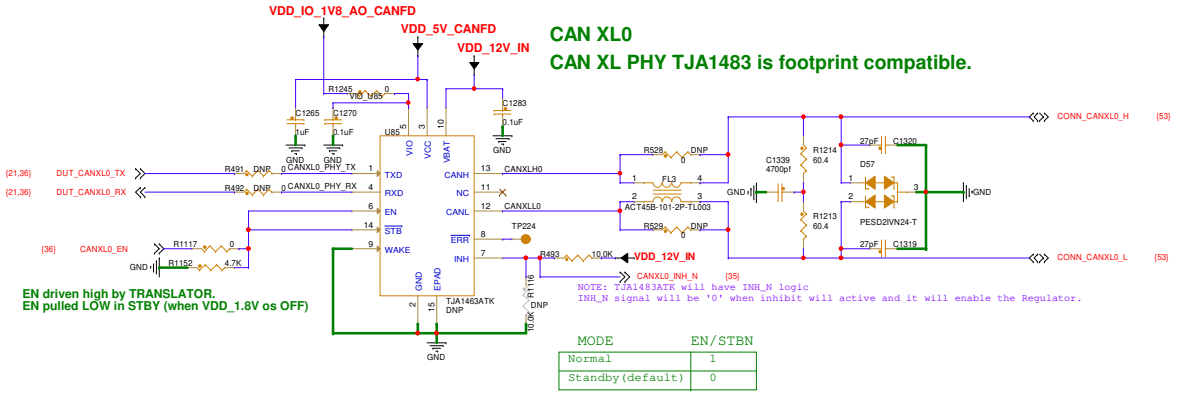
ENABLE TRANSLATION 3.3V to 1.8V



1.8V to 3.3V CONVERSION FOR ALBI

CAN XL

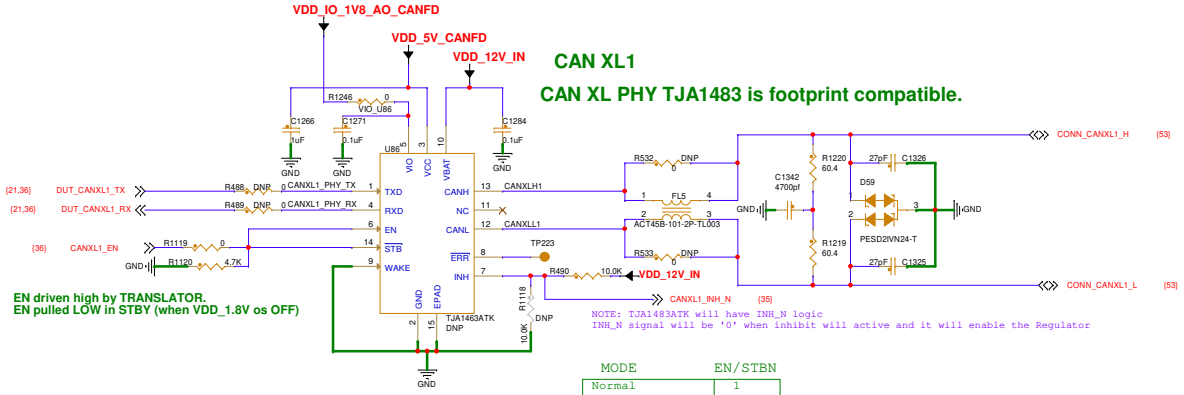




CAN XL0

CAN XL PHY TJA1483 is footprint compatible.

MODE	EN/STBN
Normal	1
Standby(default)	0



CAN XL1

CAN XL PHY TJA1483 is footprint compatible.

MODE	EN/STBN
Normal	1
Standby(default)	0

<Core Design>

Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **30-CANXL**

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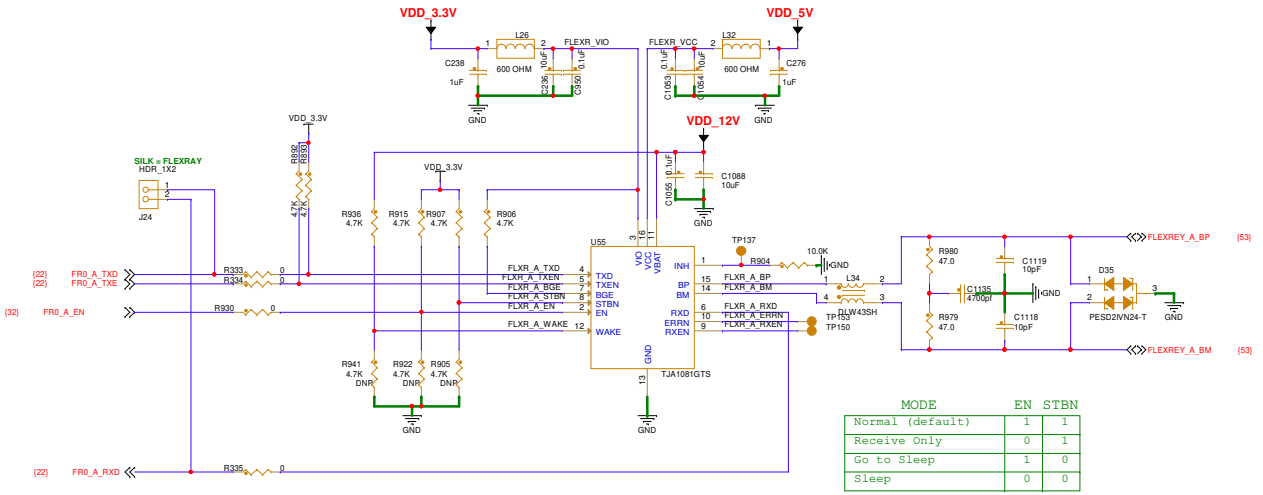
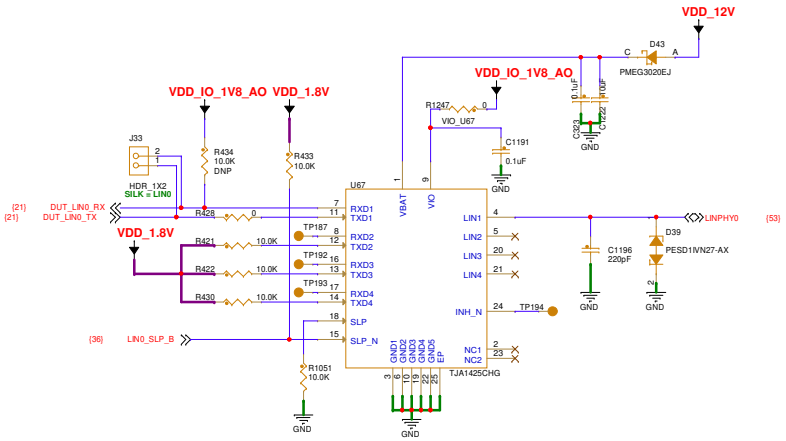
FLEXRAY

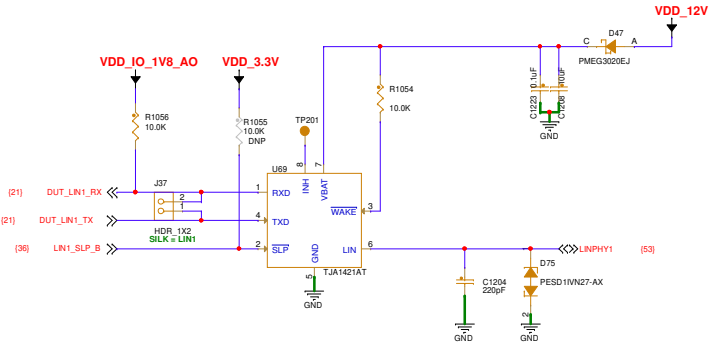


MODE	EN	STBN
Normal (default)	1	1
Receive Only	0	1
Go to Sleep	1	0
Sleep	0	0

LIN MASTER (LIN0)



LIN Slave (LIN1)



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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **31-LIN & FLEXRAY**

Size	Document Number	Rev
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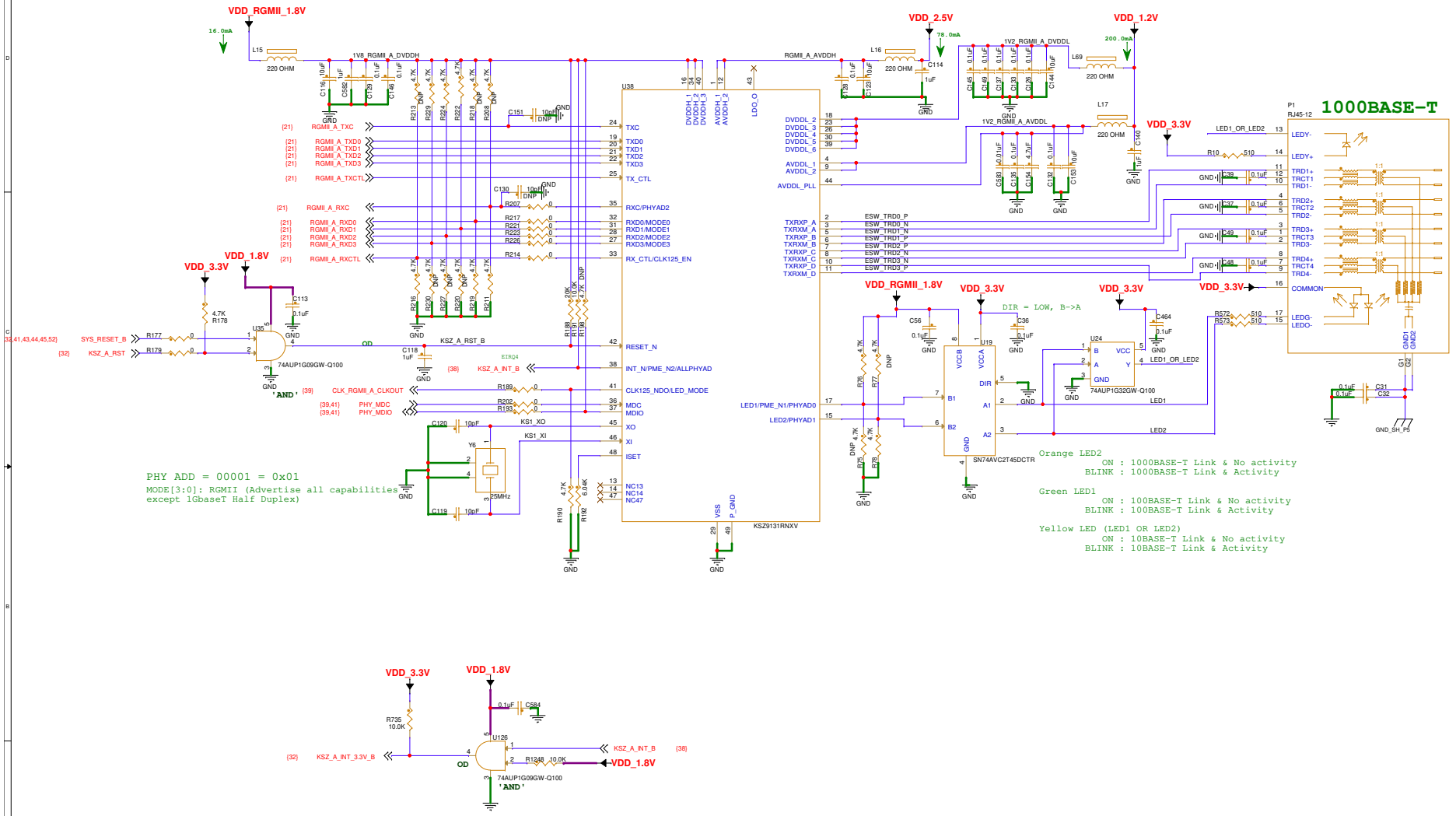
RGMII PHY



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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **32-RGMII-A KSZ9131 PHY**

Size	Document Number	Rev
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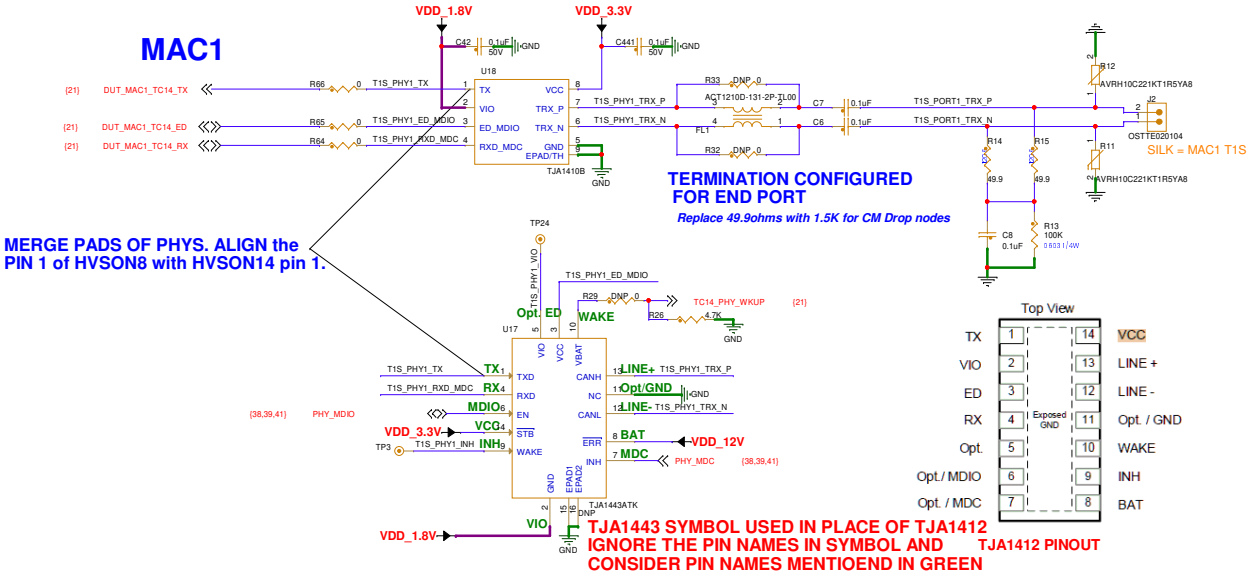
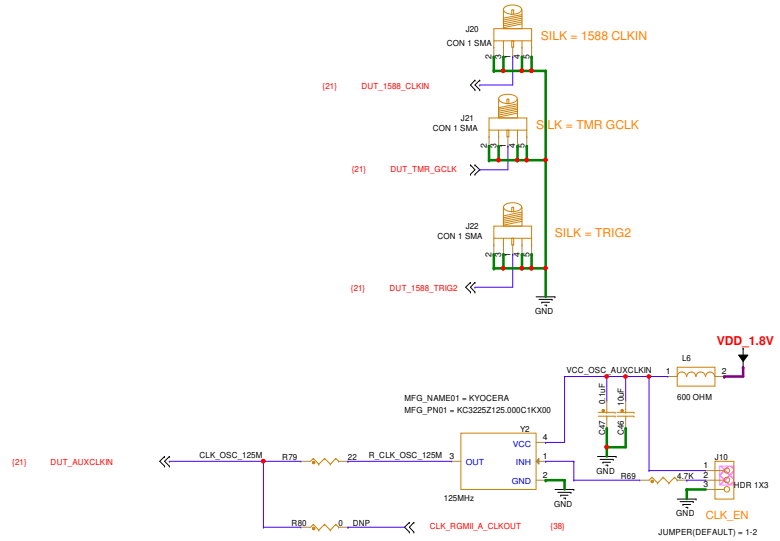
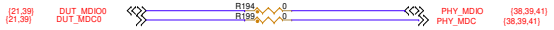
MAC1



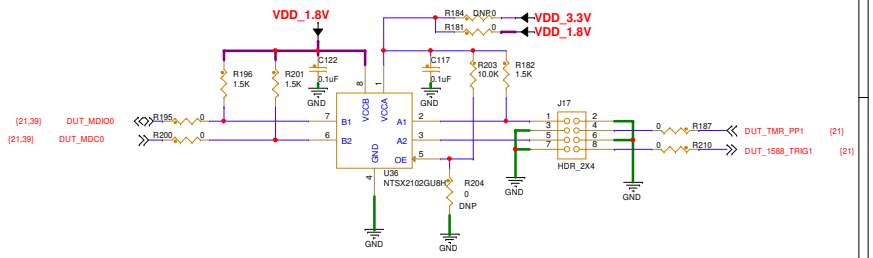
1588 & AUXCLK



MDC-MDIO MUX



ETHERENT HEADER



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Classification:
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Page Title: **33-10BaseT1S TJA1410/2**

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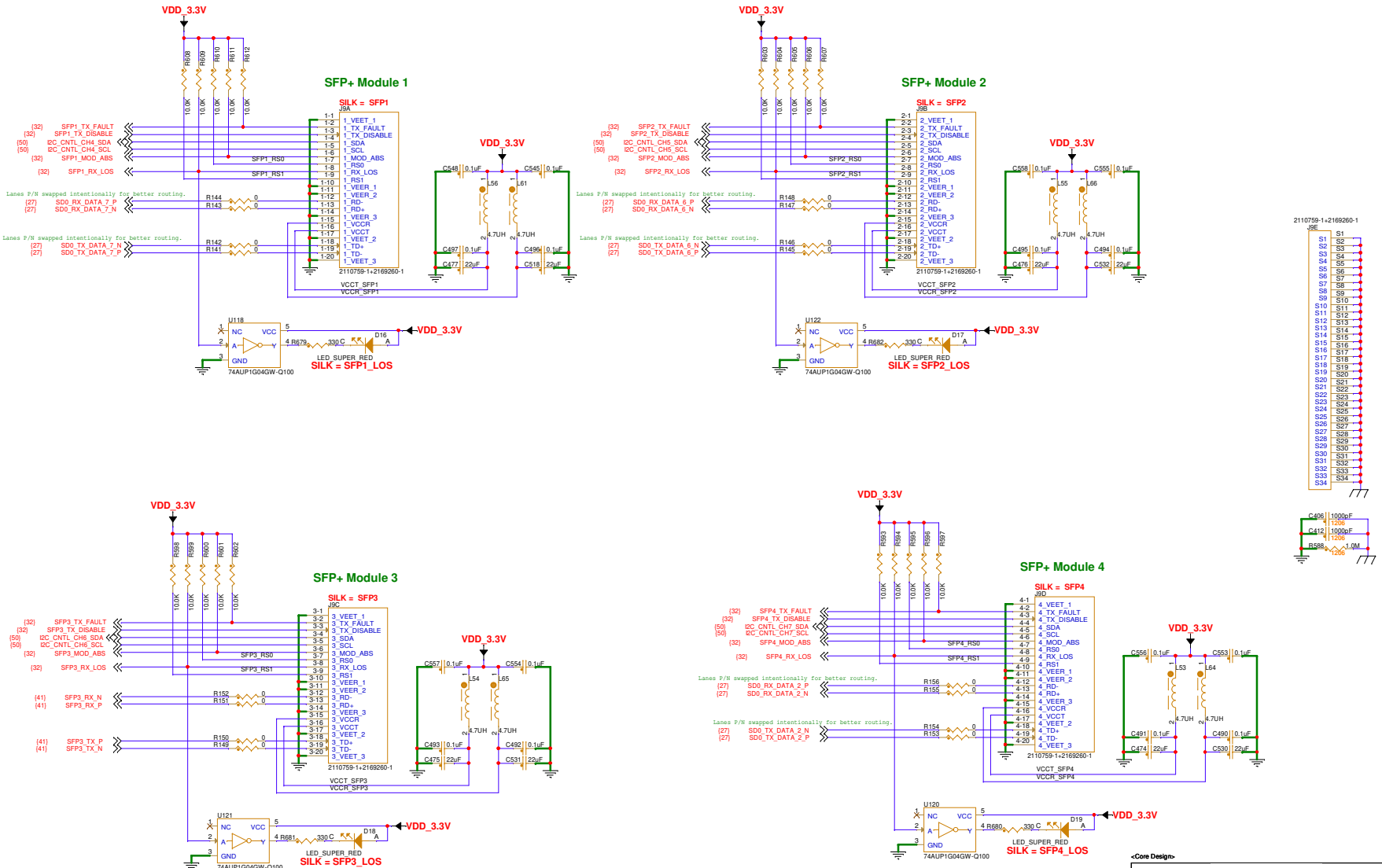
1x4 SFP+ CONNECTOR



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Drawing Title:	S32N79-RDB-S
Page Title:	35-SFP+ [1]
Size C	Document Number SCH-96624 PDF: SPF-96624 Rev A
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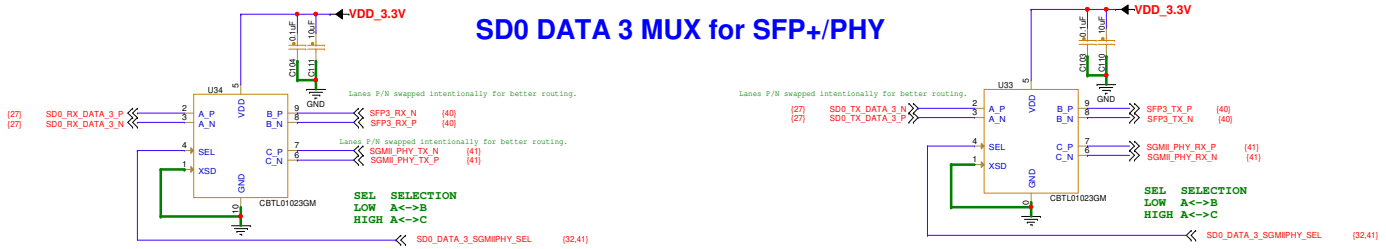
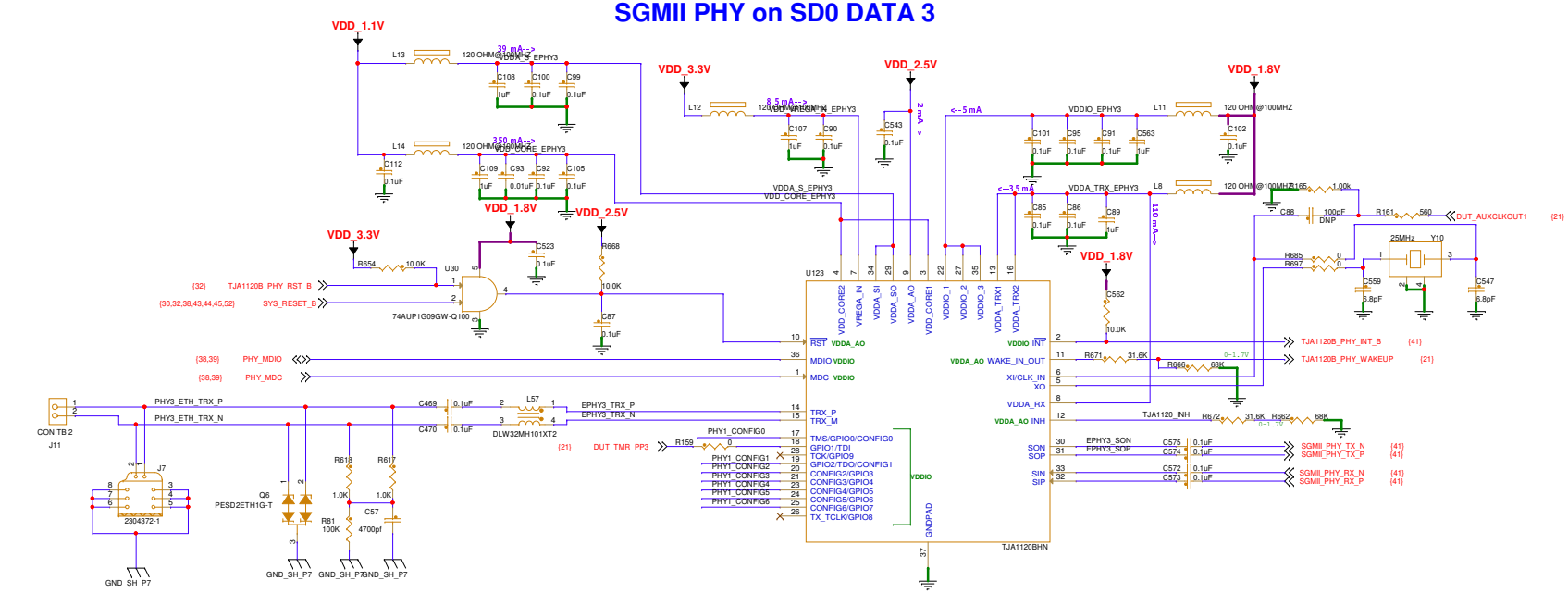
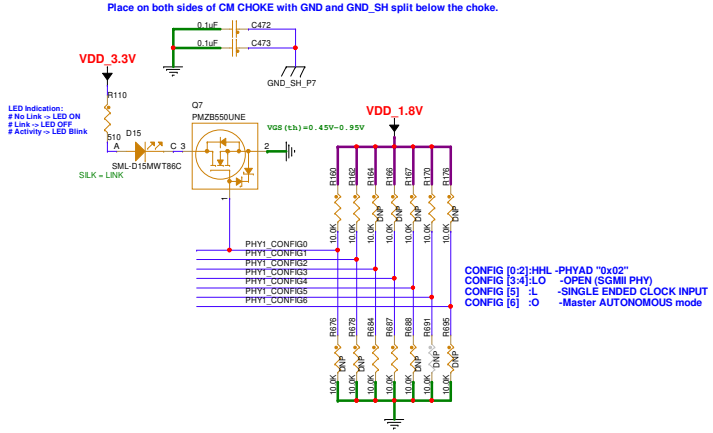
SD0 DATA 3 MUX for SFP+/PHY

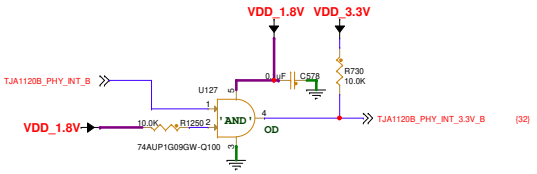


SGMII PHY on SD0 DATA 3



Place on both sides of CM CHOKE with GND and GND_SH split below the choke.





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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **36-TJA1120B 1GT1 PHY**

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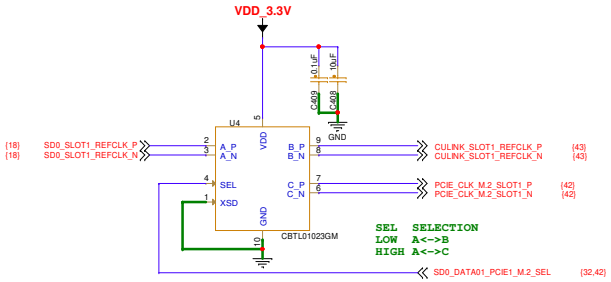
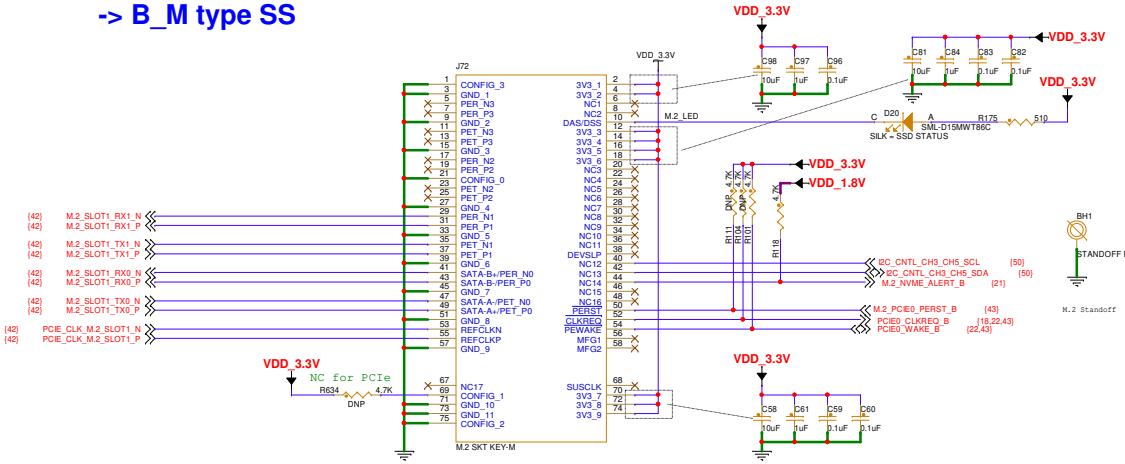
SD0 DATA0/1 MUX for CULINK/M.2

REFCLK MUX for CULINK/M.2



M.2 (2280) NVMe PCIe SSD using Gen 4.0 x 2 PCIe interface [Type M Key]
-> NVMe SSD
-> B_M type SS



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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **37-SD0 MUX & M.2 NVMe**

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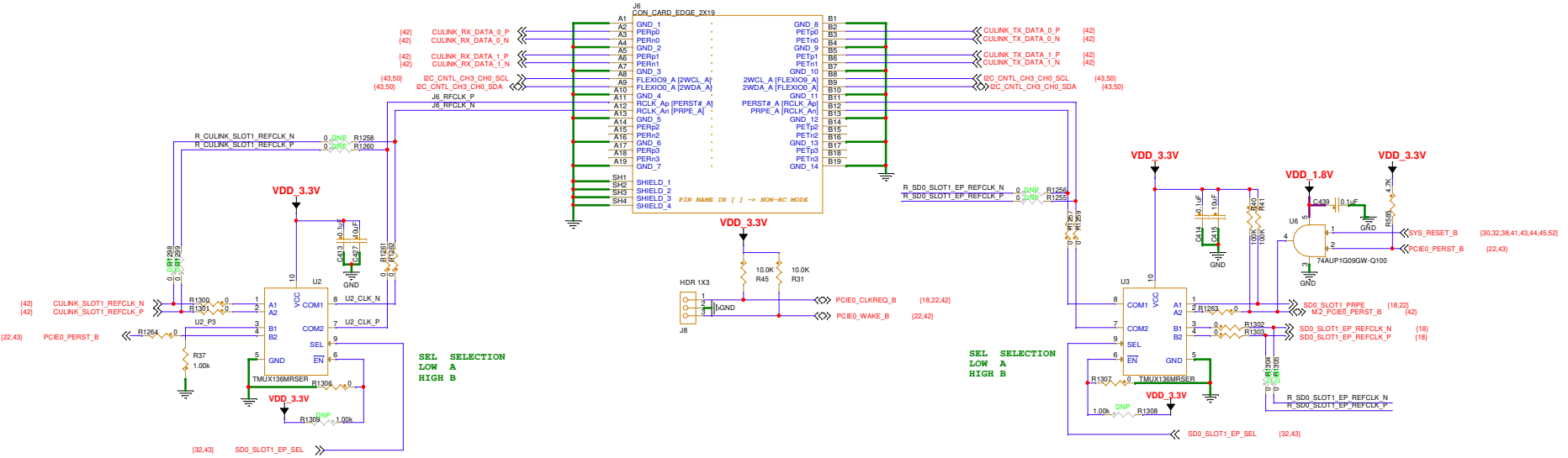
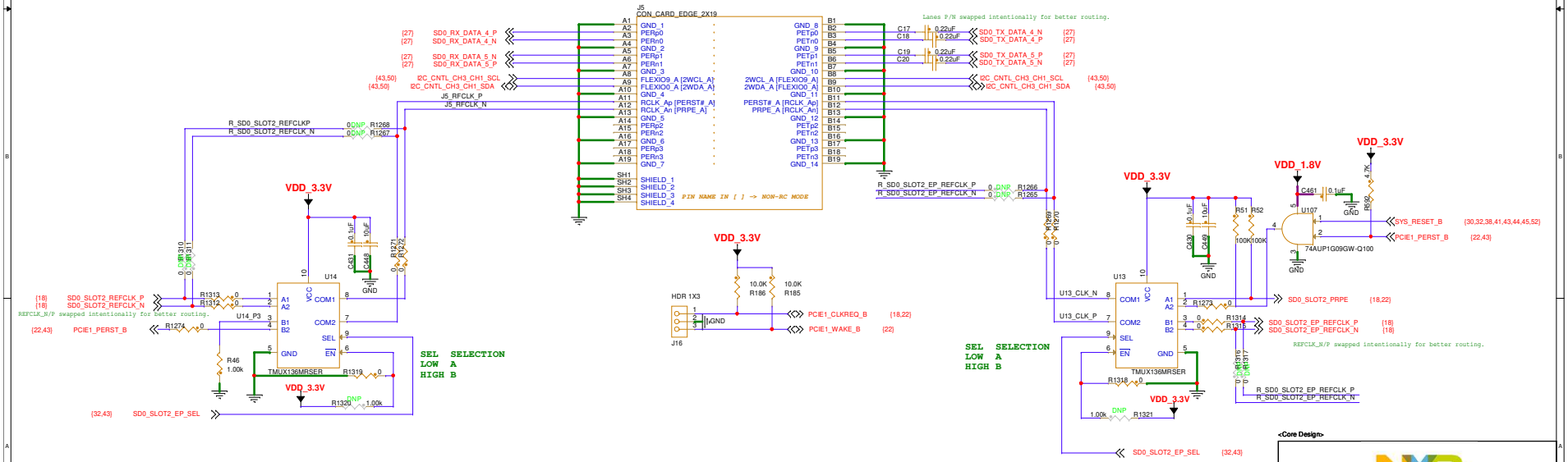
PCIe CuLINK x2 (SD0 PCIE CONTROLLER 0)



PCIe CuLINK x2 (SD0 PCIE CONTROLLER 1)



<Core Design>

Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **38-SD0 Culink**

Size	Document Number	Rev
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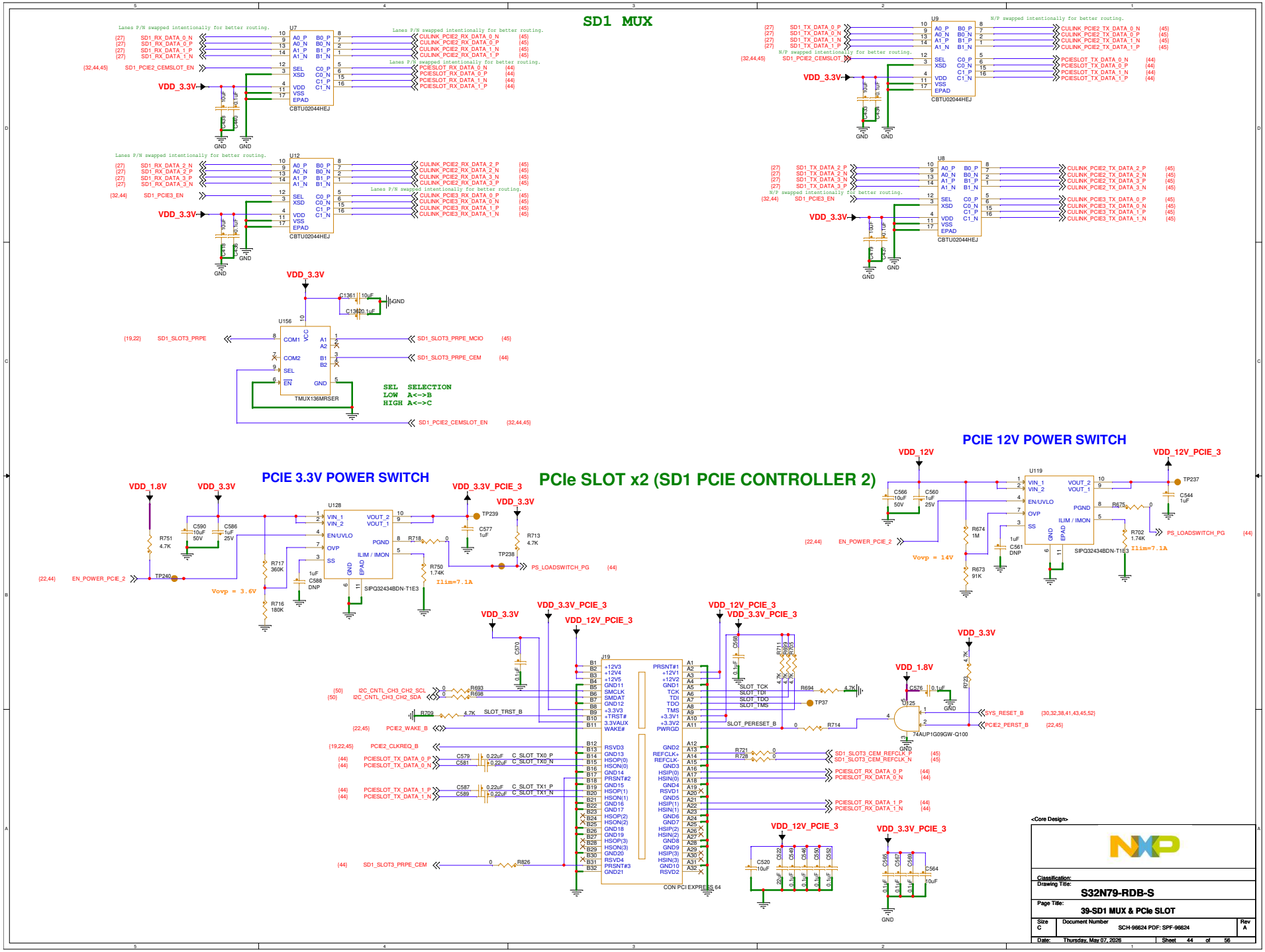
SD1 MUX
PCIE 12V POWER SWITCH
PCIE 3.3V POWER SWITCH
PCIe SLOT x2 (SD1 PCIE CONTROLLER 2)
SEL SELECTION
LOW A<->B
HIGH A<->C
Vovp = 14V
Vovp = 3.6V
Ilim=7.1A
CON PCI EXPRESS 64
<Core Design>
Classification:
Drawing Title:
S32N79-RDB-S
Page Title:
39-SD1 MUX & PCIe SLOT
Size C
Document Number SCH-96824 PDF: SPF-96824
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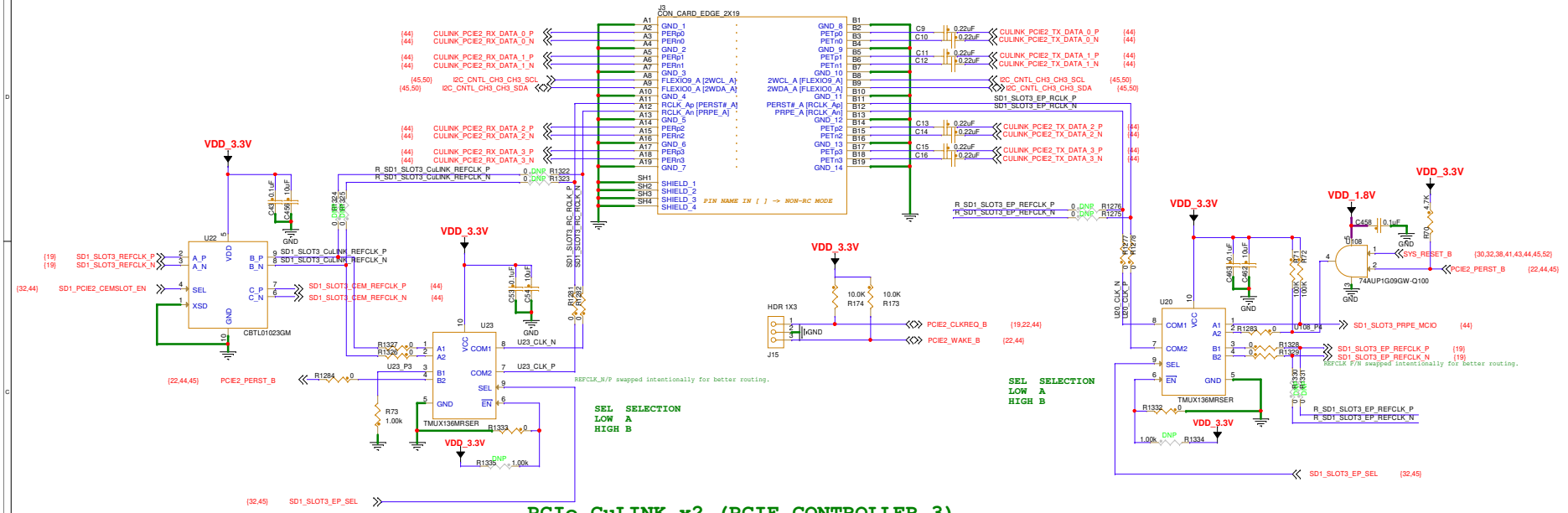
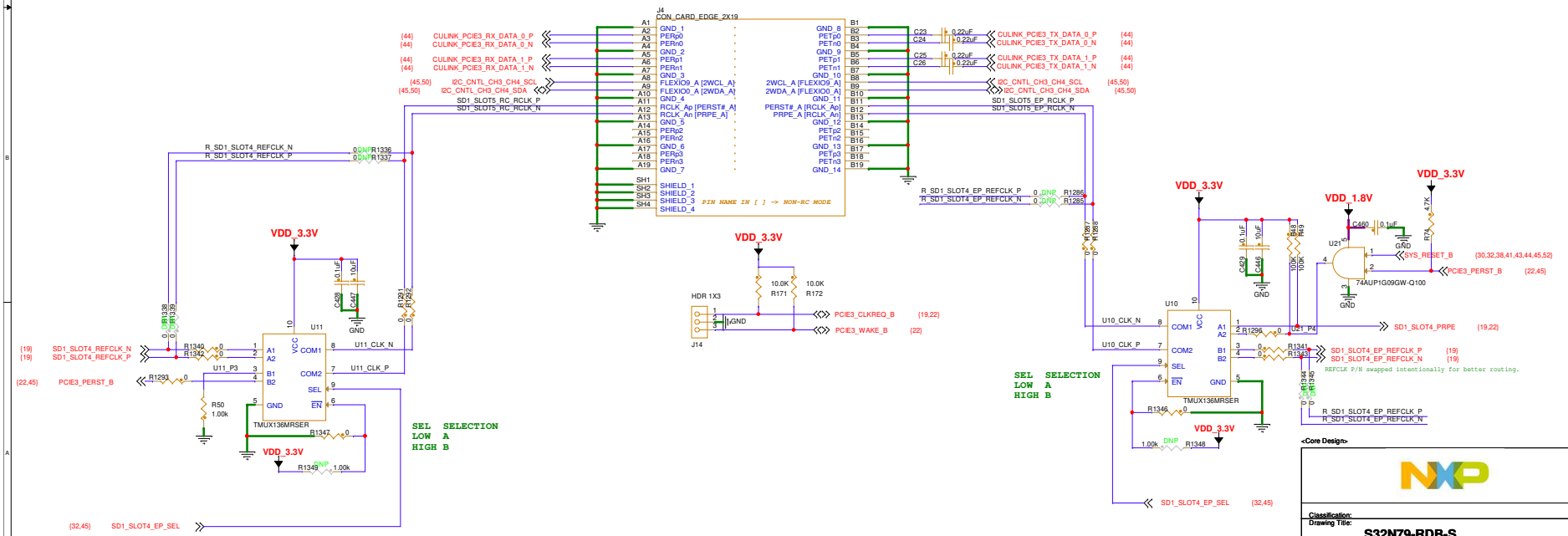
Date: Thursday, May 07, 2026
Sheet 44 of 56

PCIe CuLINK x2/x4 (PCIE CONTROLLER 2)



PCIe CuLINK x2 (PCIE CONTROLLER 3)



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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **40-SD1 CuLink**

Size	Document Number	Rev
C	SCH-96624 PDF: SPF-96624	A

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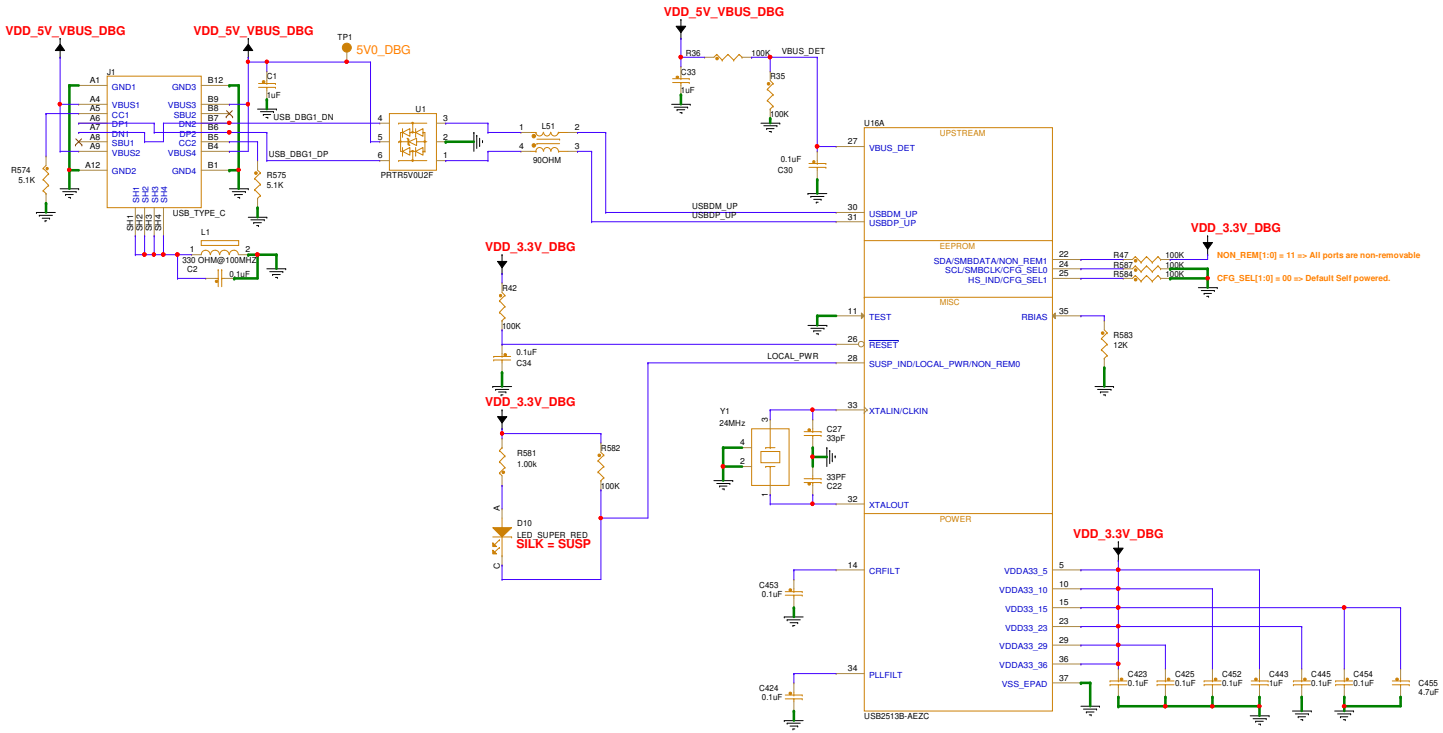
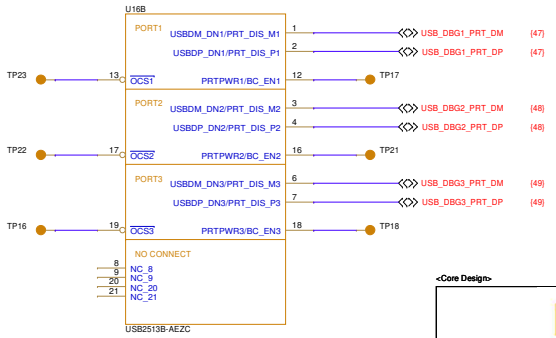
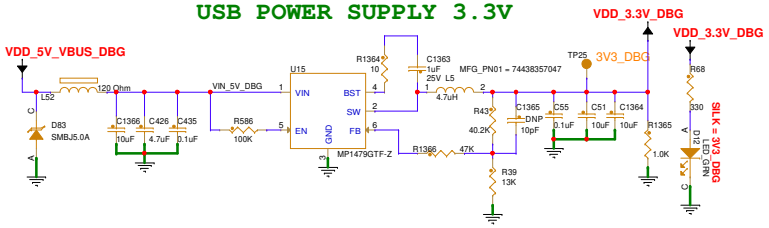
USB HUB UPSTREAM & CONFIGURATION



USB HUB DOWNSTREAM PORTS



USB POWER SUPPLY 3.3V



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Classification:

Drawing Title: **S32N79-RDB-S**

Page Title: **41-USB HUB**

Size	Document Number	Rev
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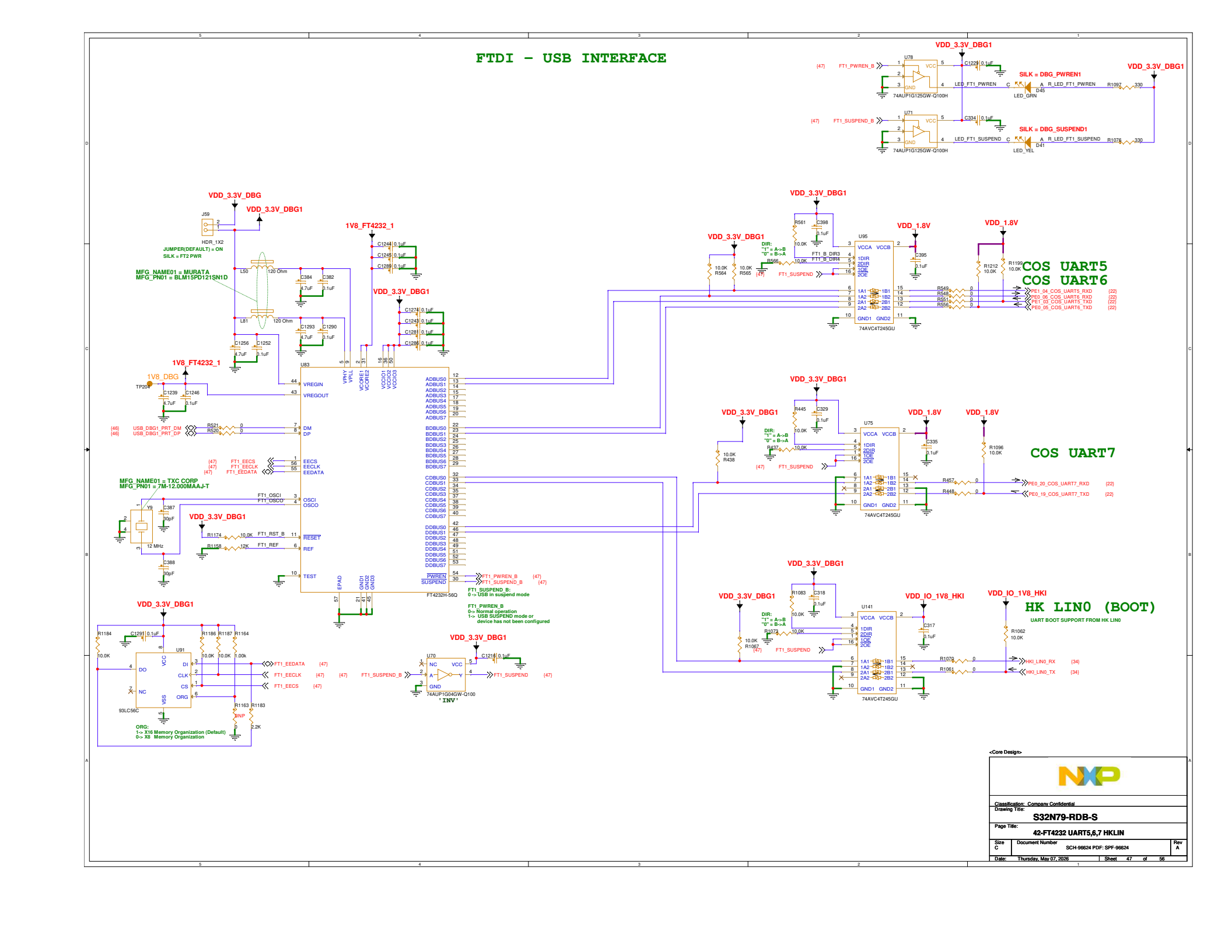
FTDI – USB INTERFACE



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Classification: Company Confidential

Drawing Title: **S32N79-RDB-S**

Page Title: **42-FT4232 UART5,6,7 HKLIN**

Size	Document Number	Rev
C	SCH-96624 PDF: SPF-96624	A

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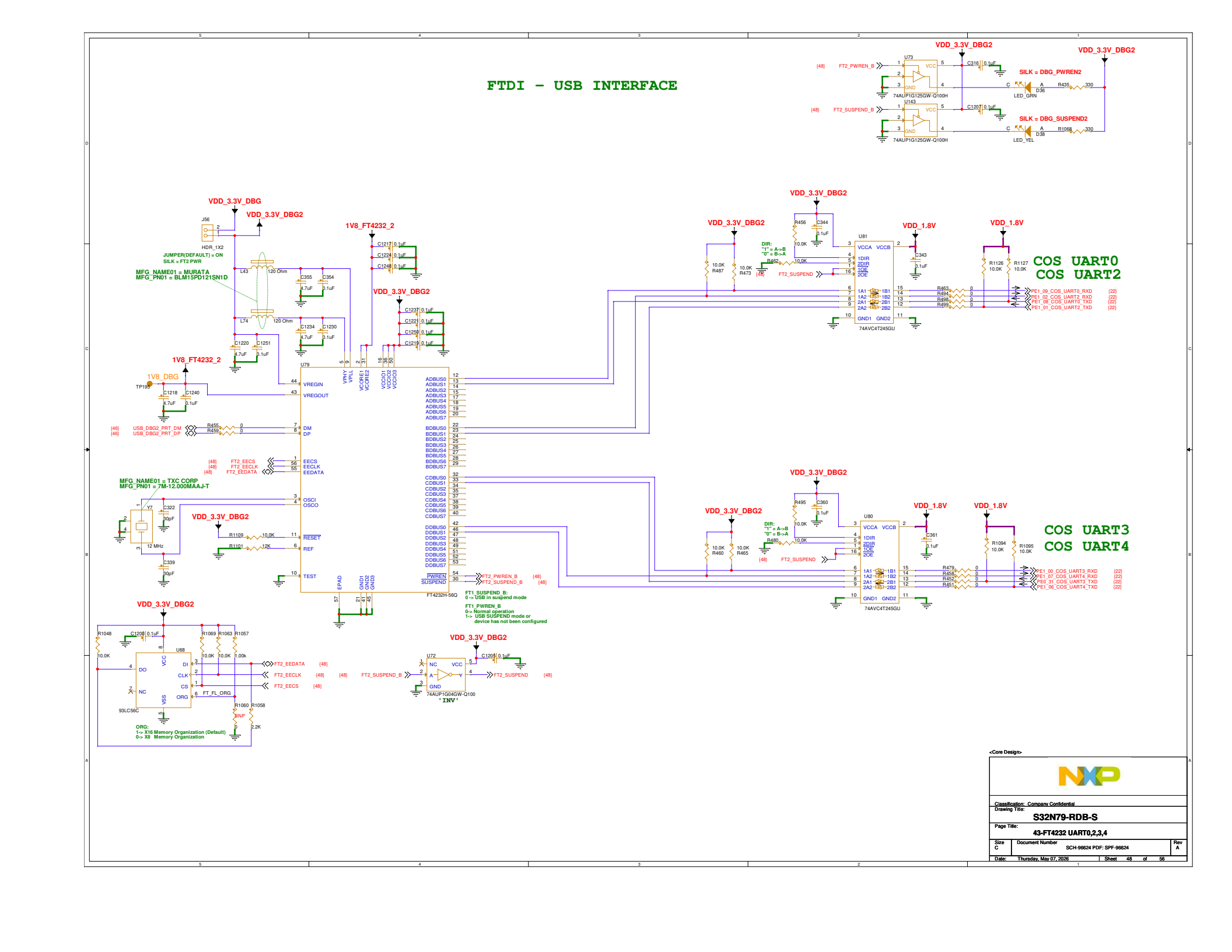
FTDI – USB INTERFACE



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Classification: Company Confidential

Drawing Title: S32N79-RDB-S

Page Title: 43-FT4232 UART0,2,3,4

Size	Document Number	Rev
C	SCH-96624 PDF: SPF-96624	A

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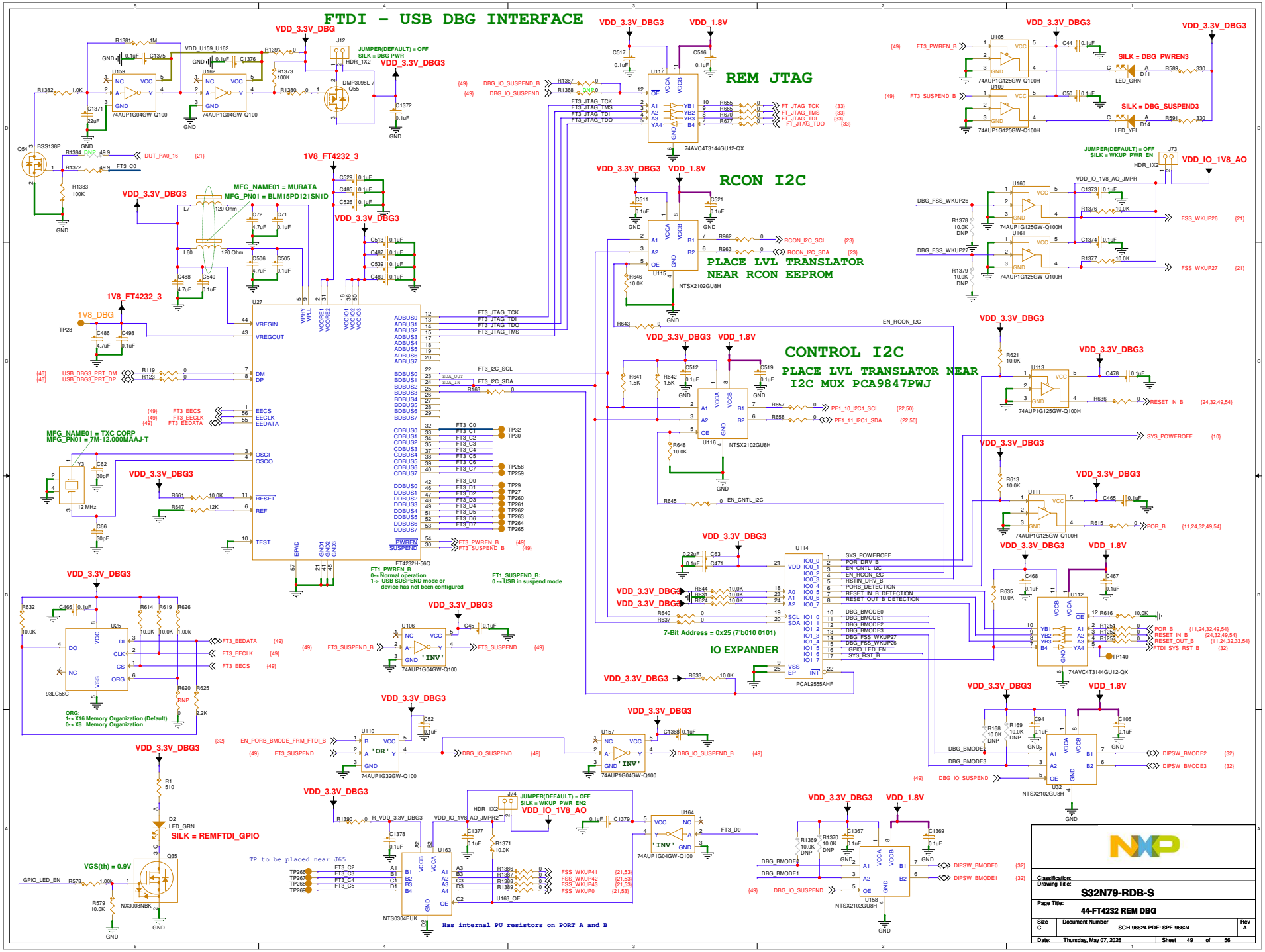

Classification:	
Drawing Title:	S32N79-RDB-S
Page Title:	44-FT4232 REM DBG
Size: C	Document Number: SCH-96624 PDF: SPF-96624
Rev: A	
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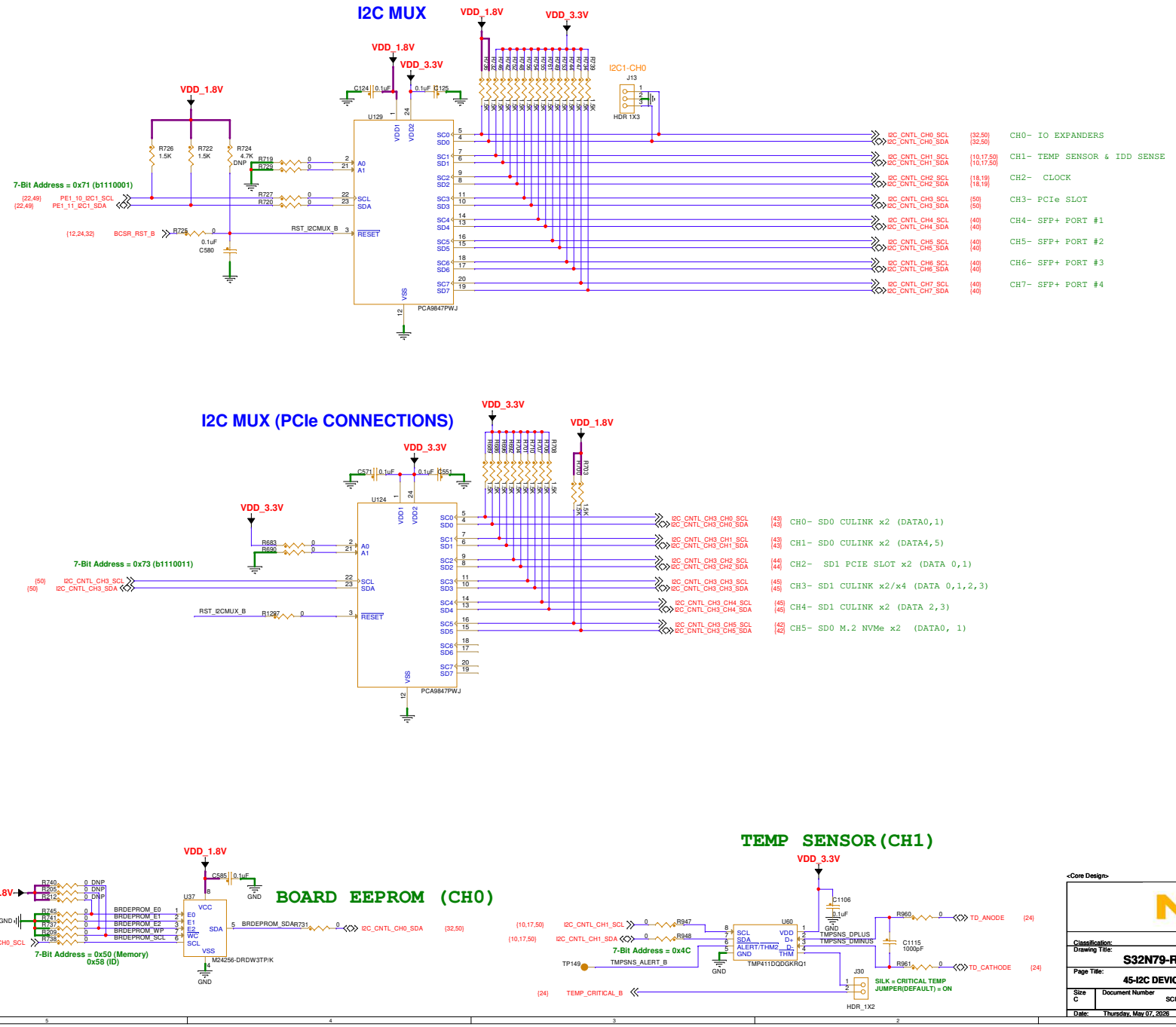

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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **45-I2C DEVICES**

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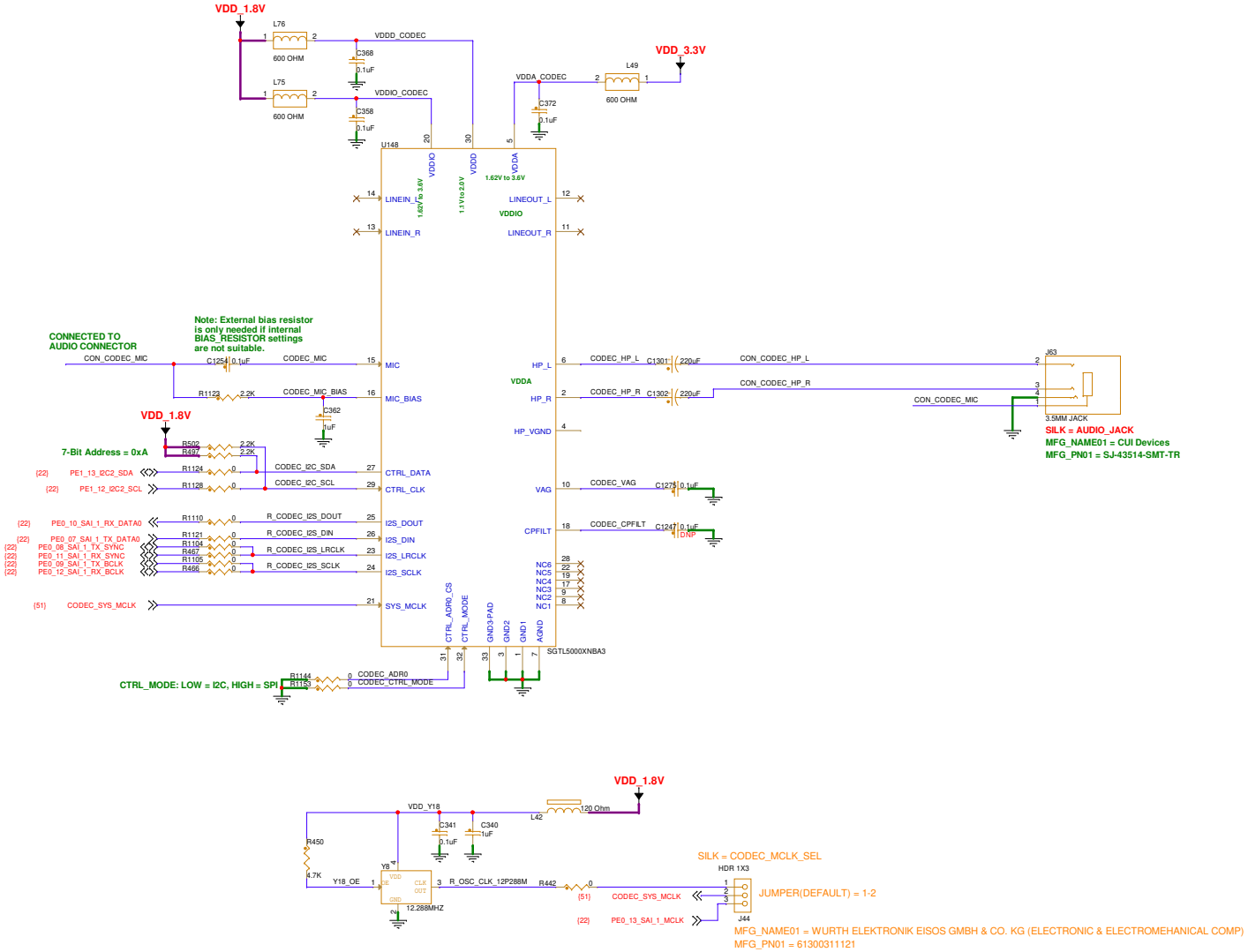
AUDIO CODEC



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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **46-AUDIO SGTL5000**

Size	Document Number	Rev
C	SCH-96624 PDF: SPF-96624	A

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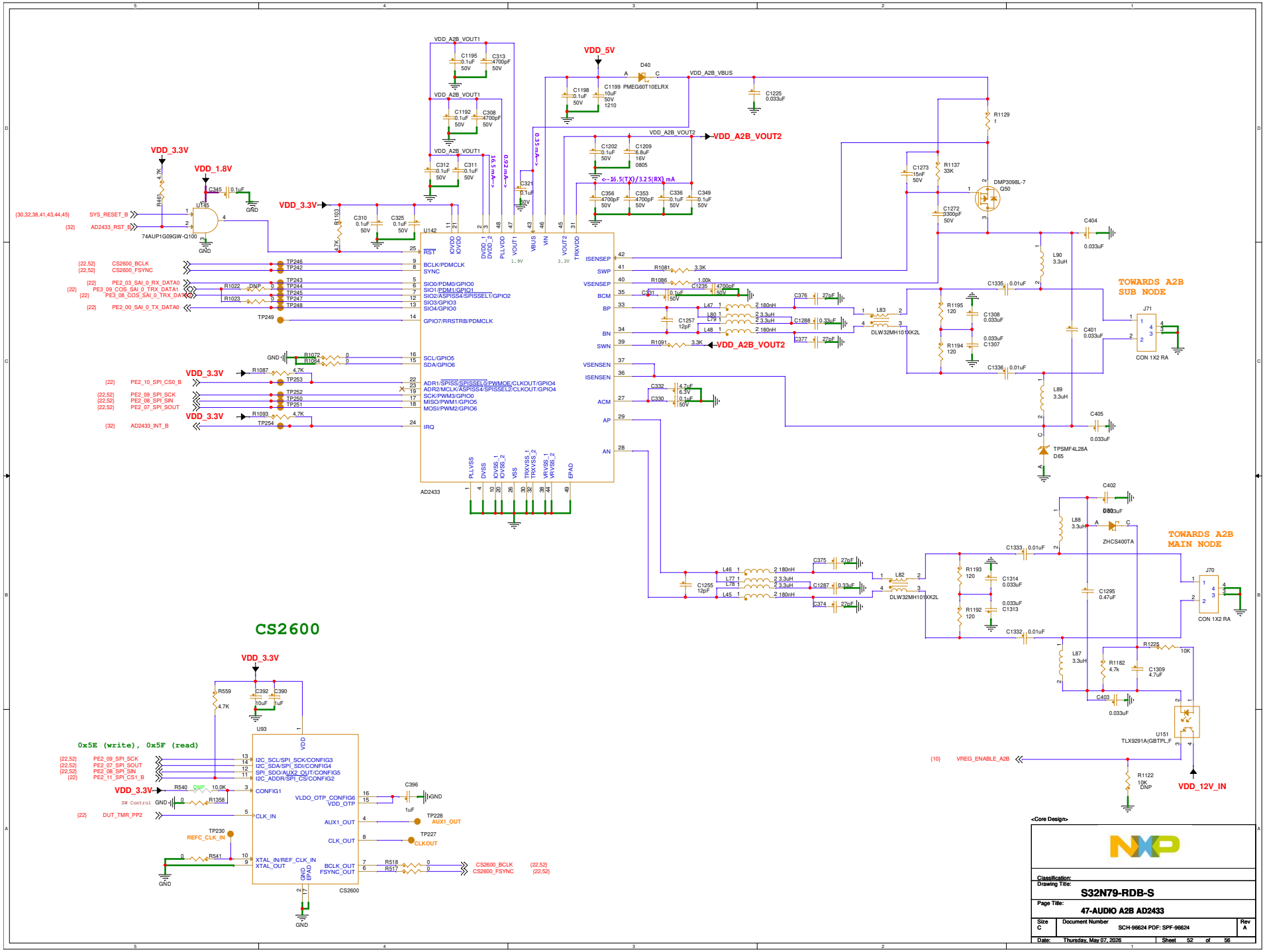
VDD_5V
VDD_3.3V
VDD_1.8V
VDD_A2B_VOUT1
VDD_A2B_VOUT2
VDD_A2B_VBUS
D40
PMEG60T10ELRX
DMP3098L-7
Q50
74AUP1G09GW-C100
U145
AD2433
U142
ISENSE
SWP
VSENSE
BCM
BP
BN
SWN
VSENSEN
ISENSEN
ACM
AP
AN
DLW32MH101XK2L
TPSMF4L28A
D65
ZHCS400TA
CON 1X2 RA
J71
J70
TOWARDS A2B
SUB NODE
TOWARDS A2B
MAIN NODE
CS2600
U93
CS2600
I2C_SCL/SPI_SCK/CONFIG3
I2C_SDA/SPI_SDI/CONFIG4
SPI_SDO/AUX2_OUT/CONFIG5
I2C_ADDR/SPI_CS/CONFIG2
CONFIG1
CLK_IN
XTAL_IN/REF_CLK_IN
XTAL_OUT
VLDO_OTP_CONFIG6
VDD_OTP
AUX1_OUT
CLK_OUT
BCLK_OUT
FSYNC_OUT
0x5E (write), 0x5F (read)
TLX9291A(GBTPL_F
VREG_ENABLE_A2B
VDD_12V_IN
SYS_RESET_B
AD2433_RST_B
AD2433_INT_B
CS2600_BCLK
CS2600_FSYNC
PE2_03_SAI_0_RX_DATA0
PE3_09_COS_SAI_0_TRX_DATA1
PE3_08_COS_SAI_0_TRX_DATA0
PE2_00_SAI_0_TX_DATA0
PE2_10_SPI_CS0_B
PE2_09_SPI_SCK
PE2_08_SPI_SIN
PE2_07_SPI_SOUT
PE2_11_SPI_CS1_B
DUT_TMR_PP2
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Drawing Title:
S32N79-RDB-S
Page Title:
47-AUDIO A2B AD2433
Size C
Document Number SCH-96624 PDF: SPF-96624
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Date: Thursday, May 07, 2026
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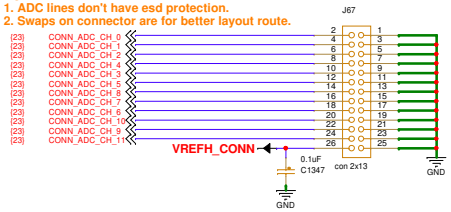
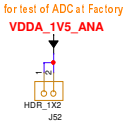
ADC

SILK = ADC PORT

NOTE:
1. ADC lines don't have esd protection.
2. Swaps on connector are for better layout route.

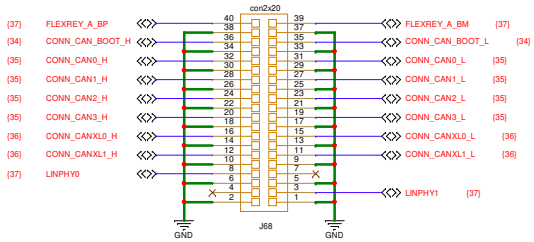


ONLY for test of ADC at Factory



COMM CONNECTOR @ BOARD EDGE

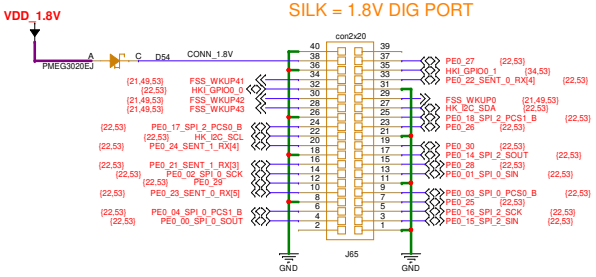
SILK = COMM PORT



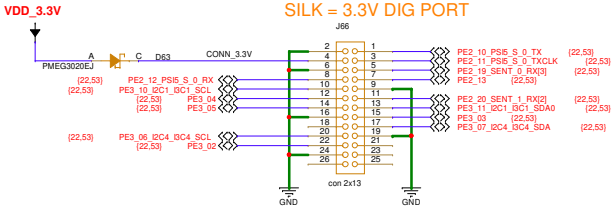
Multi Function Ports @ BOARD EDGE

NOTE: Refer to IOMUX table for more details on mux options

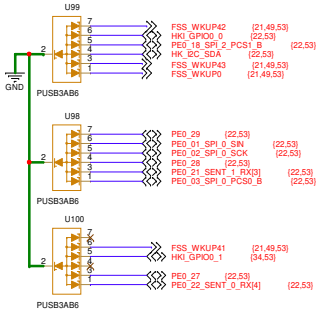
SILK = 1.8V DIG PORT

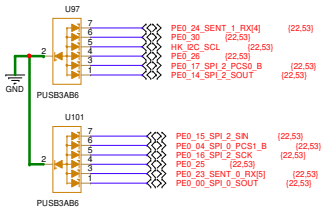


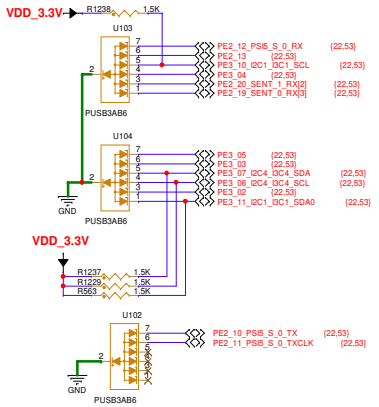
SILK = 3.3V DIG PORT

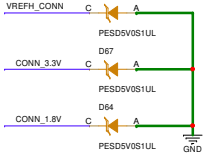


ESD DIODES FOR IOS and POWER









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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **48-CONNECTORS**

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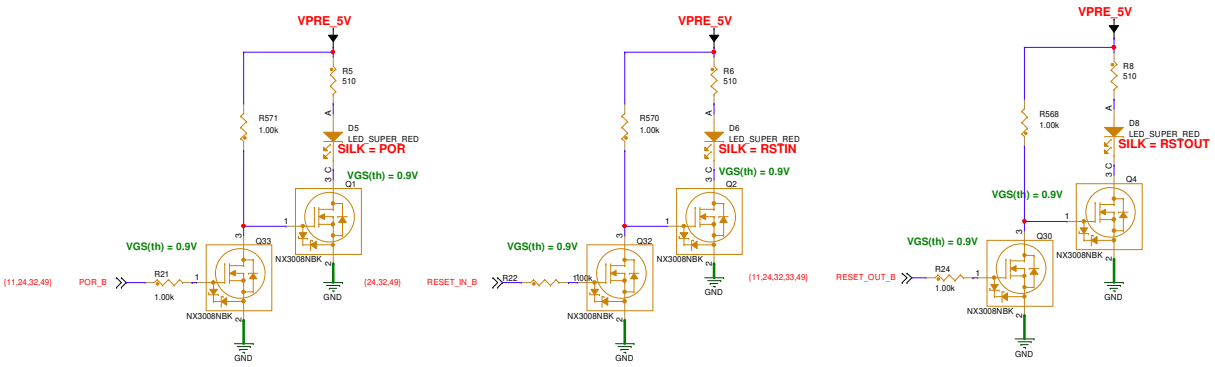
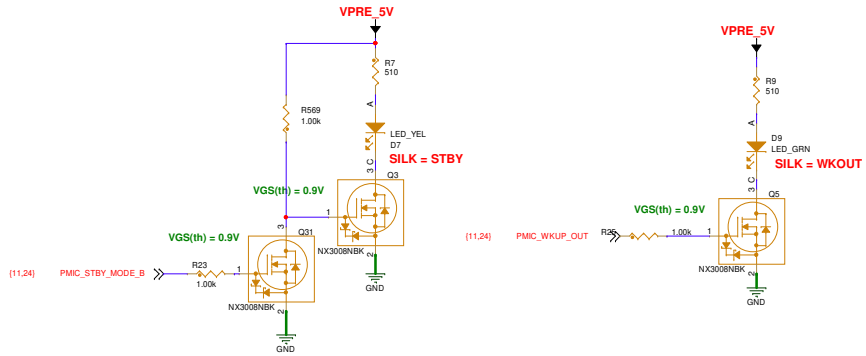
LED FOR RESET

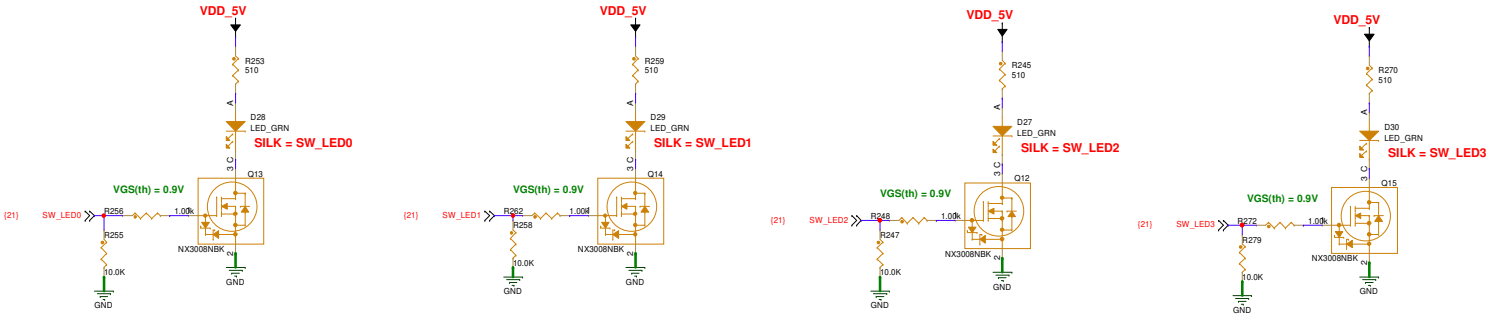


LED FOR STBY



LED	SILK	LED COLOR	DEFAULT STATE
D1	BRD 3.3V	LED_GRN	ON
D2	REMFTDI_GPIO	LED_GRN	OFF(FTDI SW Control)
D3	BRD 1.8V	LED_GRN	ON
D4	VDD_IO_1V8	LED_GRN	ON
D5	POR	LED_SUPER_RED	OFF
D6	RSTIN	LED_SUPER_RED	OFF
D7	STBY	LED_YEL	OFF
D8	RSTOUT	LED_SUPER_RED	OFF
D9	WKOUT	LED_GRN	OFF
D10	SUSP	LED_SUPER_RED	OFF
D11	DBG_PWREN3	LED_GRN	ON
D12	3V3_DBG	LED_GRN	ON(If Debug USB is connected), Else Off
D13	RGMII PG	LED_GRN	ON
D14	DBG_SUSPEND3	LED_YEL	OFF
D15	LINK	SML-D15MWT86C	OFF
D16	SFP1_LOS	LED_SUPER_RED	OFF
D17	SFP2_LOS	LED_SUPER_RED	OFF
D18	SFP3_LOS	LED_SUPER_RED	OFF
D19	SFP4_LOS	LED_SUPER_RED	OFF
D20	SSD STATUS	SML-D15MWT86C	OFF
D21	BRD 5V	LED_GRN	ON
D22	VDD_1.1V	LED_GRN	ON
D23	VDD_2.5V	LED_GRN	ON
D24	VDD_SW_0.77V	LED_GRN	ON
D25	VDD_RGMII_1.8V	LED_GRN	ON
D26	VDD_1.2V	LED_GRN	ON
D27	SW_LED2	LED_GRN	SW/GPIO Control
D28	SW_LED0	LED_GRN	SW/GPIO Control
D29	SW_LED1	LED_GRN	SW/GPIO Control
D30	SW_LED3	LED_GRN	SW/GPIO Control
D31	VDD_SW_APS_0.77V	LED_GRN	ON
D32	VDD_0.77V	LED_GRN	ON
D33	SYS RESET	LED_SUPER_RED	OFF
D34	SYS_PG	LED_GRN	ON
D36	DBG_PWREN2	LED_GRN	ON
D38	DBG_SUSPEND2	LED_YEL	OFF
D41	DBG_SUSPEND1	LED_YEL	OFF
D42	VDD2_LP5X	LED_GRN	ON
D45	DBG_PWREN1	LED_GRN	ON
D46	VDD_AO_0V77	LED_GRN	ON
D48	VDD_IO_1V8_AO	LED_GRN	ON
D49	12V IN	LED_GRN	ON
D50	VDD_IO_1V8_HK	LED_GRN	ON
D51	VDDA_1V5_ANA	LED_GRN	ON
D52	VDD_IO_1V2_UFS	LED_GRN	ON
D53	VDD_IO_3V3	LED_GRN	ON
D56	12V	LED_GRN	ON

LED FOR SOFTWARE



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Classification:
Drawing Title: **S32N79-RDB-S**
Page Title: **49-LED**

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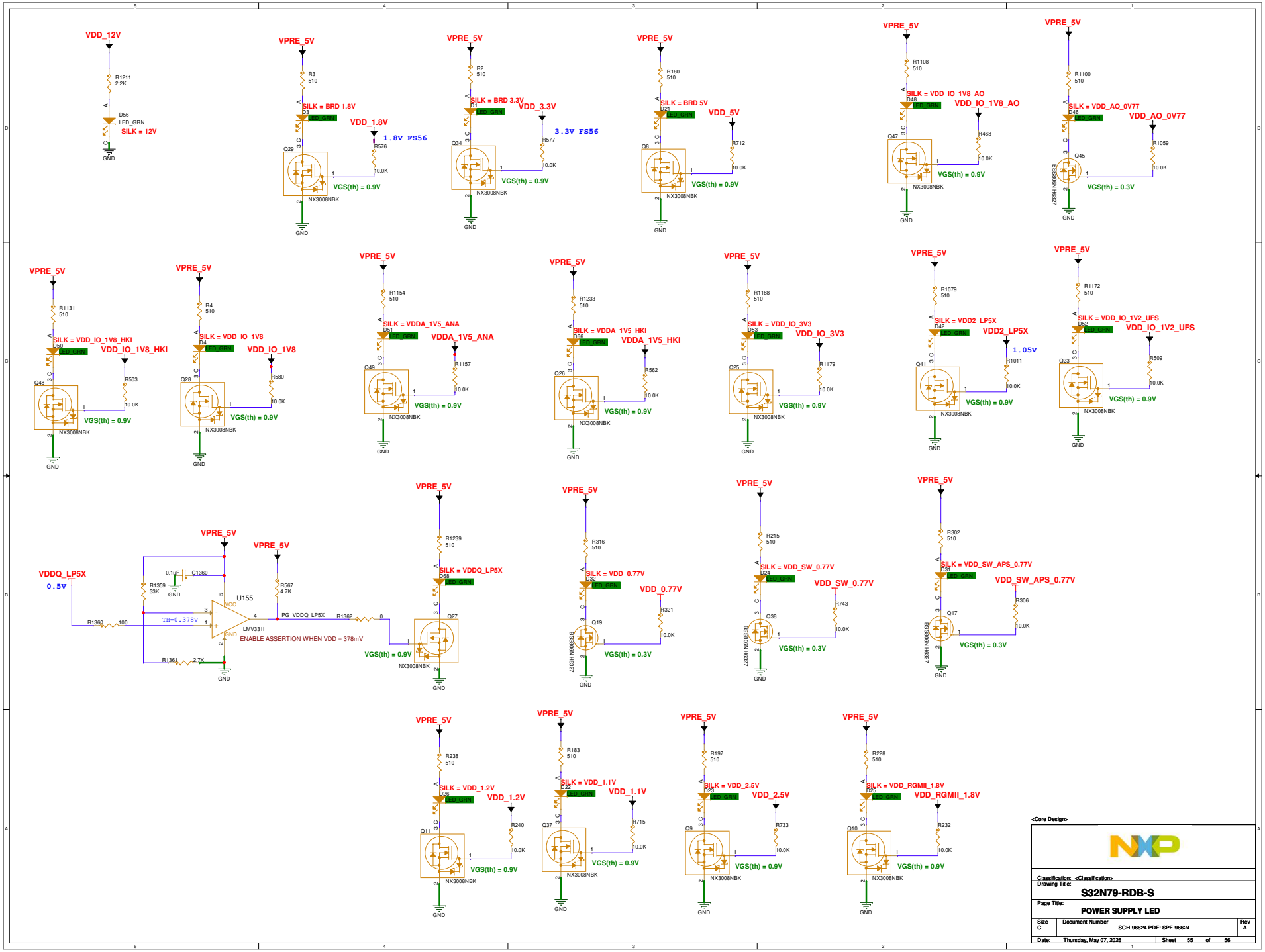
VDD_12V
R1211 2.2K
D56 LED_GRN
SILK = 12V
GND
VPRE_5V
R3 510
SILK = BRD 1.8V
D3 LED_GRN
VDD_1.8V
1.8V FS56
R576 10.0K
Q29 NX3008NBK
VGS(th) = 0.9V
R2 510
SILK = BRD 3.3V
D1 LED_GRN
VDD_3.3V
3.3V FS56
R577 10.0K
Q34 NX3008NBK
R180 510
SILK = BRD 5V
D21 LED_GRN
VDD_5V
R712 10.0K
Q8 NX3008NBK
R1108 510
SILK = VDD_IO_1V8_AO
D48 LED_GRN
VDD_IO_1V8_AO
R468 10.0K
Q47 NX3008NBK
R1100 510
SILK = VDD_AO_0V77
D46 LED_GRN
VDD_AO_0V77
R1059 10.0K
Q45 BSS816N H6327
VGS(th) = 0.3V
R1131 510
SILK = VDD_IO_1V8_HKI
D50 LED_GRN
VDD_IO_1V8_HKI
R503 10.0K
Q48 NX3008NBK
R4 510
SILK = VDD_IO_1V8
D4 LED_GRN
VDD_IO_1V8
R580 10.0K
Q28 NX3008NBK
R1154 510
SILK = VDDA_1V5_ANA
D51 LED_GRN
VDDA_1V5_ANA
R1157 10.0K
Q49 NX3008NBK
R1233 510
SILK = VDDA_1V5_HKI
D66 LED_GRN
VDDA_1V5_HKI
R562 10.0K
Q26 NX3008NBK
R1188 510
SILK = VDD_IO_3V3
D53 LED_GRN
VDD_IO_3V3
R1179 10.0K
Q25 NX3008NBK
R1079 510
SILK = VDD2_LP5X
D42 LED_GRN
VDD2_LP5X
1.05V
R1011 10.0K
Q41 NX3008NBK
R1172 510
SILK = VDD_IO_1V2_UFS
D52 LED_GRN
VDD_IO_1V2_UFS
R509 10.0K
Q23 NX3008NBK
VDDQ_LP5X
0.5V
0.1uF C1360
R1359 33K
R1360 100
R1361 2.7K
U155
LMV331I
TH=0.378V
ENABLE ASSERTION WHEN VDD = 378mV
R567 4.7K
PG_VDDQ_LP5X
R1362 0
R1239 510
SILK = VDDQ_LP5X
D68 LED_GRN
Q27 NX3008NBK
R316 510
SILK = VDD_0.77V
D32 LED_GRN
VDD_0.77V
R321 10.0K
Q19 BSS806N H6327
R215 510
SILK = VDD_SW_0.77V
D24 LED_GRN
VDD_SW_0.77V
R743 10.0K
Q38 BSS806N H6327
R302 510
SILK = VDD_SW_APS_0.77V
D31 LED_GRN
VDD_SW_APS_0.77V
R306 10.0K
Q17 BSS806N H6327
R238 510
SILK = VDD_1.2V
D26 LED_GRN
VDD_1.2V
R240 10.0K
Q11 NX3008NBK
R183 510
SILK = VDD_1.1V
D22 LED_GRN
VDD_1.1V
R715 10.0K
Q37 NX3008NBK
R197 510
SILK = VDD_2.5V
D23 LED_GRN
VDD_2.5V
R733 10.0K
Q9 NX3008NBK
R228 510
SILK = VDD_RGMII_1.8V
D25 LED_GRN
VDD_RGMII_1.8V
R232 10.0K
Q10 NX3008NBK
VGS(th) = 0.9V
GND
<Core Design>
Classification: <Classification>
Drawing Title:
S32N79-RDB-S
Page Title:
POWER SUPPLY LED
Size C
Document Number
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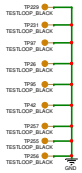
REF DES	ASSY_OPT	PAGE NAME
R488,R489,R491,R492,R528, R529,R532,R533,R1116,R1118, U85,U86	DNP	36-CANXL
C1201,C1249,C1292,C1358,R308, R344,R346,R367,R848,R880, R881,R925,R939,R1350,R1351, R1352	DNP	10-12V PWRIN & FS56
J43	DNP	11-PMIC FS04
C843,R929	DNP	12-PMIC PF09
C115,C121,C127,C131,C134, C136,C138,C139,C141,C142, C143,C150,C152,C157,C158, C159,C164,C166,C167,C176, C179,C187,C189,C190,C192, C194,C195,C199,C200,C208, C211,C213,C215,C219,C222, C226,C232,C242,C245,C252, C253,C254,C256,C259,C261, C280,C282,C283,C593,C642, C1030,L19,L22,L31,R231,R236, R237,R239,R241,R242,R243, R244,R249,R250,R251,R271, R273,R277,R278,R282,R283, R285,R286,R290,R291,R292, R352,R356,R359,R363,R370, R371,R373,R375,R377,R378, R379,R791,R853,R927,R934,SJ1, SJ2,SJ3,SJ4,SJ5,SJ6,TP44, TP45,TP48,TP77,TP79,TP94, TP145,TP146,TP157,U41,U46,U56	DNP	13-PMIC PF53
C169,C172,C175,C177,C184, C186,C240,C244,C249,C255, C268,C271,C596,C638,C995, C1067,R821,R822,R829,R842, R859,R860,R888,R937,R953,R954	DNP	14-PMIC PF5400
R16,R20,R53,R54	DNP	15-PMIC PF5020
C608,R206,R225,R257,R266, R305,R310,R323,R349,R633, R864,R865,R874,R896,R908, R932,R940,R956,R990,R1034, R1040,R1042,R1049,R1363	DNP	16-POWER RMUX & CUR SENSE R
C193,C201,C203,C205,C209, C214,C218,C220,C380,C385, C389,C393,C1084,C1099,C1125, C1145,R281,R295,R543,R958	DNP	17-POWER & CUR MONITORS
R84,R85,R96,R97,R98,R103, R107,R113,R116,R120,R121, R124,R128,R131,R133,R157, R627,R628,R638,R651,R653, R656,R659,R663,R667	DNP	18-CLOCK GENERATOR I
R82,R83,R92,R93,R94,R95,R99, R105,R109,R115,R117,R122, R127,R132,R134,R137,R140, R158,R622,R623,R629,R630, R639,R650,R660,R664,R669	DNP	19-CLOCK GENERATOR II
J57,J58,R557,R989	DNP	20-UFS CLOCK GENERATOR
R550,R965,R998,R999,R1085, R1112,R1113	DNP	22-S32N79 PD-PE
R895,R900,R910,R920,R996, R1025,R1046,R1254	DNP	24-S32N79 JTAG & MISC
R260,R264,R341,R383	DNP	28-S32N79 POWER
C1160	DNP	29-S32N79 DECAPS
R1071,R1078,R1099,R1106,R1107	DNP	31-SD CARD & UFS MEM
R233,R234,R235,R789,R790, R794,R795,R801,R802,R803,R804	DNP	32-RESET & BCSR (DIP, IOEX)
R1090,R1130,R1138	DNP	33 JTAG & TPIU
R402,R405,R408,R411,R523, R524,R1132,R1168	DNP	34-CAN BOOT & HIZ BUFF
R530,R531,R534,R535,R536, R537,R538,R539,R1115,R1133, R1134,R1135,R1136,R1162, R1200,R1201,R1202,R1203, R1204,R1205,R1206,R1207	DNP	35-CAN FD
R434,R905,R922,R941,R1055	DNP	37-LIN & FLEXRAY
C130,C151,R75,R77,R198,R208, R213,R218,R220,R227,R230	DNP	38-RGMII-A KSZ9131 PHY
R29,R32,R33,R80,R184,R204,U17	DNP	39-10BaseT1S TJA1410/2
C88,R164,R166,R167,R176,R676, R678,R688,R691,R695	DNP	41-TJA1120B 1GT1 PHY
R104,R111,R634	DNP	42-SD0 MUX & M.2 NVMe
R1255,R1256,R1258,R1260, R1265,R1266,R1267,R1268, R1298,R1299,R1304,R1305, R1308,R1309,R1310,R1311, R1316,R1317,R1320,R1321	DNP	43-SD0 Culink
C561,C588	DNP	44-SD1 MUX & PCIe SLOT
R1275,R1276,R1285,R1286, R1322,R1323,R1324,R1325, R1330,R1331,R1334,R1335, R1336,R1337,R1338,R1339, R1344,R1345,R1348,R1349	DNP	45-SD1 CuLink
C1365	DNP	46-USB HUB
R1163	DNP	47-FT4232 UART5,6,7 HKLIN
R1060	DNP	48-FT4232 UART0,2,3,4
R168,R169,R620,R1368,R1369, R1370,R1378,R1379,R1384	DNP	49-FT4232 REM DBG
R205,R212,R724,R740	DNP	50-I2C DEVICES
C1247	DNP	51-AUDIO SGTL5000
R540,R1022,R1122	DNP	52-AUDIO A2B AD2433

REF DES	SHORT	PAGE NAME
SJ1,SJ2,SJ3,SJ4,SJ5,SJ6	OPEN	13-PMIC PF53

REF DES	SWITCH(DEFAULT)	PAGE NAME
SW1	OFF: DISABLE 12V	10-12V PWRIN & FS56
S4	1-ON; 2-ON; 3-OFF; 4-ON	20-UFS CLOCK GENERATOR
S1	1-OFF; 2-OFF; 3-OFF; 4-ON	32-RESET & BCSR (DIP, IOEX)
S2	1-OFF; 2-OFF; 3-OFF;4-OFF;5-OFF ;6-OFF;7-ON;8-ON;9-ON;10-ON	32-RESET & BCSR (DIP, IOEX)
S3	1-OFF; 2-OFF; 3-OFF;4-ON;5-ON;6 -OFF;7-OFF;8-ON;9-OFF;10-ON	32-RESET & BCSR (DIP, IOEX)

REF DES	JUMPER(DEFAULT)	PAGE NAME
J62	7-8	10-12V PWRIN & FS56
J45	ON	11-PMIC FS04
J28	1-2	24-S32N79 JTAG & MISC
J26	ON	24-S32N79 JTAG & MISC
J34	OFF	29-S32N79 DECAPS
J42	OFF	33 JTAG & TPIU
J60,J61	OPEN	34-CAN BOOT & HIZ BUFF
J38,J47,J48,J49	OPEN	35-CAN FD
J10	1-2	39-10BaseT1S TJA1410/2
J59	ON	47-FT4232 UART5,6,7 HKLIN
J56	ON	48-FT4232 UART0,2,3,4
J12,J73,J74	OFF	49-FT4232 REM DBG
J30	ON	50-I2C DEVICES
J44	1-2	51-AUDIO SGTL5000



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Page Title:	49-LED
Size: D	Document Number: SCH-99824 PDF: SPF-99824
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